

Benchmark Investigation of Panda: A Pretrained Forecast Model for Chaotic Dynamics

Experiment Log and Analysis

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Abstract

This document records all benchmark experiments conducted on Panda (Lai, Bao, Gilpin, ICLR 2026), a pretrained forecast model for chaotic dynamical systems. The investigation was motivated by Prof. Salim’s direction to test Panda’s emergent capabilities on standard time series and spatiotemporal benchmarks. Each experiment is reported with its motivation, methodology, methodological limitations, direct observations, and competing explanations. Claims are explicitly labelled as: **[OBS]** observed in data, **[PAT]** empirical pattern across multiple conditions, **[HYP]** proposed explanation, **[SPEC]** speculative, or **[EST]** established by prior theory or paper. This labelling convention was decided to maintain distinction between observation and possible interpretations rather than mixing them. Revision convention (July 2026): conclusions shipped at the time are never silently rewritten. Where later analysis revises an interpretation, the original section is preserved and a clearly marked *Revision (July 2026)* addendum records the update and its evidence.

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1 Experimental Setup

1.1 Models

Two models are compared throughout all experiments.

Panda 21M. The base checkpoint from `GilpinLab/panda` on HuggingFace. A 21 million parameter encoder-only transformer trained exclusively on 20,000 synthetic chaotic ordinary differential equations (ODEs). Prediction horizon 128 steps. Context length 512 steps. For horizons exceeding 128 steps, window autoregression is used: predictions are fed back as context for subsequent rollouts.

Chronos 20M. The `amazon/chronos-t5-small` checkpoint. A 20 million parameter causal univariate decoder model trained on a large corpus of real-world time series. Chronos forecasts each channel independently.

The two models are matched in parameter count, making comparisons attributable to architecture and training distribution rather than scale.

Training-corpus contamination check (added July 2026). [\[EST\]](#) The Chronos paper partitions its 55 datasets into training-only (13), Benchmark I (15, used for training and in-domain evaluation), and Benchmark II (27, used solely for zero-shot evaluation). ETT (Hourly), from which ETTh1/ETTh2 derive, is a Benchmark II zero-shot dataset and is not in the `chronos-t5-small` training corpus. The Weather dataset used throughout this log is the Jena/Max-Planck 21-channel 10-minute dataset (Autoformer benchmark), which does not appear in the Chronos corpus in any split; the “Weather” dataset in the Chronos collection is the Monash daily Australian rainfall dataset, a different dataset. Consequently both models are genuinely zero-shot on all real-world evaluations in this log, and the Weather/ETTh asymmetry cannot be explained by Chronos having memorised either dataset. Panda, trained exclusively on synthetic ODEs, cannot be contaminated by construction.

1.2 Evaluation Protocol

Unless stated otherwise, the following protocol is used across all experiments.

Normalisation. Each context window is normalised independently using its own mean and standard deviation per channel (per-window instance normalisation). This prevents global statistics from leaking future information into each evaluation window.

Evaluation windows. A sliding window of context length 512 is moved across the time series. The number of windows is specified per experiment. In early experiments (pre-presentation), $n_{\text{windows}} = 6\text{--}10$ was used. In the fixed experiment set, $n_{\text{windows}} = 8$ or $n_{\text{windows}} = 20$ is used as noted.

Primary metric. Mean Absolute Error (MAE) is the primary metric throughout. Symmetric Mean Absolute Percentage Error (sMAPE) was used in earlier experiments but was found to be unreliable on instance-normalised data where values near zero cause the denominator to vanish, artificially inflating or deflating the metric.

Statistical test. Where stated, the Wilcoxon signed-rank test (one-sided, alternative: model A better than model B) is applied to the per-window MAE values. With $n_{\text{windows}} = 8$,

the minimum achievable p-value is 0.004. With $n_{\text{windows}} = 20$, the minimum achievable p-value is approximately 0.0001. All p-values should be interpreted in light of these floors.

Advantage. Panda advantage is defined as $\text{MAE}_{\text{Chronos}} - \text{MAE}_{\text{Panda}}$. Positive values indicate Panda is better.

Relative skill (added July 2026). Absolute advantage conflates task headroom with model skill when compared across systems of different intrinsic difficulty: a system where both models are near-perfect cannot produce a large advantage regardless of relative skill, and a system where one model collapses produces a large advantage regardless of the other model’s behaviour. The ratio $\text{MAE}_{\text{Chronos}}/\text{MAE}_{\text{Panda}}$ is therefore reported alongside absolute advantage wherever cross-system comparisons are drawn. Ratios of near-zero MAE values are unstable and are always reported with IQRs.

Advantage decomposition (added July 2026). Every advantage value is a difference of two model behaviours. Wherever a condition change moves the advantage, the movement is decomposed into the change in Panda MAE and the change in Chronos MAE before interpretation. Several earlier misreadings (Experiments 12, 16, 19, 20) are traceable to interpreting advantage movements as Panda properties when Chronos was the moving component.

Statistical policy (July 2026 revision). This log contains on the order of 60 or more Wilcoxon tests with no multiplicity correction. Isolated significant results at moderate p-values (for example ETTh2 H = 336, $p = 0.013$ in Experiment 8) are therefore treated as exploratory regardless of direction, and the same standard is applied symmetrically to results that favour and disfavour any hypothesis. Confirmatory status is reserved for results that are (a) pre-registered, (b) run at $n_{\text{windows}} \geq 20$, and (c) replicated or part of a consistent multi-condition pattern. Headline claims currently resting on $n_{\text{windows}} = 8$ (the heterogeneity chain, the complexity continuum, the surrogate control) are scheduled for confirmatory $n = 20$ reruns on the GPU pipeline.

Raw prediction retention (adopted July 2026). Earlier campaigns saved only aggregated MAE values, which means any metric revision requires rerunning inference. All runs from July 2026 onward save raw per-window predictions (context, target, forecast) so that future metric changes are re-scoring operations rather than recomputation.

Estimator validation rule (adopted July 2026). Every analysis estimator is validated on a case with known ground truth before being applied to unknown data. This rule was adopted after two estimator failures (the Rosenstein λ_1 estimator, Experiments 3 and 15, and the correlation dimension estimator, which returns ≈ 0.86 on Lorenz against a literature value of ≈ 2.05) and was applied prospectively in the structure statistic investigation (Section 10), where it caught three further pipeline failures before any real data was interpreted.

1.3 Hardware

Inference experiments were run on a CPU-only laptop; runtimes are consequently long for experiments involving many channels or long horizons. From late June 2026, a Kaggle T4 retraining pipeline is operational (custom PyTorch loop on the GilpinLab/skew40 corpus, throughput approximately 1.4 it/s), enabling the retraining ablations in Section 8 and the planned confirmatory reruns.

2 Pre-Presentation Experiments

These experiments were conducted prior to the group meeting and form the basis of the initial findings presented to Prof. Salim.

2.1 Experiment 1: Standard Time Series Benchmarks (Initial)

2.1.1 Motivation

To establish whether Panda has any advantage on non-chaotic standard benchmarks. ETTh1, ETTh2, and Weather were chosen as the canonical time series forecasting benchmarks used by TimesNet, PatchTST, and Chronos comparisons in the literature.

2.1.2 Method

Datasets: ETTh1 (7 channels, 17,420 timesteps), ETTh2 (7 channels, 17,420 timesteps), Weather (21 channels, 52,696 timesteps). Prediction horizon: 128 steps. $n_{\text{windows}} = 10$. Primary metric: sMAPE. Instance normalisation applied globally to the full dataset before windowing (not per-window).

2.1.3 Methodological Limitations

1. sMAPE is unreliable on instance-normalised data where channel values are near zero. Results are not quantitatively trustworthy.
2. Global normalisation leaks future statistics into evaluation windows.
3. Single prediction horizon (128 steps). Results may not generalise to other horizons.
4. No statistical significance tests.
5. $n_{\text{windows}} = 10$ provides limited statistical power.

2.1.4 Observations

[OBS] Both models produced sMAPE approximately 0.8–0.9 on all three datasets. **[OBS]** Panda slightly worse than Chronos on ETTh1. **[OBS]** Chronos sMAPE = 0.009 on Lorenz rho = 15 was flagged as a metric artifact rather than genuine model performance.

2.1.5 Explanation Shipped at the Time

The conclusion presented was: “Panda fails on non-chaotic data because it was trained on stationary chaotic ODEs with no seasonal inductive bias.” This was stated as a finding despite the invalid metric and the limitation to a single prediction horizon.

2.1.6 Alternative Explanations Not Presented

1. **[HYP]** The result may be specific to $H = 128$. Later fixed experiments showed Panda wins on Weather at all tested horizons and on ETTh1 and ETTh2 at longer horizons.
2. **[HYP]** The sMAPE artifact may have masked the true comparison entirely, making the result uninterpretable.
3. **[HYP]** Weather and ETT differ substantially in channel count and coupling structure. Grouping them as “non-chaotic data” ignored this distinction.

Confidence: Very Low. The original conclusion is largely contradicted by the fixed Experiment 8.

2.2 Experiment 2: Double Pendulum Graded Noise

2.2.1 Motivation

To quantify the cost of measurement noise on a chaotic system and test whether Panda’s advantage over Chronos is robust to noise. This also served as a proxy for the gap between clean synthetic data and noisy real experimental data.

2.2.2 Method

The double pendulum ODE was simulated with Gaussian observation noise at six levels: $\sigma \in \{0, 0.01, 0.05, 0.10, 0.25, 0.50\}$ times the signal standard deviation. One random seed, one initial condition. $n_{\text{windows}} = 8$. sMAPE and MAE both reported.

2.2.3 Methodological Limitations

1. Single seed and single initial condition. One realisation of the trajectory.
2. The real experimental dataset (Asseman et al. 2018) was not obtained. The experiment was described as “synthetic versus real” but was in practice “synthetic versus synthetic with Gaussian noise.” Gaussian additive noise is not the same type of nonstationarity as real experimental measurement error.
3. Noise levels were not tested beyond $\sigma = 0.50$.
4. No statistical significance tests.

2.2.4 Observations

[OBS] Panda MAE at $\sigma = 0$: 0.62. Panda MAE stays approximately flat (0.63–0.67) across all tested noise levels up to $\sigma = 0.50$. **[OBS]** Chronos MAE at $\sigma = 0$: 0.71. Chronos MAE peaks at approximately 0.77 at high noise. **[OBS]** Panda consistently lower MAE than Chronos at every tested noise level. **[OBS]** sMAPE panel showed erratic non-monotone behaviour, correctly identified as a metric artifact on normalised data.

2.2.5 Explanation Shipped at the Time

“The random Fourier and polynomial features are smooth functions of the input, making the embedding noise-robust. Takens’ theorem is noise-tolerant.” This was stated as a conclusion rather than a hypothesis.

2.2.6 Alternative Explanations Not Presented

1. **[HYP]** Chronos uses a discrete tokenisation scheme. Continuous Gaussian noise may specifically disrupt this tokenisation, independently of any property of Panda’s architecture.
2. **[HYP]** Panda’s predictions may have lower variance by default. A flat MAE noise curve could reflect low-variance predictions rather than genuine noise robustness.
3. **[HYP]** The tested noise levels may be below the breakdown threshold for both models. The claim cannot be extrapolated to higher noise regimes.

Confidence: Medium. for the observation. **Confidence: Low.** for the proposed mechanism.

Revision note (July 2026). The early-campaign Hellinger preview (Section 1.2, statistical policy; data from the original benchmark run) shows that on Double Pendulum rows the MAE winner and the Hellinger winner disagree in 6 of 8 cases, with Chronos winning under Hellinger. Double Pendulum conclusions in this section are therefore among those most at risk under the planned distributional-metric revalidation and should not be cited until it completes.

2.3 Experiment 3: Lorenz Rho Sweep

2.3.1 Motivation

To test whether Panda’s advantage specifically tracks the onset of chaos. The Lorenz parameter ρ controls the transition from periodic ($\rho < 24.74$) to chaotic ($\rho \geq 24.74$) dynamics. This provides a controlled test of the chaos-threshold hypothesis.

2.3.2 Method

$\rho \in \{10, 15, 20, 24, 24.74, 26, 28, 35, 45, 60\}$. One trajectory per ρ value, one initial condition. $n_{\text{windows}} = 8$. MAE as primary metric. Maximum Lyapunov exponent estimated using the Rosenstein method. The estimator required three implementation attempts before producing physically reasonable output.

2.3.3 Methodological Limitations

1. The Lyapunov estimator systematically overestimates. For the Lorenz system at $\rho = 28$, the literature value is $\lambda_1 \approx 0.9$ but the estimator returned 1.66. The x-axis of Lyapunov-based figures is not quantitatively reliable.
2. Single trajectory and single seed per ρ value.
3. $n_{\text{windows}} = 8$ with no confidence intervals on advantage values.
4. Results at $\rho = 10, 15$ showed Chronos sMAPE near zero in the periodic regime, correctly identified as an artifact.

2.3.4 Observations

ρ	Regime	$\hat{\lambda}_1$	Panda MAE	Chronos MAE	Advantage
10.00	Periodic	-0.29	≈ 0.02	≈ 0.02	≈ 0
15.00	Periodic	-0.33	≈ 0.04	≈ 0.04	≈ 0
20.00	Near transition	0.20	≈ 0.09	≈ 0.31	+0.22
24.00	Near transition	0.33	≈ 0.10	≈ 0.44	+0.34
24.74	Onset of chaos	1.75	≈ 0.18	≈ 0.65	+0.47
28.00	Chaotic	1.66	≈ 0.25	≈ 0.89	+0.64
60.00	Chaotic	1.63	≈ 0.48	≈ 1.17	+0.69

Table 1: Lorenz ρ sweep results (approximate values from $n_{\text{windows}} = 8$, one seed).

[OBS] Estimated λ_1 is negative at $\rho = 10, 15$ and jumps to 1.75 at $\rho = 24.74$, consistent with the theoretical bifurcation point. **[OBS]** Panda MAE advantage is approximately

zero at $\rho = 10, 15, 20$. The advantage jumps near $\rho = 24.74$ and remains positive across all tested chaotic ρ values. **[OBS]** No tested chaotic ρ value produced a result where Chronos had lower MAE than Panda. **[PAT]** The pattern of advantage increasing with ρ is consistent across the chaotic range, though not strictly monotone.

2.3.5 *Explanation Shipped at the Time*

“Panda’s advantage is specifically activated by the chaotic transition. The positive Lyapunov exponent is necessary and sufficient for Panda’s advantage.” This was stated as a conclusion.

2.3.6 *Alternative Explanations Not Presented*

1. **[HYP]** Signal statistics (variance, spectral content, amplitude) change discontinuously at the bifurcation. Any model better suited to high-variance aperiodic signals would show this pattern, independently of any chaos-specific mechanism. This alternative is not yet ruled out.
2. **[HYP]** Chronos tokenisation may fail specifically on aperiodic continuous signals, making Chronos worse in the chaotic regime regardless of Panda’s capabilities.
3. **[HYP]** The advantage may be driven primarily by the Koopman dynamics embedding rather than any chaos-specific learning. This is not testable without an ablation removing the dynamics embedding.

What would falsify the claim: A chaotic system where Panda does not win, or a non-chaotic system where Panda wins substantially. The Weather and Burgers sweep results (Experiments 8 and 10) partially address this.

Confidence: Medium. for the observation. **Confidence: Low.** for the chaos-specific mechanism hypothesis.

Revision note (July 2026). The Hellinger preview flags $\rho = 10$: Panda’s Hellinger distance (0.500) is far worse than Chronos’s (0.083) despite near-zero MAE for both. Point-metric ties in the periodic regime may conceal distributional failure. Periodic-regime claims from this sweep await the distributional revalidation. The $\hat{\lambda}_1$ jump at exactly 24.74 in the table above should also be treated cautiously given the estimator’s documented bias; the third alternative explanation (Koopman embedding as driver) is now under direct test in Experiment 28.

2.4 Experiment 4: dysts Systems and Advantage versus Lambda1

2.4.1 *Motivation*

To test Panda’s advantage across a broader set of known chaotic systems from the **dysts** library and to measure whether a Spearman correlation between estimated λ_1 and Panda advantage reaches statistical significance.

2.4.2 *Method*

Systems tested: Lorenz, Rossler, Chua, Duffing, Halvorsen, SprottB, Thomas, Rucklidge, Dadras, Bouali, DequanLi. $n_{\text{windows}} = 8$ per system. Lyapunov estimates via Rosenstein

method. Additional systems were added iteratively until the Spearman p-value crossed 0.05.

2.4.3 Methodological Limitations

1. **P-hacking.** Systems were added iteratively until $p < 0.05$ was achieved. The reported Spearman $\rho = 0.370$, $p = 0.044$ is not a valid pre-registered result. The true p-value corrected for sequential testing is unknown but higher.
2. Systems in the **dysts** library are not independent (several share structural similarities).
3. Lyapunov estimates are noisy and systematically biased (see Experiment 3).
4. Single seed per system. No cross-system multiple comparison correction.

2.4.4 Observations

[OBS] Panda wins (lower MAE) on every tested **dysts** chaotic system. **[OBS]** Spearman $\rho = 0.370$, $p = 0.044$ after iterative addition of systems. This p-value is not reliable due to p-hacking. **[OBS]** Two visual sub-clusters: Lorenz family above the trend line, double pendulum below the trend line.

2.4.5 Explanation Shipped at the Time

“Panda has learned something universal about chaotic dynamics. Chaos is necessary and sufficient. In-distribution systems show larger advantage.” Overstated given the methodological problems.

2.4.6 Alternative Explanations Not Presented

1. **[HYP]** The correlation may be driven entirely by the Lorenz family cluster. Removing those points may eliminate significance entirely.
2. **[HYP]** The sub-cluster structure (Lorenz family above trend, double pendulum below) suggests training-distribution effects rather than universality. This directly challenges the “universal chaos learning” claim.

Confidence: Low. due to p-hacking. The observation that Panda wins on every tested chaotic system is more robust (**Confidence: Medium.**) but still limited to $n = 8$ windows per system.

2.5 Experiment 5: Burgers Viscosity Sweep (First Version)

2.5.1 Motivation

To test whether the chaos-threshold pattern found in the Lorenz rho sweep extends to partial differential equations (PDEs). Burgers equation viscosity ν controls the transition from smooth diffusion-dominated to chaotic shock-dominated dynamics, analogous to ρ in Lorenz.

2.5.2 Method

$\nu \in \{0.1, 0.05, 0.02, 0.01, 0.005\}$. PCA reduction to 16 channels. $n_{\text{windows}} = 8$. Lyapunov estimated from first PCA component. Solver diverged at $\nu = 1.0$, leaving no non-chaotic

baseline.

2.5.3 Methodological Limitations

1. No non-chaotic baseline. Without $\nu \geq 0.5$, it is impossible to verify a threshold effect.
2. Lyapunov estimation returned approximately zero for all conditions. The chaos-threshold connection could not be verified from the data.
3. Non-monotone reversal of advantage at $\nu = 0.02 \rightarrow 0.01$ was not discussed.
4. PCA representation changes character as ν changes, confounding the comparison across conditions.

2.5.4 Observations

[OBS] Panda advantage positive at all tested ν values. **[OBS]** Advantage roughly increases with decreasing ν but with a non-monotone reversal at $\nu = 0.02 \rightarrow 0.01$.

Current status: Superseded by the fixed Experiment 10, which includes a non-chaotic baseline and a stable solver.

2.6 Experiment 6: PCA versus Spatial Subsampling (First Version)

2.6.1 Motivation

To compare two methods for converting a 1D spatiotemporal PDE field into channels for Panda: global PCA modes versus raw spatial location subsampling.

2.6.2 Method

Single ν value (0.005). Single spatial location ($x = 0$) for subsampling. $n_{\text{windows}} = 6$.

2.6.3 Critical Flaw

The spatial location $x = 0$ was a near-nodal point of the Burgers field with near-constant dynamics. Both models trivially predicted near-zero values and scored well. No general conclusion about spatial subsampling is possible from one degenerate location. This experiment was rerun in fixed Experiment 12 with four methods including a variance-stratified control.

2.6.4 Observations

[OBS] Both models performed approximately $8\times$ better in absolute MAE on spatial subsampling than on PCA. **[OBS]** Panda relative advantage larger with PCA (0.093) than subsampling (0.044).

Confidence: Very Low. for any conclusion. Single degenerate location.

3 Fixed Experiments

These experiments address the methodological problems identified in the pre-presentation set. A new notebook (`fixed_experiments.ipynb`) was created. Key improvements: per-window normalisation, increased n_{windows} , Wilcoxon signed-rank tests, multiple prediction horizons, non-chaotic baselines, and removal of oracle leakage.

3.1 Experiment 7: Standard Horizon Evaluation (Week 1, Invalid)

The Week 1 version of standard horizon evaluation retained global normalisation leakage and used $n_{\text{windows}} = 6$ with sMAPE as primary metric. Results were superseded by Experiment 8 and are not reported here.

3.2 Experiment 8: Standard Horizon Evaluation (Fixed)

3.2.1 Motivation

To establish the true performance gap between Panda and Chronos on standard non-chaotic benchmarks across the full range of prediction horizons used in the literature.

3.2.2 Method

Datasets: ETTh1 (7 channels), ETTh2 (7 channels), Weather (21 channels). Horizons: $H \in \{96, 192, 336, 720\}$. $n_{\text{windows}} = 20$. Per-window normalisation. Wilcoxon signed-rank tests. Weather $H = 720$ was not completed due to hardware time constraints.

3.2.3 Observations

Dataset	H	Panda MAE	IQR	Chronos MAE	IQR	Adv (p)
ETTh1	96	0.7269	0.2203	0.6633	0.2388	−0.064 (0.844)
ETTh1	192	0.8185	0.3046	0.7825	0.2742	−0.036 (0.826)
ETTh1	336	0.8571	0.1971	0.9013	0.1859	+0.044 (0.649)
ETTh1	720	0.9921	0.3226	1.0189	0.3845	+0.027 (0.774)
ETTh2	96	0.8736	0.4790	0.9494	0.5583	+0.076 (0.478)
ETTh2	192	0.9697	0.5380	0.9505	0.5730	−0.019 (0.147)
ETTh2	336	0.9255	0.3860	1.1101	0.3831	+0.185 (0.013)*
ETTh2	720	1.1139	0.2194	1.1027	0.3862	−0.011 (0.392)
Weather	96	0.6378	0.1723	0.8115	0.2036	+0.174 (0.000)*
Weather	192	0.7224	0.2385	0.9582	0.2089	+0.236 (0.001)*
Weather	336	0.8481	0.2898	1.0843	0.3507	+0.236 (0.000)*

Table 2: Fixed Experiment 8. * denotes $p < 0.05$ (Wilcoxon, $n = 20$ windows).

[OBS] On ETTh1, no horizon shows statistically significant difference in either direction. All p-values exceed 0.6. The direction of advantage is inconsistent across horizons. **[OBS]** On ETTh2, one horizon ($H = 336$) shows significant Panda advantage ($p = 0.013$). Other horizons are not significant and inconsistent in direction. **[OBS]** On Weather, Panda has statistically significant advantage at $H = 96, 192, 336$ (all $p \leq 0.001$). Advantage magnitude is 0.17–0.24, consistent across horizons. **[PAT]** Weather is the only dataset where Panda shows consistent, statistically significant advantage across multiple horizons.

3.2.4 Competing Explanations for the Weather Result

1. **[HYP]** Weather has 21 channels versus 7 for ETT. More channels provide more information regardless of coupling. Any multivariate model might benefit.
2. **[HYP]** Weather has genuine multivariate physical coupling (pressure drives wind drives temperature). Channel attention may capture this.
3. **[HYP]** Chronos is specifically poorly suited to Weather’s complex multi-scale spectral structure (10-minute sampling, 144-step daily cycle).
4. **[SPEC]** Panda handles quasi-chaotic components of Weather (atmospheric turbulence at synoptic scales) better than Chronos due to its chaotic training distribution.

Confidence: High. for the Weather observation ($n = 20$, $p < 0.001$). **Confidence: Low.** for any proposed mechanism.

3.2.5 Revision (July 2026): Relative Skill, Multiplicity, Contamination

[OBS] Relative skill on Weather is strikingly stable across horizons: $\text{MAE}_C/\text{MAE}_P = 1.272, 1.326, 1.279$ at $H = 96, 192, 336$. Panda’s edge is a roughly constant multiplicative factor rather than a horizon-dependent effect. Under the July 2026 statistical policy, the isolated ETTh2 $H = 336$ result ($+0.185$, $p = 0.013$; one of eight ETT tests, uncorrected) is reclassified as exploratory. It is neither cited as evidence of ETTh transfer nor silently discarded; it is queued for replication in the confirmatory campaign. The contamination check (Section 1.1) closes the alternative explanation that the Weather/ETTh asymmetry reflects Chronos training exposure: both datasets are zero-shot for Chronos. Channel-count (hypothesis 1 above) is partially addressed by the heterogeneity series, where 7-channel Weather subsets reproduce large advantages; hypothesis 3 is directly addressed and rejected in its sampling-rate form by Experiment 32 only for the structure statistic, not for model behaviour — the corresponding model-side intervention (Panda and Chronos on downsampled Weather) is designed and queued.

3.3 Experiment 9: Univariate Ablation on Weather

3.3.1 Motivation

To directly test whether channel attention drives Panda’s advantage on Weather. If channel attention is responsible, removing it (by forecasting each channel independently) should substantially reduce Panda’s advantage over Chronos.

3.3.2 Method

Multivariate Panda (all 21 channels processed jointly) was compared against univariate Panda (each channel processed independently, suppressing cross-channel attention) at $H \in \{96, 336\}$. $n_{\text{windows}} = 8$.

3.3.3 Observations

[OBS] At both horizons, univariate Panda has lower MAE than multivariate Panda.

[OBS] Neither result is statistically significant at $\alpha = 0.05$ ($p = 0.074$, $p = 0.371$). **[OBS]** Effect size is small (0.03–0.06 MAE difference).

H	panda_uni	panda_multi	Adv (uni better)	p
96	0.5541	0.6113	+0.057	0.074
336	0.8467	0.8762	+0.030	0.371

Table 3: Univariate ablation. Positive advantage means univariate Panda has lower MAE than multivariate Panda.

3.3.4 What This Establishes and Does Not Establish

The data is inconsistent with the hypothesis that channel attention drives Panda’s Weather advantage. Channel attention does not help and may marginally hurt on Weather. However the effect is small and not significant at $n = 8$ windows. A definitive conclusion requires more windows.

3.3.5 Competing Explanations

1. **[HYP]** The temporal architecture (non-causal encoder, Koopman patch embedding, fixed prediction head) is sufficient to beat Chronos. Channel attention adds noise on non-chaotic data. **Confidence: Medium.:** consistent with data.
2. **[HYP]** $n = 8$ is too small to detect a real but small positive channel attention effect. The true effect may be positive but undetectable. **Confidence: Medium.:** cannot rule out.
3. **[SPEC]** Channel attention trained on chaotic ODE coupling learns patterns counterproductive on periodic non-chaotic data. **Confidence: Low.:** speculative.

Confidence: Medium. for the observation. The channel attention hypothesis is not supported but not conclusively falsified.

3.4 Experiment 10: Burgers Viscosity Sweep (Fixed)

3.4.1 Motivation

To test the chaos-threshold hypothesis for PDEs with a proper non-chaotic baseline. The stable solver prevents divergence at high ν , enabling comparison across the full smooth-to-chaotic range.

3.4.2 Method

$\nu \in \{2.0, 1.0, 0.5, 0.1, 0.05, 0.02, 0.01, 0.005\}$. Adaptive timestep solver (CFL stability condition satisfied at all ν). PCA to 16 channels. $n_{\text{windows}} = 8$. $H = 128$.

3.4.3 Observations

[OBS] At $\nu = 2.0$, Panda advantage is +0.004, $p = 0.273$. Not significant. This is the strongly diffusion-dominated regime. **[OBS]** At $\nu = 1.0$, $p = 0.004$. Significant. Burgers at $\nu = 1.0$ is diffusion-dominated and non-chaotic by standard criteria ($\lambda_1 \leq 0$). **[OBS]** Panda wins significantly at $\nu = 1.0$ and $\nu = 0.5$, which are not chaotic regimes. **[OBS]** Advantage is not monotonically increasing with decreasing ν . It peaks near $\nu = 0.05$ (0.149), drops at $\nu = 0.02$ (0.120) and $\nu = 0.01$ (0.095), then rises at $\nu = 0.005$ (0.122).

ν	Physical regime	Panda MAE	Chronos MAE	Adv (p)
2.000	Diffusion dominated	0.0152	0.0191	+0.004 (0.273)
1.000	Diffusion dominated	0.0186	0.0568	+0.038 (0.004)*
0.500	Transition	0.0402	0.1023	+0.062 (0.004)*
0.100	Weakly nonlinear	0.1127	0.2239	+0.111 (0.004)*
0.050	Shock forming	0.1374	0.2865	+0.149 (0.004)*
0.020	Strong shocks	0.1394	0.2593	+0.120 (0.004)*
0.010	Strong shocks	0.1375	0.2325	+0.095 (0.004)*
0.005	Strong shocks	0.1509	0.2732	+0.122 (0.004)*

Table 4: Fixed Burgers viscosity sweep. * denotes $p = 0.004$, the minimum achievable with $n = 8$ windows.

3.4.4 Critical Finding

[PAT] The chaos-threshold pattern observed in the Lorenz ρ sweep does not replicate cleanly in the Burgers viscosity sweep. Panda wins significantly at non-chaotic viscosity values ($\nu = 1.0$, $\nu = 0.5$) where a chaos-specific mechanism would predict no advantage.

3.4.5 Competing Explanations

1. **[HYP]** Even at $\nu = 1.0$, the PCA channels of the Burgers field are spatially coupled through the diffusion operator. Panda’s channel attention may capture coupling that is present even in non-chaotic smooth flows. **Confidence: Medium..**
2. **[HYP]** Chronos is poorly suited to PCA modal time series regardless of chaoticity. The advantage reflects Chronos weakness rather than Panda strength in the non-chaotic regime. **Confidence: Medium.:** consistent with decomposition results.
3. **[HYP]** The single non-significant result at $\nu = 2.0$ versus the significant result at $\nu = 1.0$ may be noise. The true threshold may lie between 1.0 and 2.0 rather than at the expected chaos onset. **Confidence: Medium.:** cannot rule out with $n = 8$.
4. **[SPEC]** Burgers PCA modes may have spectral characteristics that look dynamically complex to Panda even at non-chaotic ν , due to the mode-mixing nature of PCA on a spatially heterogeneous field.

What would distinguish these: Estimating λ_1 from the PCA component time series at each ν value using the corrected Rosenstein estimator. If $\lambda_1 < 0$ at $\nu = 1.0$ but Panda still wins significantly, explanations 1, 2, or 3 are supported over a chaos-specific mechanism.

Confidence: Medium. for the observation. **Confidence: Low.** for any specific mechanism.

Note: explanation 1 (channel attention captures diffusion coupling) was subsequently tested and not supported by the univariate ablation at these viscosities (Experiment 27).

3.4.6 Revision (July 2026): Relative Skill Inverts the Trend

[OBS] Recomputing this table as relative skill $\text{MAE}_C/\text{MAE}_P$: $\nu = 2.0$: 1.26 (n.s.); $\nu = 1.0$: **3.06**; $\nu = 0.5$: 2.54; $\nu = 0.1$: 1.99; $\nu = 0.05$: 2.09; $\nu = 0.02$: 1.86; $\nu = 0.01$: 1.69; $\nu = 0.005$: 1.81. **[PAT]** Under relative skill there is no advantage-grows-with-chaos trend. Relative skill peaks at the non-chaotic $\nu = 1.0$ and is roughly flat at 1.7–2.1

throughout the chaotic regime. The absolute-advantage trend in the original table reflects growing task difficulty (both MAEs grow as ν falls), not growing relative capability. This strengthens the anti-chaos-specificity finding beyond the original framing. Caveat: at $\nu \geq 1.0$ both MAEs are near zero and the ratio is unstable; the confirmatory rerun carries IQRs on the ratio. The PCA-representation confound (explanation 4) remains open and is the subject of the queued representation-arm experiment (raw spatial versus PCA versus Fourier channels at fixed ν).

3.5 Experiment 11: FFT Decomposition Preprocessing (Fixed)

3.5.1 Motivation

To test whether removing periodic and trend components from a time series before presenting it to Panda changes Panda’s relative advantage. An earlier version of this experiment (Week 1) contained oracle leakage: the future trend and seasonal components were extracted from the full series and given directly to both models. The fixed version uses only context-window statistics.

3.5.2 Method

FFT decomposition applied to each channel of the context window only. Deterministic component (trend plus seasonal harmonics) projected forward naively: linear trend extrapolation plus repetition of the last full seasonal cycle. Both models forecast the residual; the projected deterministic is added back before evaluation. Comparison against vanilla (no decomposition) run in the same evaluation loop. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

3.5.3 Note on the Invalid Earlier Version

The oracle-leaking version of this experiment reported a 65% reduction in MAE for both models after decomposition. The corrected version shows a 25% increase in Panda MAE on ETTh1 at $H = 336$. The two results are in opposite directions. This confirms that the earlier result was entirely an artifact of providing future periodic information to both models.

3.5.4 Observations

[OBS] Decomposition increases absolute MAE for both models on ETTh1 at $H = 336$ (Panda +25%, Chronos +15%) and on Weather at $H = 96$ (Panda +11%, Chronos +0.1%). **[OBS]** On Weather, decomposition collapses Panda’s advantage. Vanilla $H = 96$: advantage = +0.080, $p = 0.020$. Decomp $H = 96$: advantage = +0.010, $p = 0.230$. **[OBS]** At $H = 336$ on Weather, the decomp advantage remains statistically significant ($p = 0.020$) but the effect size is very small (+0.015 versus vanilla +0.114).

3.5.5 Competing Explanations

1. **[HYP]** Panda’s advantage on Weather comes from better handling of the periodic and trend components. On the residual, both models are comparable. **Confidence: Medium.**: consistent with all available data.
2. **[HYP]** The naive FFT projection introduces large errors in the projected deterministic component, degrading both models. The result reflects projection error

Dataset	Cond.	H	Panda	Chronos	Adv (p)
ETTh1	Vanilla	96	0.7205	0.8157	+0.095 (0.422)
ETTh1	Decomp	96	0.8032	0.8293	+0.026 (0.578)
ETTh1	Vanilla	336	0.8407	0.8856	+0.045 (0.273)
ETTh1	Decomp	336	1.0474	1.0174	−0.030 (0.770)
ETTh2	Vanilla	96	0.9329	1.0409	+0.108 (0.230)
ETTh2	Decomp	96	0.8856	0.9640	+0.078 (0.273)
Weather	Vanilla	96	0.6082	0.6880	+0.080 (0.020)*
Weather	Decomp	96	0.6771	0.6871	+0.010 (0.230)
Weather	Vanilla	336	0.8762	0.9987	+0.114 (0.004)*
Weather	Decomp	336	1.0609	1.0760	+0.015 (0.020)*

Table 5: Fixed FFT decomposition experiment. * denotes $p < 0.05$.

rather than genuine comparison on the residual. **Confidence: Medium.:** the 25% MAE increase on ETTh1 is suspicious and may reflect projection error.

3. **[SPEC]** The periodic component of Weather contains genuine multivariate coupling. Removing it leaves a weakly coupled residual where channel attention is irrelevant.

Confidence: High. for the observation. **Confidence: Low.** for any mechanism.

3.6 Experiment 12: Subsampling Methods (Fixed)

3.6.1 Motivation

To test whether dynamical diversity subsampling of spatial locations gives Panda a larger advantage than uniform or variance-stratified subsampling. The variance-stratified baseline was introduced to control for the confound in the earlier experiment, where a degenerate near-zero-variance location artificially improved both models.

3.6.2 Method

Four methods: (1) Uniform spacing, (2) Variance-stratified uniform (excludes bottom 10% variance locations before uniform spacing), (3) PCA (global spatial modes), (4) Diversity (farthest-point sampling in a dynamical feature space: standard deviation, mean absolute value, 90th percentile amplitude, spectral entropy). $\nu \in \{0.05, 0.005\}$. $N_{\text{channels}} = 16$. $n_{\text{windows}} = 8$. $H = 128$.

3.6.3 Observations

[OBS] All results reach $p = 0.004$, the minimum achievable Wilcoxon p-value with $n = 8$ windows. The p-values are therefore uninformative for comparing methods against each other. **[OBS]** Panda absolute MAE across Uniform, Stratified, and Diversity methods is nearly identical at both ν values. At $\nu = 0.005$: Uniform = 0.0288, Stratified = 0.0301, Diversity = 0.0300. **[OBS]** Chronos MAE varies substantially across methods. Diversity locations produce higher Chronos MAE (0.2475) than Uniform locations (0.1696) at $\nu = 0.005$. **[OBS]** Diversity advantage (0.218) exceeds Stratified advantage (0.148) at $\nu = 0.005$. However, the difference is entirely attributable to Chronos MAE (0.2475 vs 1.779). Panda MAE is the same (0.0300 vs 0.0301).

ν	Method	Panda MAE	Chronos MAE	Adv	p
0.05	Uniform	0.0317	0.0765	+0.045	0.004
0.05	Stratified	0.0296	0.0736	+0.044	0.004
0.05	PCA	0.1374	0.2915	+0.154	0.004
0.05	Diversity	0.0317	0.1132	+0.082	0.004
0.005	Uniform	0.0288	0.1696	+0.141	0.004
0.005	Stratified	0.0301	0.1779	+0.148	0.004
0.005	PCA	0.1509	0.3029	+0.152	0.004
0.005	Diversity	0.0300	0.2475	+0.218	0.004

Table 6: Fixed subsampling comparison. All results $p = 0.004$ (minimum achievable with $n = 8$ windows).

3.6.4 Critical Observation

[PAT] Panda’s absolute forecasting performance is invariant to the spatial subsampling method (Uniform, Stratified, Diversity). Diversity subsampling increases Panda’s relative advantage over Chronos by making the problem harder for Chronos, not easier for Panda.

3.6.5 Competing Explanations for the Diversity Effect

1. **[HYP]** Diversity locations are more dynamically informative. Chronos cannot handle them but Panda can, demonstrating a genuine advantage of Panda’s architecture. **Confidence: Low.:** speculative, not supported by Panda’s flat absolute MAE.
2. **[HYP]** Diversity sampling selects spatially extreme or high-variance locations that break Chronos pattern matching without providing additional information to Panda. **Confidence: Medium.:** consistent with the observation that Panda MAE is unchanged.
3. **[HYP]** PCA gives the largest relative advantage because global modes are harder for Chronos (high complexity, global structure) while Panda handles them adequately due to the Koopman embedding. **Confidence: Medium..**

Confidence: Medium. for the invariance of Panda absolute MAE observation. **Confidence: Low.** for any proposed mechanism.

4 Mechanistic Investigation Experiments

These experiments were designed to address the open scientific questions identified after the fixed experiment set. The primary questions were: (1) what drives Panda’s Weather advantage, (2) whether signal statistics confound the Lorenz result, (3) whether $\lambda_1 < 0$ at non-chaotic Burgers viscosities, (4) whether diversity subsampling conclusions are seed-stable, and (5) whether improved period projection changes the decomposition result.

4.1 Experiment 13: Periodic Component Forecasting on Weather (Constructed Target)

4.1.1 Motivation

To test whether Panda’s Weather advantage is specifically located in the periodic/deterministic component of the signal. If Panda’s advantage amplifies when both models are given only the periodic component as input and evaluated against a periodic target, this suggests Panda handles periodicity better than Chronos.

4.1.2 Method

FFT extraction of the top-5 frequency components (excluding DC) from each context window channel. The resulting periodic context is presented to both models. The target is constructed as the FFT sinusoidal extrapolation of those 5 harmonics into the prediction window — not the real future values of the series. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

4.1.3 Critical Methodological Note

The target in this experiment is a constructed FFT projection, not the real future values. Both models are therefore evaluated against a deterministic sinusoidal extrapolation rather than ground truth. This conflates “which model better reproduces FFT sinusoidal extrapolation” with “which model better forecasts the actual periodic component of the future.” This confound was identified after the run and led directly to Experiment 18 (Option A), which corrects it.

4.1.4 Observations

Condition	H	Panda advantage	p
Full signal (reference)	96	+0.1893	0.004
Periodic only (constructed target)	96	+0.5220	0.004
Full signal (reference)	336	+0.1031	0.020
Periodic only (constructed target)	336	+0.5014	0.004

Table 7: Experiment 13. Periodic-only advantage computed against FFT-constructed target, not real future values. Result superseded by Experiment 18.

[OBS] Panda advantage is approximately $2.8\times$ larger on the periodic-only condition than the full signal at $H = 96$, and $4.9\times$ larger at $H = 336$. **[OBS]** Both conditions are statistically significant ($p = 0.004$ and $p = 0.020$).

Status: Confounded by the constructed target. Superseded by Experiment 18. The amplified advantage is not interpretable in isolation.

Confidence: Very Low. for any conclusion about Panda’s periodic handling ability from this experiment alone.

4.2 Experiment 14: Lorenz Phase Surrogate Control

4.2.1 Motivation

To test whether Panda’s advantage on chaotic Lorenz ($\rho = 28$) is specific to chaotic dynamical structure, or whether it is driven by signal statistics alone (variance, power spectrum). A phase-randomization surrogate matches the power spectrum of the original signal exactly while destroying all phase relationships and temporal ordering.

4.2.2 Method

Phase-randomization surrogate applied to the Lorenz $\rho = 28$ trajectory: FFT magnitudes are preserved; phases are replaced with uniform random draws on $[0, 2\pi]$; inverse FFT is applied. The surrogate is rescaled to exactly match the original mean and standard deviation. Both models are run on the original Lorenz and on the surrogate. $n_{\text{windows}} = 8$. $H = 96$.

4.2.3 Methodological Limitations

1. $n_{\text{windows}} = 8$ provides low statistical power. Minimum achievable p-value is 0.004.
2. Phase shuffling introduces endpoint discontinuities and artificial stationarity not present in the original signal. These properties differ from the original for reasons beyond merely removing dynamical structure.
3. The log proposed matching variance and spectral entropy to the $\rho = 15$ periodic baseline. The implemented control is a phase surrogate of $\rho = 28$ itself. These are related but distinct controls.

4.2.4 Observations

Condition	Panda advantage	p
Lorenz $\rho = 28$ (chaotic)	+0.3835	0.004
Phase-shuffled surrogate	+0.1715	0.320

Table 8: Experiment 14. Phase surrogate control for Lorenz $\rho = 28$. $n_{\text{windows}} = 8$.

[OBS] Panda advantage drops by approximately 55% on the surrogate ($0.3835 \rightarrow 0.1715$).

[OBS] Panda advantage on the surrogate is not statistically significant ($p = 0.320$).

[OBS] The residual advantage of 0.1715 is not negligible in absolute terms.

4.2.5 Competing Explanations

1. **[HYP]** Panda’s advantage is specific to chaotic dynamical structure. Phase shuffling destroys temporal ordering and the advantage collapses. **Confidence: Medium.:** direction is consistent, but see alternative 2.

2. **[HYP]** The result is a Type II error. With $n = 8$ windows, the test has low power to detect a true advantage of 0.17. The surrogate advantage may be real but undetectable at this sample size. **Confidence: Medium.**: cannot rule out without larger n .
3. **[HYP]** Phase shuffling introduces artificial stationarity that differentially affects both models, independently of any dynamics-specific mechanism. **Confidence: Low.**: known issue with phase surrogates, magnitude unknown.

What would strengthen this: Rerun with $n_{\text{windows}} = 20$. If the surrogate advantage remains near 0.17 and still non-significant, the dynamics-specific hypothesis is strengthened.

Confidence: Medium. for the directional observation. **Confidence: Low.** for the conclusion that dynamics are the specific driver.

4.2.6 Revision (July 2026): Advantage Decomposition Upgrades This Result

The original framing (“advantage drops and loses significance”) understates the finding. Decomposing the movement of both models: **[OBS]** Panda MAE: $0.0558 \rightarrow 0.7279$, a $13.0\times$ degradation. Chronos MAE: $0.4393 \rightarrow 0.8995$, a $2.0\times$ degradation. **[PAT]** Destroying deterministic temporal structure while exactly preserving the power spectrum degrades Panda by an order of magnitude while Chronos merely doubles. The effect is large and strongly Panda-asymmetric. This is currently the strongest available evidence that Panda’s in-regime performance exploits deterministic dynamical structure rather than second-order signal statistics. The three competing explanations above stand, but explanation 2 (Type II error framing) applied only to the significance of the residual advantage, not to the $13\times$ Panda-side degradation, which is not a subtle effect. Status upgraded from directional to supporting evidence; the $n = 20$ confirmatory rerun (with distributional metrics) remains scheduled before this is cited as established. **Confidence: Medium.**, raised from low-medium.

4.3 Experiment 15: Burgers λ_1 Estimation from PCA Components

4.3.1 Motivation

Experiment 10 showed Panda wins significantly at $\nu = 1.0$ (diffusion-dominated, non-chaotic by physical criteria). To directly verify whether $\lambda_1 < 0$ at these viscosities, the corrected Rosenstein estimator was applied to the first PCA component time series at each ν value.

4.3.2 Method

Burgers equation simulated at $\nu \in \{2.0, 1.0, 0.5, 0.1, 0.05, 0.01, 0.005\}$ using the stable adaptive-timestep spectral solver. PCA reduction to 16 channels. Rosenstein estimator applied to first PCA component with embedding dimension $m = 3$, delay $\tau = 1$, Theiler exclusion window $w = \max(0.02N, 10)$. $n_{\text{windows}} = 8$. $H = 96$.

4.3.3 Critical Methodological Note on the Estimator

Two independent reasons render the λ_1 estimates unreliable for this system:

1. **Insufficient embedding dimension.** Rosenstein requires $m \geq 2d + 1$ where d is the attractor dimension. For Burgers at low ν , $d \sim \mathcal{O}(1/\nu)$. At $\nu = 0.005$, d may be $\mathcal{O}(100)$. With $m = 3$, nearest-neighbour search is performed in a 3-dimensional projection of a potentially 100-dimensional attractor. All points appear near all others in this projection, producing a flat divergence curve regardless of the true λ_1 .
2. **PCA modes are not generic Takens observables.** Takens' theorem guarantees attractor reconstruction only for a generic scalar observable. PCA modes are constructed to maximise explained variance, which means they capture the dominant stable spatial structure while discarding the high-wavenumber chaotic directions. The first PCA component may genuinely not observe the chaotic dynamics even when they are present.

These issues are not fixable by adjusting estimator parameters without changing the observable. All λ_1 values reported here should be treated as unreliable. Physical regime labels derived from the known Burgers bifurcation structure should be used as the chaos proxy instead.

4.3.4 Observations

ν	Physical regime	$\hat{\lambda}_1$ (unreliable)	Panda advantage	p
2.000	Diffusion dominated	NaN	−0.0015	0.191
1.000	Diffusion dominated	−0.011	+0.0045	0.004*
0.500	Transition	−0.007	+0.0156	0.012*
0.100	Weakly nonlinear	−0.004	+0.1010	0.004*
0.050	Shock forming	−0.003	+0.1187	0.004*
0.010	Strong shocks	−0.001	+0.1510	0.004*
0.005	Strong shocks	−0.001	+0.1003	0.004*

Table 9: Experiment 15. $\hat{\lambda}_1$ values are unreliable due to insufficient embedding dimension and non-generic PCA observable. * denotes $p = 0.004$, the minimum achievable with $n = 8$ windows. $H = 96$ throughout (versus $H = 128$ in Experiment 10).

[OBS] $\hat{\lambda}_1$ is negative or NaN at every tested ν , including the most shock-dominated regimes. Values cluster in the range $[-0.011, -0.001]$ with near-zero absolute magnitude across all ν . This is inconsistent with physical expectations and confirms estimator failure. **[OBS]** Panda advantage increases monotonically with decreasing ν from $\nu = 1.0$ to $\nu = 0.01$ (+0.0045 to +0.1510). A non-monotone reversal occurs at $\nu = 0.005$ (+0.1003 < +0.1510). **[OBS]** At $\nu = 2.0$, Panda advantage is −0.0015 ($p = 0.191$, not significant). Consistent with Experiment 10 at this regime. **[OBS]** Advantage values at $\nu = 1.0$ and $\nu = 0.5$ are substantially smaller here (+0.0045, +0.0156) than in Experiment 10 (+0.038, +0.062). This is attributable to the different prediction horizon ($H = 96$ here versus $H = 128$ in Experiment 10), not a contradiction.

4.3.5 Competing Explanations for the Monotonic ν –Advantage Relationship

1. **[HYP]** Lower ν produces more dynamically complex Burgers fields. Panda's Koopman embedding captures this complexity better than Chronos regardless of whether the field is technically chaotic by λ_1 criteria. **Confidence: Medium.:** consistent with data but mechanism not isolated.

2. **[HYP]** Lower ν produces PCA modes with higher frequency content and sharper transitions. Chronos tokenisation degrades on these modes independently of any chaos-specific factor. **Confidence: Medium.:** consistent with Chronos-weakness pattern seen in Experiment 12.
3. **[HYP]** The ν sweep is confounded by changing PCA explained variance ratios. At different ν , the first PCA mode captures a different fraction of total variance, changing the effective forecast difficulty independently of dynamics. **Confidence: Medium.:** PC1 explained variance was 0.819 at $\nu = 2.0$; values at other ν were not recorded.

Confidence: Low. for any conclusion involving $\hat{\lambda}_1$. The estimator is inappropriate for PCA modal time series of high-dimensional PDEs.

4.4 Experiment 16: Multi-Seed Subsampling Variance

4.4.1 Motivation

Experiment 12 reported diversity subsampling results from a single random seed. The critical observation—that Panda absolute MAE is invariant to subsampling method—was based on one realisation. This experiment tests whether that observation is stable across seeds and whether diversity consistently outperforms stratified subsampling.

4.4.2 Method

Diversity subsampling (farthest-point sampling in dynamical feature space) and Variance-Stratified Uniform subsampling run at 10 seeds (0–9) on Burgers $\nu = 0.05$. $N_{\text{channels}} = 16$. $n_{\text{windows}} = 8$. $H = 96$.

4.4.3 Observations

Seed	Diversity adv	Stratified adv	p (div)	p (strat)
0	+0.0677	+0.0424	0.004	0.004
1	+0.0277	+0.0455	0.004	0.004
2	+0.0849	+0.0707	0.004	0.004
3	+0.1018	+0.0362	0.004	0.004
4	+0.0310	+0.1071	0.004	0.004
5	+0.1101	+0.0610	0.004	0.004
6	+0.0716	+0.0432	0.004	0.008
7	+0.1146	+0.0729	0.004	0.004
8	+0.0826	+0.0528	0.004	0.004
9	+0.0797	+0.0324	0.004	0.004
Mean	+0.0772	+0.0564		
Std	± 0.0296	± 0.0220		

Table 10: Experiment 16. Advantage per seed for Diversity and Stratified Uniform subsampling. Burgers $\nu = 0.05$, $N_{\text{channels}} = 16$, $H = 96$.

[OBS] Panda advantage is positive at every seed and both methods (all $p \leq 0.008$). The fact of Panda winning is robust to seed choice. **[OBS]** The ordering of diversity versus stratified advantage is inconsistent across seeds. At seed 4, stratified advantage (0.1071)

exceeds diversity advantage (0.0310). At seed 7, diversity advantage (0.1146) exceeds stratified (0.0729). No consistent dominance. **[OBS]** Diversity advantage standard deviation (0.0296) is 38% of its mean (0.0772). Stratified advantage standard deviation (0.0220) is 39% of its mean (0.0564). Both methods have high relative variance across seeds.

4.4.4 Critical Observation

[PAT] The single-seed finding from Experiment 12 that diversity subsampling gives larger advantage than stratified is not reliable. Across 10 seeds the two methods are not consistently ordered. The seed variance is large relative to the mean advantage for both methods.

4.4.5 Competing Explanations for High Seed Variance

1. **[HYP]** Different seeds select genuinely different dynamical regimes from the Burgers field, producing legitimately different forecast difficulties. **Confidence: Medium.:** plausible for diversity method, less so for stratified which is deterministic.
2. **[HYP]** $n_{\text{windows}} = 8$ is insufficient to average out window-to-window variance. Seed variance and window-sampling variance are confounded. **Confidence: Medium.:** cannot be separated at this sample size.
3. **[HYP]** The diversity metric (dynamical features plus farthest-point sampling) is sensitive to its random initialisation, producing substantially different channel subsets across seeds. **Confidence: Low.:** not directly verified.

Confidence: Medium. for the seed-instability observation. The single-seed Experiment 12 conclusion about diversity subsampling superiority is not supported.

4.5 Experiment 17: Improved Period Projection in Decomposition

4.5.1 Motivation

Experiment 11 used a naive projection method (repeat last full seasonal cycle) for the deterministic component. One competing explanation for the collapse of Panda’s advantage after decomposition was that the naive projection introduces large errors, degrading both models. This experiment tests a better projection: average seasonal pattern over all complete periods in the context window, with phase-aligned tiling. No future information is used.

4.5.2 Method

FFT decomposition as in Experiment 11. For the deterministic component projection, instead of repeating the last period, the dominant period is estimated from the FFT peak of the context window and the average over all complete periods is tiled with phase alignment. Evaluated on ETTh1, ETTh2, and Weather at $H \in \{96, 336\}$. $n_{\text{windows}} = 8$.

4.5.3 Observations

[OBS] Improved projection consistently collapses or reverses Panda’s advantage relative to vanilla on Weather. At $H = 96$: vanilla $+0.1592 \rightarrow$ improved -0.0380 . Chronos wins after improved decomposition. **[OBS]** At Weather $H = 336$, the improved projection reduces

Dataset	Condition	H	Panda advantage	p
Weather	Vanilla	96	+0.1592	0.008
Weather	Improved proj	96	−0.0380	1.000
Weather	Vanilla	336	+0.1219	0.008
Weather	Improved proj	336	+0.0299	0.012*
ETTh1	Vanilla	96	+0.0618	0.039*
ETTh1	Improved proj	96	+0.0551	0.191
ETTh1	Vanilla	336	+0.1005	0.125
ETTh1	Improved proj	336	−0.0382	0.320
ETTh2	Vanilla	96	−0.0015	0.371
ETTh2	Improved proj	96	−0.0546	0.875
ETTh2	Vanilla	336	−0.0671	0.371
ETTh2	Improved proj	336	−0.1426	0.961

Table 11: Experiment 17. Improved period projection versus vanilla. * denotes $p < 0.05$.

Panda’s advantage from +0.1219 to +0.0299 ($p = 0.012$, significant but small). **[OBS]** On ETTh1 and ETTh2, improved projection reduces or reverses Panda’s advantage at all tested horizons. **[OBS]** Panda’s absolute MAE increases after improved decomposition relative to vanilla on Weather. The improved projection does not reduce overall error; it changes the relative position of the two models.

[PAT] Across Experiments 11, 17, and 18 (Option A), three independent decomposition experiments consistently show that removing or isolating the periodic component collapses or reverses Panda’s advantage. This is the most reproducible finding of the mechanistic investigation.

4.5.4 Competing Explanations

1. **[HYP]** Panda’s advantage on Weather requires the full mixed signal (trend, seasonal, and residual components jointly). Any decomposition disrupts the joint representation that Panda’s Koopman embedding exploits. **Confidence: Medium.:** consistent with all three experiments.
2. **[HYP]** Both Panda and Chronos are worse after decomposition because the improved projection still introduces errors, and Panda is more sensitive to input quality than Chronos. This makes Panda appear worse relative to Chronos, but the cause is input degradation rather than structural dependence on the full signal. **Confidence: Medium.:** the absolute MAE increase in both models is consistent with this.
3. **[HYP]** Chronos is specifically better than Panda at forecasting residuals. Once the periodic component is cleanly removed, Chronos recovers its advantage on the remaining stochastic component. **Confidence: Medium.:** consistent with the reversal at Weather $H = 96$.

Confidence: High. for the observational pattern. **Confidence: Low.** for any specific mechanism. See the Experiment 20 revision for the July 2026 re-reading of the vanilla-versus-improved deltas in this experiment.

4.6 Experiment 18: Periodic Context with Real Targets

4.6.1 Motivation

Experiment 13 reported that Panda’s advantage amplifies on periodic-only context. However, the target in that experiment was a constructed FFT projection, not the real future values. Experiment 18 corrects this: both models are given the periodic-extracted context but evaluated against the actual future values of the Weather series. This directly determines whether the amplified advantage in Experiment 13 was real or a target-construction artifact.

4.6.2 Method

FFT extraction of top-5 frequency components from each context window channel (same as Experiment 13). Context presented to both models is the periodic component. Target is the real future values from the Weather series. Normalisation is applied using the statistics of the original (not periodic) context window, so the target normalisation is consistent with vanilla evaluation. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

4.6.3 Observations

Condition	H	Panda MAE	Chronos MAE	Advantage	p
Vanilla (reference)	96	0.6128	0.8021	+0.1893	0.004
Periodic ctx, real tgt	96	1.2007	1.0473	−0.1534	0.992
Vanilla (reference)	336	0.8754	0.9786	+0.1031	0.020
Periodic ctx, real tgt	336	1.1911	1.1816	−0.0095	0.875

Table 12: Experiment 18 (Option A). Both models evaluated against real future Weather values. Periodic context constructed from top-5 FFT harmonics of context window.

[OBS] With periodic context and real targets, Panda loses to Chronos at $H = 96$ (advantage -0.1534 , $p = 0.992$) and is essentially tied at $H = 336$ (advantage -0.0095 , $p = 0.875$). **[OBS]** Both models degrade substantially in absolute MAE relative to vanilla. Panda MAE increases from 0.6128 to 1.2007 (+96%). Chronos MAE increases from 0.8021 to 1.0473 (+31%). **[OBS]** Panda degrades approximately $3\times$ more than Chronos in absolute MAE when the context is reduced to its periodic component. **[OBS]** The amplified advantage in Experiment 13 (+0.52) is fully explained by the constructed target and does not reflect a real advantage of Panda on periodic signals.

4.6.4 Critical Finding

[PAT] The advantage Panda shows on Weather requires the full mixed signal. When given only the periodic component as context and evaluated against real targets, Panda loses to Chronos. This is confirmed across three independent experiments: Experiment 11 (decomp collapses advantage), Experiment 17 (improved decomp also collapses advantage), and Experiment 18 (periodic context with real targets reverses advantage).

4.6.5 Implication for TimesNet-Style Pretraining Direction

This result deprioritises TimesNet-style explicit period decomposition as an architectural modification. The evidence shows Panda performs *worse* on periodic signals in isolation, not better. Teaching Panda explicit period decomposition would address a component where Panda already underperforms Chronos, while potentially disrupting the full-signal joint representation that appears to be the source of its advantage.

4.6.6 Competing Explanations for Panda’s Larger Degradation

1. **[HYP]** Panda’s Koopman patch embedding is degenerate on smooth periodic inputs. Patches of a sinusoid look nearly identical after normalisation; the embedding cannot distinguish position in the cycle. The forecast head defaults to a near-constant prediction. **Confidence: Medium.:** consistent with the scale of degradation.
2. **[HYP]** Chronos’s pretraining corpus is predominantly periodic and seasonal real-world data. Smooth sinusoidal inputs are near in-distribution for Chronos. **Confidence: Medium.:** consistent with Chronos degrading less.
3. **[HYP]** Panda is more sensitive to input amplitude calibration. The periodic context has lower energy than the full signal (residual energy removed), and Panda’s fixed prediction head scales poorly under this amplitude mismatch. **Confidence: Low.:** speculative, not directly tested.

Confidence: High. for the core observation. **Confidence: Low.** for any specific mechanism explaining Panda’s larger degradation.

5 Architectural Component Investigations

These experiments were designed to move from observational findings toward mechanistic diagnosis. Each experiment targets one identifiable architectural component of Panda or one structural property of the evaluation datasets, motivated by the open questions identified at the end of the mechanistic investigation.

5.1 Experiment 19: Complexity Continuum

5.1.1 Motivation

The Lorenz ρ sweep (Experiment 3) showed a pattern consistent with a chaos threshold, but confounded by simultaneously changing signal statistics (variance, spectral content, amplitude). This experiment uses five qualitatively distinct dynamical systems spanning the full range from periodic to chaotic, without the confound of continuously varying a single parameter. The goal is to test whether Panda’s advantage tracks dynamical complexity or something else.

5.1.2 Method

Five systems: Harmonic oscillator (periodic), Van der Pol oscillator (limit cycle), Duffing oscillator (weakly chaotic), Rossler (chaotic), Lorenz (chaotic). $H = 96$. $n_{\text{windows}} = 8$. Permutation entropy at order 3 computed per system as a complexity proxy.

5.1.3 Observations

System	Regime	PE	Panda MAE	Chronos MAE	Adv (p)
Harmonic	Periodic	0.438	0.0647	0.4346	+0.370 (0.004)*
Van der Pol	Limit cycle	0.431	0.0319	0.0426	+0.011 (0.027)*
Duffing	Weakly chaotic	0.476	0.5811	0.7948	+0.214 (0.055)
Rossler	Chaotic	0.442	0.0826	0.3868	+0.304 (0.004)*
Lorenz	Chaotic	0.455	0.0555	0.5314	+0.476 (0.004)*

Table 13: Experiment 19. * denotes $p < 0.05$ (Wilcoxon, $n = 8$ windows).

[OBS] Panda has a statistically significant advantage on the Harmonic oscillator (+0.370, $p = 0.004$). This is the simplest possible periodic system—a pure sinusoid. This directly contradicts the hypothesis that chaos is necessary for Panda’s advantage. **[OBS]** Van der Pol (limit cycle) shows the smallest advantage of all tested systems (+0.011, $p = 0.027$). Effect size is very small. **[OBS]** Duffing (weakly chaotic) advantage +0.214 is not statistically significant ($p = 0.055$), just above the threshold. **[OBS]** Rossler and Lorenz (both chaotic) show large, statistically significant advantages (+0.304 and +0.476). **[OBS]** Permutation entropy at order 3 ranges from 0.431 to 0.476 across all five systems. It cannot discriminate periodic from chaotic regimes in this dataset: the PE values are nearly identical across qualitatively different dynamics.

5.1.4 Critical Finding

[PAT] The advantage pattern is non-monotone with respect to dynamical complexity. The Harmonic oscillator produces the second-largest advantage (+0.370), larger than

the weakly chaotic Duffing (+0.214). The Van der Pol limit cycle produces the smallest advantage (+0.011), which is anomalously low relative to both the simpler Harmonic and the more complex chaotic systems. The chaos-specific hypothesis cannot account for either the large Harmonic advantage or the small Van der Pol advantage.

5.1.5 Competing Explanations

1. **[HYP]** Chronos specifically fails on sinusoidal inputs due to its discrete tokenisation scheme. The large Harmonic advantage reflects Chronos weakness rather than Panda strength. The small Van der Pol advantage may similarly reflect Chronos being relatively adequate for limit cycle dynamics. **Confidence: Medium.:** consistent with data; requires Chronos-specific ablation to confirm.
2. **[HYP]** Panda’s Koopman embedding represents harmonic oscillations efficiently because sinusoidal functions are eigenfunctions of the Koopman operator for linear systems. This would make the Harmonic oscillator in-distribution in a functional sense even though it is not a chaotic ODE. **Confidence: Low.:** speculative, not testable without code access to the lifting.
3. **[HYP]** The Van der Pol limit cycle has dynamical properties that are close to a transition point between two regimes. The small advantage at this transition may reflect genuine model uncertainty rather than a failure of either model. **Confidence: Low.:** speculative.
4. **[SPEC]** The Harmonic oscillator advantage may be an artifact of the specific initial conditions or trajectory length chosen, rather than a robust property of the system class. **Confidence: Very Low.:** not tested with multiple seeds.

What permutation entropy at order 3 establishes and does not establish: PE at order 3 is computed on very short embedded sequences. With embedding dimension $m = 3$, only $3! = 6$ ordinal patterns are possible. This is insufficient resolution to distinguish dynamics across qualitatively different systems that all have roughly similar power spectra at the timescales being embedded. The metric is not useful as a complexity discriminator at this order.

Confidence: Medium. for the observational pattern. **Confidence: Low.** for any proposed mechanism. The chaos-specific hypothesis is directly contradicted by the Harmonic oscillator result.

5.1.6 Revision (July 2026): Headroom Correction Under Relative Skill

Absolute advantage across systems of very different intrinsic difficulty conflates headroom with skill (Section 1.2). Relative skill $\text{MAE}_C/\text{MAE}_P$: Lorenz $9.6\times$, Harmonic $6.7\times$, Rossler $4.7\times$, Duffing $1.37\times$, Van der Pol $1.33\times$. **[OBS]** Panda’s absolute MAE is near-ceiling on four of five systems (0.032–0.083); only Duffing is hard for Panda (0.581). Chronos is near-ceiling only on Van der Pol (0.043). Cross-system variance in advantage is therefore mostly Chronos-side, plus one both-models-fail system (Duffing). Two consequences. First, the “anomalously small” Van der Pol advantage is substantially a floor effect: both models are near-perfect, so there is no headroom for advantage regardless of relative capability. Second, the correct framing of the continuum is not “on which systems is Panda strong” but “on which signals does Chronos collapse while Panda does not” — which is exactly what the queued Chronos-attribution arm (Chronos alone on Harmonic with amplitude/frequency/sampling variants) tests. Explanation 1 above rises in importance

accordingly. The non-monotonicity claim survives in weakened form (Harmonic > Duffing in both views), but the headline anomaly framing does not. **Confidence: Medium..**

5.2 Experiment 20: Chronos Residual Ablation

5.2.1 Motivation

Three decomposition experiments (11, 17, 18) showed that removing the periodic component collapses Panda’s advantage. A critical ambiguity remained: does Panda’s advantage require the full mixed signal because Panda specifically benefits from the joint structure, or because Chronos is specifically hurt by the periodic component? This experiment directly measures how Chronos performs on Weather residuals versus the full signal, without involving Panda at all.

5.2.2 Method

Chronos run on: (1) full Weather signal (vanilla), (2) FFT residual of Weather after removing top harmonics (residual). Both evaluated against the real future values. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

5.2.3 Observations

Condition	H	Chronos vanilla	Chronos residual	Δ	p
Weather	96	0.7695	1.0464	+0.277	0.016*
Weather	336	1.0012	1.2461	+0.245	0.008*

Table 14: Experiment 20. Chronos MAE on full signal versus FFT residual. $\Delta > 0$ means residual is harder for Chronos. * denotes $p < 0.05$.

[OBS] Chronos MAE increases significantly on the FFT residual relative to the full signal at both horizons (+0.277 at $H = 96$, +0.245 at $H = 336$; both $p < 0.02$). **[OBS]** The magnitude of Chronos degradation after decomposition (+0.277) is comparable to the magnitude of Panda degradation found in earlier decomposition experiments.

5.2.4 Critical Revision to Earlier Interpretation

[PAT] Both Panda and Chronos degrade approximately equally when the periodic component is removed. The earlier interpretation from Experiments 11 and 17 — that Panda’s advantage requires the full mixed signal specifically — must be revised. The decomposition results are equally consistent with projection error in the FFT decomposition degrading both models proportionally. The signal-dependence of Panda’s advantage is not specifically established by the decomposition experiments; both models depend on the full signal.

5.2.5 Competing Explanations

1. **[HYP]** FFT decomposition introduces substantial projection error in the deterministic component. Both models receive a degraded context and their MAE increases

for this reason, not because either model specifically depends on the periodic component. **Confidence: Medium.**: the similar magnitude of degradation in both models supports this.

2. **[HYP]** Both Panda and Chronos leverage the periodic component for forecasting. The Weather residual is genuinely harder for both models because it contains less predictable structure. **Confidence: Medium.**: also consistent with the data; cannot be distinguished from hypothesis 1 without a perfect oracle decomposition.
3. **[SPEC]** The periodic component of Weather carries the dominant predictable signal (daily cycle, annual seasonality). Removing it leaves a near-unpredictable residual. Both models degrade because the task becomes near-impossible, not because of any model-specific failure. **Confidence: Low.**: speculative without measuring residual autocorrelation structure.

Confidence: High. for the observation. **Confidence: Medium.** for the revision to the earlier interpretation. The decomposition story is now less diagnostic than previously believed.

5.2.6 Revision (July 2026): The Revision Above Overcorrected

Re-reading the Experiment 17 raw values with advantage decomposition: **[OBS]** Weather, vanilla $\rightarrow$ improved projection. $H = 96$: Panda $0.6128 \rightarrow 1.0848$ ($\Delta = +0.472$); Chronos $0.7720 \rightarrow 1.0468$ ($\Delta = +0.275$). $H = 336$: Panda $0.8754 \rightarrow 1.2408$ ($\Delta = +0.365$); Chronos $0.9974 \rightarrow 1.2707$ ($\Delta = +0.273$). **[PAT]** The degradation is not equal. There is a large shared component (approximately $+0.27$ in both models at both horizons, consistent with projection error) plus a Panda-specific additional component ($\approx +0.20$ at $H = 96$, $\approx +0.09$ at $H = 336$). The “both degrade equally” revision above was itself an overcorrection: it correctly identified the shared projection-error component but discarded the Panda-specific excess. The current reading: projection error explains most of the collapse, and a residual Panda-specific full-signal dependence remains, strongest at short horizon. This partially rehabilitates hypothesis 1 of Experiment 17 in weakened form. Single runs at $n = 8$; the effect is directionally consistent across two horizons but confirmatory status awaits the $n = 20$ campaign. **Confidence: Medium.**

5.3 Experiment 21: Permutation Entropy as Complexity Predictor

5.3.1 Motivation

To test whether a single complexity metric—permutation entropy at order 3—could predict Panda’s advantage across all tested systems simultaneously, including Lorenz, Burgers, ETTh1, ETTh2, and Weather. This would provide a single quantitative criterion for predicting when Panda generalises.

5.3.2 Method

PE at order 3 computed per dataset (or per ρ value for Lorenz, per ν value for Burgers). Advantage values taken from previously completed experiments. A visual and Spearman correlation analysis was performed across the combined dataset.

5.3.3 Critical Methodological Note

The PE values separate into two disjoint regimes with no overlap: Burgers PE ranges from 0.024 to 0.132; real datasets (ETTh1, ETTh2, Weather) range from 0.756 to 0.955; Lorenz values cluster at 0.460–0.494. These three groups do not share a common scale. Any correlation computed across all three groups is dominated by between-group separation rather than within-group structure and cannot be interpreted as a genuine predictor.

5.3.4 Observations

[OBS] Burgers PE values (0.024–0.132) are an order of magnitude lower than real dataset PE values (0.756–0.955). The metric operates in fundamentally different regimes for PDE-derived PCA modal time series versus raw sensor data. **[OBS]** Lorenz PE values (0.460–0.494) are insensitive to ρ , ranging from $\rho = 10$ (periodic, PE = 0.494) to $\rho = 60$ (strongly chaotic, PE = 0.473). PE at order 3 cannot discriminate the periodic from chaotic Lorenz regimes. **[OBS]** Within the Lorenz sweep, Panda advantage varies from 0 to 0.69 while PE varies by less than 0.05. There is no within-group correlation. **[OBS]** Experiment 19 independently confirmed that PE at order 3 cannot discriminate Harmonic, Van der Pol, Duffing, Rossler, and Lorenz systems, which have PE values ranging only from 0.431 to 0.476.

5.3.5 Conclusion

[PAT] Permutation entropy at order 3 is not a useful predictor of Panda’s advantage. The metric fails on three independent grounds: (1) it cannot discriminate periodic from chaotic Lorenz dynamics, (2) it cannot discriminate across systems in the complexity continuum, and (3) it operates in incomparable numerical regimes across different data types. A higher-order PE (order 5 or 6) on longer time series with more stringent embedding parameters might partially address (1) and (2), but the between-group scale problem (3) would persist without normalisation.

Confidence: High. for the failure of PE order 3 as a discriminator. The finding is consistent and replicated across three data contexts.

Deprecation status (July 2026). This direction is formally retired. Its lesson (estimator validation before deployment) is encoded in the Section 1.2 estimator validation rule. The auxiliary correlation dimension analysis run alongside this experiment is likewise retired: the estimator returns $\hat{d} \approx 0.86$ on Lorenz against a literature value of ≈ 2.05 , failing its own ground truth, so the previously reported Weather–Lorenz clustering at $\hat{d} \approx 0.86$ –0.89 is uninterpretable even directionally (a shared preprocessing artifact is an undischarged competing explanation). The structure statistic investigation (Section 10) replaces this line of work with a gate-validated instrument.

5.4 Experiment 22: Node Identity Embeddings (Scalar Offset Proxy)

5.4.1 Motivation

The heterogeneity stratification results (Section 6) identified sensor heterogeneity as a factor correlated with reduced Panda advantage. One architectural hypothesis is that Panda lacks sensor identity information: it treats all channels as interchangeable state variables, whereas real weather sensors have distinct physical meanings. This experiment

tests a minimal version of sensor identity: adding a learned scalar offset per channel (a per-sensor bias term) and measuring whether this modifies Panda’s relative advantage on homogeneous versus heterogeneous subsets.

5.4.2 Method

A scalar offset is fitted per channel using the context window mean of each channel. This is not a full node embedding; it is the minimal possible channel identity signal (a scalar constant per channel). Two Weather subsets were used: (1) `homo_matched`, a homogeneous subset with very low heterogeneity (0.086), and (2) `hetero_controlled`, a heterogeneous subset with high heterogeneity (0.853) and controlled difficulty variance ($CV = 0.11$). $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

5.4.3 Observations

Subset	H	Panda base	Panda offset	Chronos	Adv base	Adv offset	Δ Panda
homo_matched	96	0.328	0.412	0.714	+0.386	+0.439	−0.085
homo_matched	336	0.839	0.882	1.207	+0.368	+0.144	−0.043
hetero_controlled	96	0.605	0.568	0.847	+0.242	+0.277	+0.037
hetero_controlled	336	0.867	0.870	0.954	+0.088	+0.146	−0.003

Table 15: Experiment 22. Δ Panda = Panda offset − Panda base. Negative means offset hurts Panda. Chronos MAE is unchanged across conditions (Chronos does not use offsets).

[OBS] On the homogeneous subset at $H = 96$, the scalar offset hurts Panda MAE (+0.085, Panda gets worse). At $H = 336$, Panda also gets worse (+0.043). **[OBS]** On the heterogeneous subset at $H = 96$, the scalar offset marginally improves Panda MAE (−0.037, Panda gets better). At $H = 336$, essentially no change (−0.003). **[OBS]** The direction of the offset effect is opposite on homogeneous versus heterogeneous subsets at $H = 96$: hurts on homogeneous, helps on heterogeneous. This is directionally consistent with the hypothesis that channel identity information is only beneficial when channels are genuinely distinguishable. **[OBS]** Effect sizes are small throughout. The largest effect ($\Delta = -0.085$ on `homo_matched` $H = 96$) is in the wrong direction for the hypothesis; the small positive effect on `hetero_controlled` $H = 96$ ($\Delta = +0.037$) is marginal.

5.4.4 Competing Explanations

1. **[HYP]** A scalar offset is too minimal to encode channel identity in any useful sense. The null result reflects the weakness of the proxy rather than the absence of a channel identity effect. **Confidence: Medium.**: consistent with the small effect sizes.
2. **[HYP]** Channel identity information is not what is missing from Panda on heterogeneous data. The heterogeneity bottleneck identified in Section 6 may be due to interaction structure (graph topology) rather than per-sensor identity. **Confidence: Medium.**: cannot distinguish these from this experiment.
3. **[HYP]** Adding a per-channel mean offset disrupts the per-window normalisation scheme, which already removes per-channel mean by construction. The offset may be double-correcting and introducing noise on homogeneous channels that have

well-defined means. **Confidence: Low.**: requires analysis of the normalisation interaction to confirm.

Confidence: Low. for any conclusion about sensor identity. The scalar offset is too minimal a proxy and the effect sizes are too small to support strong claims.

Deprecation status (July 2026). Retired as a test of the sensor identity hypothesis: the proxy cannot distinguish the hypotheses in either direction. The identity-versus-interaction question is now assigned to the univariate-on-heterogeneous-subset discriminator (Section 13), which is decisive and cheaper than a proper node-embedding implementation.

5.5 Experiment 23: Prediction Head Fine-Tuning

5.5.1 Motivation

Panda’s prediction head was pretrained on chaotic ODE trajectories and is fixed at inference. A hypothesis is that the head imposes a chaotic prior on the output distribution, limiting performance on non-chaotic data. This experiment tests whether fine-tuning only the prediction head (leaving the encoder frozen) on Weather improves Panda’s performance.

5.5.2 Method

The prediction head alone was fine-tuned on Weather context-forecast pairs for 50 gradient steps. Encoder weights were held fixed. Performance compared against the base Panda (no fine-tuning) at $H \in \{96, 336\}$. $n_{\text{windows}} = 8$.

5.5.3 Observations

H	Panda base	Panda FT	Adv base	Adv FT	Δ Panda
96	0.609	0.699	+0.154	+0.071	−0.091
336	0.870	0.881	+0.189	+0.198	−0.012

Table 16: Experiment 23. Δ Panda = Panda FT – Panda base. Negative means fine-tuning hurts. Adv = $\text{MAE}_{\text{Chronos}} - \text{MAE}_{\text{Panda}}$.

[OBS] Head fine-tuning at $H = 96$ worsens Panda MAE by 0.091. This is a substantial degradation (approximately 15% relative increase in MAE). **[OBS]** At $H = 336$, the degradation is small (−0.012) and likely not significant at $n = 8$ windows. **[OBS]** Panda’s relative advantage over Chronos decreases after fine-tuning at $H = 96$ ($+0.154 \rightarrow +0.071$). **[OBS]** The direction of the $H = 96$ result is strongly negative. Fine-tuning the prediction head without fine-tuning the encoder worsens performance.

5.5.4 Critical Finding

[PAT] Prediction head fine-tuning at $H = 96$ produces a negative result: Panda performance degrades. This rules out the prediction head as an easily recoverable bottleneck through lightweight adaptation.

5.5.5 Competing Explanations

1. **[HYP]** The encoder and head co-adapted during pretraining. Fine-tuning the head alone without updating the encoder introduces a mismatch between encoder representations and head expectations. The head adapts to the target distribution but the encoder still produces representations calibrated for chaotic ODE distributions. **Confidence: Medium.:** consistent with the magnitude of degradation.
2. **[HYP]** 50 gradient steps is insufficient for genuine adaptation but sufficient to corrupt the pretrained head. The degradation reflects the head moving away from its pretrained state without reaching a new stable optimum. **Confidence: Medium.:** cannot rule out without testing more gradient steps.
3. **[HYP]** The prediction head is not the bottleneck. The representation upstream of the head (encoder output) contains the relevant distributional mismatch, and adapting the head cannot address this. **Confidence: Medium.:** consistent with the overall pattern across experiments that no single downstream component is responsible.

Confidence: Medium. for the observational result. **Confidence: Low.** for any specific mechanism. The head-fine-tuning direction is not supported as a lightweight adaptation strategy. The negative result is additionally confounded by optimisation choices (single learning rate, 50 steps, no significance test); it rules out the tested configuration, not head adaptation in general.

6 Sensor Heterogeneity Investigation

These experiments address the hypothesis that Panda’s advantage on Weather is partially limited by sensor heterogeneity—the degree to which Weather channels represent qualitatively different physical processes with different dynamical properties—whereas Panda’s pretraining distribution consists of homogeneous state variables from single dynamical systems.

6.1 Experiment 24: Heterogeneity Stratification on Weather

6.1.1 Motivation

To test whether Panda’s advantage over Chronos systematically decreases as the dynamical heterogeneity of the channel subset increases. Weather has 21 channels spanning temperature, humidity, pressure, wind speed, and other variables with distinct dynamical properties. Subsets of Weather channels can be constructed with controlled levels of heterogeneity.

6.1.2 Method

Three subsets of 7 Weather channels each were constructed by clustering channels on dynamical feature vectors (standard deviation, lag-1 autocorrelation, spectral entropy, dominant frequency, percentile amplitude): (1) **homogeneous** (all channels from the same cluster, heterogeneity index 0.035), (2) **mixed** (channels from two clusters, heterogeneity index 0.668), (3) **heterogeneous** (channels from maximally distinct clusters, heterogeneity index 0.947). $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

6.1.3 Observations

Subset	Het.	H	Panda MAE	Chronos MAE	Adv	p
homogeneous	0.035	96	0.318	0.689	+0.371	0.004*
homogeneous	0.035	336	0.817	1.183	+0.365	0.004*
mixed	0.668	96	0.581	0.786	+0.205	0.039*
mixed	0.668	336	0.925	0.982	+0.056	0.371
heterogeneous	0.947	96	0.618	0.697	+0.079	0.074
heterogeneous	0.947	336	1.123	1.163	+0.040	0.191

Table 17: Experiment 24. Het. = heterogeneity index. * denotes $p < 0.05$.

[OBS] Panda advantage on the homogeneous subset at $H = 96$ is +0.371 ($p = 0.004$). On the heterogeneous subset at $H = 96$, advantage is +0.079 ($p = 0.074$, not significant). Advantage drops by approximately $5\times$ from homogeneous to heterogeneous at the same horizon. **[OBS]** Panda absolute MAE increases monotonically with heterogeneity at $H = 96$ (0.318, 0.581, 0.618). Chronos MAE also increases (0.689, 0.786, 0.697) but non-monotonically. **[OBS]** At $H = 336$, the homogeneous subset shows significant advantage (+0.365, $p = 0.004$). Mixed and heterogeneous subsets are not significant ($p = 0.371$, $p = 0.191$). **[OBS]** The confound is that homogeneous, mixed, and heterogeneous subsets may also differ in difficulty: the homogeneous subset may consist of easier-to-forecast channels.

6.2 Experiment 25: Difficulty-Matched Heterogeneity Control

6.2.1 Motivation

The stratification result in Experiment 24 is confounded by channel difficulty. The homogeneous subset channels may be intrinsically easier to forecast, regardless of heterogeneity. This experiment creates a difficulty-matched version of the comparison by ensuring all three subsets have approximately equal individual-channel Chronos MAE.

6.2.2 Method

Difficulty was estimated per channel as Chronos univariate MAE. Channels were selected for each heterogeneity level such that the mean difficulty was matched across subsets. Three matched subsets: **homo_matched** (heterogeneity 0.086, mean difficulty 0.800), **mixed_matched** (heterogeneity 0.602, mean difficulty 0.839), **heterogeneous** (heterogeneity 0.947, mean difficulty 0.839). $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

6.2.3 Observations

Subset	Het.	H	Panda MAE	Chronos MAE	Adv	p
homo_matched	0.086	96	0.331	0.700	+0.369	0.004*
homo_matched	0.086	336	0.841	1.147	+0.305	0.004*
mixed_matched	0.602	96	0.475	0.808	+0.333	0.004*
mixed_matched	0.602	336	0.856	0.946	+0.090	0.020*
heterogeneous	0.947	96	0.618	0.836	+0.218	0.074
heterogeneous	0.947	336	1.123	1.243	+0.120	0.125

Table 18: Experiment 25. Difficulty-matched heterogeneity control. * denotes $p < 0.05$.

[OBS] After difficulty matching, the pattern from Experiment 24 persists. Panda advantage on homo_matched at $H = 96$ is +0.369 ($p = 0.004$). Heterogeneous advantage at $H = 96$ is +0.218 ($p = 0.074$, not significant). Advantage drops by approximately $1.7\times$. **[OBS]** Panda absolute MAE still increases monotonically with heterogeneity (0.331, 0.475, 0.618 at $H = 96$) despite matched Chronos difficulty. Chronos MAE is approximately constant across subsets (0.700, 0.808, 0.836) after difficulty matching. **[OBS]** The heterogeneity effect on Panda MAE remains after controlling for individual channel difficulty. Chronos MAE is relatively stable; Panda MAE increases strongly with heterogeneity.

6.2.4 Critical Finding

[PAT] After controlling for channel-level forecasting difficulty, Panda MAE increases monotonically with sensor heterogeneity while Chronos MAE remains approximately stable. This pattern cannot be explained by difficulty alone. Sensor heterogeneity has a specific negative effect on Panda that is not shared by Chronos.

6.2.5 Competing Explanations

1. **[HYP]** Panda treats channels as interchangeable state variables. On homogeneous subsets, this is appropriate. On heterogeneous subsets, cross-channel attention

produces incorrect coupling estimates between physically unrelated sensors. **Confidence: Medium.**: directionally consistent with difficulty-matched result and with the univariate ablation result (Experiment 9) showing univariate Panda competitive with multivariate.

2. **[HYP]** Difficulty matching by individual Chronos MAE does not fully control for interaction complexity. Heterogeneous subsets may have higher interaction complexity, requiring more sophisticated cross-channel reasoning that neither model handles well but that specifically breaks Panda’s joint attention. **Confidence: Medium.**: cannot distinguish from hypothesis 1 with current data.
3. **[HYP]** Heterogeneous subsets have higher variance in their dynamical feature distributions. Per-window normalisation removes per-channel mean and variance but not higher-order distributional differences. Panda’s fixed embedding may be more sensitive to these residual differences than Chronos. **Confidence: Low.**: speculative.

Confidence: Medium. for the core observational finding. The difficulty-matched control substantially strengthens the heterogeneity pattern but does not establish a mechanism.

6.3 Experiment 26: Variance-CV Heterogeneity Control

6.3.1 Motivation

A remaining confound is that difficulty-matched heterogeneous subsets may still contain channels with higher within-subset variance of difficulty (more difficult channels spread more unevenly). This experiment adds a second-order difficulty control by matching the coefficient of variation (CV) of difficulty across subsets.

6.3.2 Method

The heterogeneous controlled subset (**hetero_controlled**) was constructed to have both high heterogeneity (0.853) and low difficulty CV (0.110), matching the difficulty CV of the homogeneous subsets. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

6.3.3 Observations

Subset	Het.	H	Panda MAE	Chronos MAE	Adv	p
hetero_controlled	0.853	96	0.605	0.866	+0.261	0.004*
hetero_controlled	0.853	336	0.867	1.175	+0.309	0.012*

Table 19: Experiment 26. Heterogeneous subset with controlled difficulty variance (CV = 0.11). * denotes $p < 0.05$.

[OBS] Panda still wins on the hetero_controlled subset (+0.261 at $H = 96$, +0.309 at $H = 336$; both significant). The advantage is smaller than on the homo_matched subset (+0.369 and +0.305 respectively) but remains statistically significant. **[OBS]** Comparing hetero_controlled against homo_matched from Experiment 25: at $H = 96$, Panda MAE is 0.605 (hetero) versus 0.331 (homo). Panda MAE nearly doubles despite matched difficulty CV. **[OBS]** Chronos MAE increases proportionally from homo to hetero ($0.700 \rightarrow 0.866$), while Panda MAE increases more steeply ($0.331 \rightarrow 0.605$). Panda degrades approximately $1.8\times$ more than Chronos.

[PAT] After controlling for both mean difficulty and difficulty variance, Panda absolute MAE is substantially higher on heterogeneous Weather channels than on homogeneous channels at matched difficulty. Chronos does not show this differential sensitivity. This is the most controlled evidence that sensor heterogeneity is a specific bottleneck for Panda.

6.3.4 Competing Explanations

1. **[HYP]** Panda’s joint channel attention actively harms performance on heterogeneous channels by averaging incompatible channel representations. The channel attention was pretrained on homogeneous ODE state variables and imports coupling patterns that are not present in heterogeneous sensor data. **Confidence: Medium..**
2. **[HYP]** The difficulty matching controls are incomplete. Some third property of heterogeneous channels (multi-scale frequency content, non-stationarity patterns, distributional shift) is not captured by Chronos univariate MAE or its CV, and this third property is the true cause of Panda degradation. **Confidence: Medium..**

Confidence: Medium. for the heterogeneity bottleneck pattern. The three-experiment heterogeneity series (24, 25, 26) provides consistent convergent evidence. **Confidence: Low.** for the proposed mechanism (joint attention importing incorrect coupling).

6.4 Chronos Heterogeneity Calibration Experiment

6.4.1 Motivation

To separately characterise how Chronos MAE behaves across heterogeneity levels, to confirm the interpretation that heterogeneity differentially affects Panda and not Chronos.

6.4.2 Method

Chronos run independently on the three difficulty-matched subsets from Experiment 25. Chronos forecasts each channel univariately and has no cross-channel attention, so it cannot be directly harmed by channel heterogeneity per se.

6.4.3 Observations

Subset	Het.	H	Chronos MAE	Difficulty
homo_matched	0.086	96	0.776	0.800
homo_matched	0.086	336	1.077	0.800
mixed_matched	0.602	96	0.817	0.839
mixed_matched	0.602	336	0.867	0.839
heterogeneous	0.947	96	0.687	0.839
heterogeneous	0.947	336	1.089	0.839

Table 20: Chronos heterogeneity calibration. Chronos MAE is approximately stable across heterogeneity levels at matched difficulty, as expected for a univariate model.

[OBS] Chronos MAE is approximately stable across heterogeneity levels at matched difficulty (0.776, 0.817, 0.687 at $H = 96$ across homo, mixed, heterogeneous). No monotone trend. **[OBS]** As expected, Chronos MAE scales with difficulty rather than heterogeneity, since Chronos has no cross-channel mechanism. **[OBS]** The contrast with Panda from

Experiment 25 is direct: Panda MAE increases monotonically (0.331, 0.475, 0.618 at $H = 96$) while Chronos MAE does not (0.776, 0.817, 0.687). The divergence is attributable to Panda’s multivariate architecture responding to heterogeneity.

Confidence: High. for the observation that Chronos is insensitive to heterogeneity.
Confidence: Medium. for the implication that Panda’s sensitivity is architectural.

6.4.4 Revision (July 2026): Registered Caveats and the Missing Discriminator

Three caveats are registered so that the heterogeneity claim is cited at its correct strength:

1. **[OBS]** The difficulty matching is asymmetric: homo_matched mean difficulty is 0.800 versus 0.839 for mixed and heterogeneous. Only the mixed–heterogeneous step is truly matched; the homo–hetero comparison (the “MAE nearly doubles” step) retains a residual difficulty gap.
2. **[OBS]** The Chronos-flatness contrast is clean primarily at $H = 96$. At $H = 336$ in the difficulty-matched design, the Chronos spread across subsets (0.296: 1.147, 0.946, 1.243) is comparable to the Panda spread (0.282: 0.841, 0.856, 1.123), although the Chronos pattern is non-monotone in heterogeneity while the Panda pattern is monotone.
3. What is established is Panda’s absolute degradation with heterogeneity under controls; the persistence of a significant *advantage* on the heterogeneous subset is not established ($p = 0.074$ and 0.125 in Experiment 25).

The attribution question also has an unrun discriminator. The two leading explanations — joint channel attention corrupted by incompatible channels (hypothesis 1) versus per-channel signal character of heterogeneous channels (a variant of hypothesis 2) — are in tension with Experiment 9, which found channel attention approximately irrelevant on full Weather. The decisive test is univariate Panda on the heterogeneous subsets: if univariate Panda recovers toward homogeneous-level MAE, joint attention is confirmed as the bottleneck and the node-embedding/graph direction is motivated; if univariate Panda degrades equally, the bottleneck is per-channel signal character and the graph direction loses its motivation. This experiment is queued in the confirmatory campaign (Section 13), which also fixes the difficulty asymmetry and reruns the chain at $n = 20$ with a DLinear control arm.

7 Koopman Lifting Ablation

7.1 Experiment 28: Koopman Ablation at 50k Steps (Preliminary)

7.1.1 Status of This Section

This entire section is preliminary and ungated. The in-distribution sanity check (Panda’s own convergence rule, Section 1.2 policy: OOD claims require an in-distribution check to pass first) has not been completed within the data available to this log revision. Numbers reported below are point estimates from a single 50k-step training run per condition, with *no direct significance test between the retrained baseline and the ablated model* — the only Wilcoxon tests computed are each condition against the published checkpoint, which is a different comparison. A 100k-step rerun with a matched-steps baseline and a direct baseline-versus- ablation test is queued but not started as of this writing. Nothing in this section should be cited as an established result.

7.1.2 Motivation

The Koopman random-feature lifting (random Fourier and polynomial features of the input patch, motivated by Takens-embedding theory) is the last untested candidate positive mechanism for Panda’s advantage, after channel attention (Experiments 9, 27) and the prediction head (Experiment 23) were both ruled out or found non-contributory. This requires retraining with the lifting replaced by a linear projection of matched output dimension, since it cannot be tested at inference time on the fixed checkpoint.

7.1.3 Method

Two conditions trained from scratch on the GilpinLab/skew40 corpus for 50,000 steps each on a Kaggle T4 GPU: **retrained_base** (architecture unchanged) and **koopman_ablation** (lifting replaced by a linear projection of equal output dimension). Both compared against the fixed published checkpoint. Evaluated out-of-distribution on Weather and Burgers ($\nu = 1.0$) and on three complexity-continuum systems (Van der Pol, Duffing, Harmonic) at $H \in \{96, 192, 336\}$, $n_{\text{windows}} = 8$.

7.1.4 Observations

[OBS] The retrained baseline at 50k steps is substantially worse than the published checkpoint on every dataset (e.g. Weather $H = 96$: 0.824 vs 0.635, a 30% gap). Both conditions being compared are undertrained relative to the published model. **[OBS]** Ablation is worse than the matched-steps baseline at every horizon on Burgers $\nu = 1.0$, by 23–67%. **[OBS]** Ablation is better than or approximately equal to the baseline on Van der Pol, Duffing, and Harmonic at nearly every horizon, in some cases by a large margin (Harmonic $H = 96$: 41% better). **[OBS]** Weather is mixed and roughly flat across horizons (ratios 0.80, 1.00, 1.07).

7.1.5 Competing Explanations

1. **[HYP]** Koopman lifting is genuinely useful for aperiodic/PDE-like dynamics (Burgers) and neutral-to-harmful for simple periodic systems (Van der Pol, Duffing,

Dataset	H	Published	Retrained base	Koopman ablation	Ablation/base ratio
Weather	96	0.635	0.824	0.657	0.80
Weather	192	0.723	0.868	0.869	1.00
Weather	336	0.828	0.929	0.995	1.07
Burgers $\nu=1.0$	96	0.0155	0.0571	0.0705	1.23
Burgers $\nu=1.0$	192	0.0167	0.0772	0.1286	1.67
Burgers $\nu=1.0$	336	0.0253	0.1931	0.2586	1.34
Van der Pol	96	0.0368	0.0982	0.1063	1.08
Van der Pol	192	0.0511	0.1961	0.1713	0.87
Van der Pol	336	0.0769	0.2389	0.1985	0.83
Duffing	96	0.394	0.788	0.761	0.97
Duffing	192	0.747	1.046	0.917	0.88
Duffing	336	0.886	0.996	0.988	0.99
Harmonic	96	0.069	0.348	0.205	0.59
Harmonic	192	0.118	0.529	0.477	0.90
Harmonic	336	0.307	0.719	0.512	0.71

Table 21: Experiment 28. Ablation/base ratio > 1 means the ablated model is worse than the matched-steps baseline (Koopman lifting was helping); < 1 means the ablated model is better (lifting was neutral or harmful). No significance test is computed between these two conditions.

Harmonic). **Confidence: Low.:** directionally consistent with the point estimates, but see alternatives below.

2. **[HYP]** Training-instability artifact. At 50k steps the two architectures may sit at different, noisy points on a non-monotone training trajectory; the ablated model may simply converge faster or slower per-dataset for reasons unrelated to any inductive-bias story. **Confidence: Medium.:** cannot be excluded without loss curves, which were not retained for this run per the pre-July-2026 raw-output-retention gap.
3. **[HYP]** Burgers $\nu = 1.0$ is the dataset where the baseline itself is furthest from the published checkpoint (0.057 vs 0.015, $3.7\times$), i.e. the dataset where 50k steps is least sufficient overall. The ablation may suffer disproportionately from that same general insufficiency rather than from losing the lifting specifically. **Confidence: Medium.:** plausible given the scale of the baseline gap on this dataset.

Confidence: Low.–Confidence: Medium. (low-medium) on any interpretation. This section exists to record the point estimates and their preliminary status, not to support a conclusion. The 100k rerun (Section 13) is designed specifically to resolve explanations 2 and 3 by matching both conditions’ training steps and by testing convergence in-distribution before any OOD comparison is drawn.

7.1.6 What Is Missing to Gate This Result

Per the Section 1.2 estimator/model validation rule, an in-distribution sanity check (Panda’s own training distribution, e.g. Lorenz) at matched steps for both conditions is required before any OOD comparison in this section can be cited. This check has been referenced informally (published ≈ 0.021 , retrained baseline ≈ 0.275 , ablation ≈ 0.617 at 50k on Lorenz, i.e. ablation $2.2\times$ worse than baseline in-distribution) but the underlying

run artefacts were not available for this log revision and are not included in the table above. This in-distribution figure should be treated as anecdotal until re-derived and saved alongside the 100k campaign.

7.1.7 *Revision (July 2026): 100k Matched-Steps Rerun Launched*

[OBS] The 100k rerun (ablation arm only, per the revised design principle that a fresh baseline retrain adds cost without discriminating power once the published checkpoint and the 50k baseline already anchor the comparison) is launched on the Kaggle T4 pipeline as of this revision. The saved checkpoint artefact set from the run contains four files with no `pytorch_model.bin`; this is expected for the sharded/safetensors save format in use and is not a run failure. The pre-registered protocol (Section 13, node A1) applies unchanged: in-distribution Lorenz convergence against the published checkpoint is the gating check, evaluated before any OOD table from this run is interpreted. Status: **in progress, not yet gated**. Nothing in Section 8 is superseded by the mere fact of launching; the 50k point estimates above remain preliminary until the 100k in-distribution gate reports.

7.1.8 *Correction (July 2026): The Launched Run Is the Baseline, Not the Ablation*

The preceding revision paragraph stated the 100k run was launched as “ablation arm only.” This was an inference, not a confirmed fact, and it is incorrect. **[OBS]** The training script’s own run header reports `Run: baseline (segment 2), use_dynamics_embedding: True`, confirming this run is the matched-steps `retrained_base` condition (Koopman lifting present), not `koopman_ablation`. The “ablation-only, cost-saving” design principle described in the preceding paragraph was not actually adopted; the original Research Tree A1 specification (both conditions retrained to 100k with a direct paired test between them) is the one in effect. Consequently: (a) once this baseline run completes at 100k, a separate full `koopman_ablation` run from step 0 is still required before A1’s direct significance test can be computed; (b) the compute budget for A1 is therefore larger than the July 2026 revision above implied (two full 100k runs, not one); (c) the baseline run’s remaining segment ($\approx 20k$ steps from the 80k checkpoint, $\approx 3.9h$ at the observed 1.44 it/s) is not wasted effort — a fully matched-steps baseline is required either way and is in fact more rigorous than the 50k-vs-100k comparison the mistaken “ablation-only” framing would have produced. Status: baseline segment 2 in progress (resumed from checkpoint-80000 after a session reset with no step loss); ablation run not yet started.

7.1.9 *Correction 2 (July 2026): Ablation Can Also Resume from Its Existing 50k Checkpoint*

Point (a) above, stating the ablation run needs to start “from step 0,” is itself incorrect and is corrected here rather than edited in place, consistent with this log’s append-only convention. **[OBS]** Both conditions in Experiment 28 (`retrained_base` and `koopman_ablation`) were trained to 50k steps, and the baseline segment-2 log confirms it resumed from that existing checkpoint (`Global: 50000 -> 100000`), not from step 0. If `koopman_ablation`’s `checkpoint-50000` from Experiment 28 is still available, the ablation run can resume from it the same way the baseline did, requiring only the remaining 50k steps ($\approx 50000/1.44 \text{ it/s} \approx 9.6h$) rather than a full 100k-step run from scratch ($\approx 19.3h$). This halves the previously stated remaining compute for A1 (baseline $\approx 3.9h$ + ablation $\approx 9.6h \approx 13.5h$ total, versus the $\approx 23h$ implied by Correction 1), **conditional on**

the Experiment 28 ablation checkpoint still being retrievable — checkpoint/raw-output retention only became formal policy in July 2026, after Experiment 28 ran, so this is not guaranteed and is unverified as of this revision. If the checkpoint is unavailable, the from-scratch estimate in Correction 1 applies.

7.1.10 Progress Note (July 2026): Baseline Segment 3 Launched

Baseline segment 3 (checkpoint-80000 $\rightarrow$ 100000, 20,000 steps) started following the same fresh-optimizer convention as segment 2. The segment-3 warmup/LR schedule was not pre-registered before this segment started (no prior decision existed to reuse); two options were discussed (reuse segment 2’s flat 2000-step warmup, or scale proportionally to segment length, 800 steps). **[OBS]** The schedule actually in effect was confirmed from the observed LR trajectory: step 80500, $\text{lr} = 3.13 \times 10^{-5}$; step 81000, $\text{lr} = 5.00 \times 10^{-5}$. This matches a 800-step linear warmup to a 5×10^{-5} peak ($500/800 \times 5 \times 10^{-5} = 3.125 \times 10^{-5}$ at step 500 into the segment) followed by cosine decay, not the 2000-step warmup that had been intended as the default. The discrepancy is recorded rather than corrected mid-run: restarting to change an 800-versus-2000-step warmup out of a 20,000-step segment, both reaching the same peak LR, is not expected to materially change the step-100k outcome, and stopping a run already in progress to fix it would cost more wall-clock time than the difference is worth. **Segment-3 schedule as actually run:** fresh AdamW, $\text{lr}_{\text{peak}} = 5 \times 10^{-5}$, 800-step linear warmup, cosine decay to 0 over the remaining 19,200 steps, seed 99 (unchanged across all segments and arms).

7.2 100k Evaluation: Gate, Recomputed Reference, Held-Out Systems, Significance, and OOD (July 2026)

Both arms reached 100k steps (baseline: segments 1–3, the last two documented above; ablation: an analogous multi-segment path, resuming from Experiment 28’s checkpoint-50000 through to 100k). This subsection reports the full evaluation campaign run against both final checkpoints, using a consolidated, self-contained notebook (`panda_100k_eval_clean.ipynb`) built after an earlier fragmented sequence of gate/diagnostic/recompute cells became difficult to track reliably.

7.2.1 Pre-Registration Compromise

The original design called for a blind, pre-registered in-distribution gate: decide a pass/fail threshold, then look at the number once. This was not achieved. The first gate check (a single Lorenz protocol, $H=96$) was run and observed before the full set of protocol/horizon combinations was decided, and baseline_100k’s numbers were seen across multiple protocols before any of them was selected as “the” gate criterion. **Every result in this subsection is therefore exploratory re-analysis, not a confirmatory pre-registered test.** This is stated plainly here, and in the evaluation notebook itself, rather than left implicit. Any future checkpoint pair should use a freshly pre-registered gate, informed by the protocol choices this campaign settled (see below).

7.2.2 Gate FAIL and Protocol Diagnostic

The original gate (Lorenz, $H=96$, 3-channel, threshold $\leq 50\%$ of the anecdotal 50k figures: baseline ≤ 0.1375 , ablation ≤ 0.3085) **[OBS]** failed for both conditions: baseline 100k

MAE = 0.5647 (worse than its own 50k anecdotal value), ablation 100k MAE = 0.3462 (44% improvement, short of the 50% bar).

To understand this, three Lorenz protocols were tested, isolating one variable at a time: `gate_3ch` (fixed initial condition, manual RK4, 3 channels), `gate_1ch` (identical trajectory to `gate_3ch`, x-component only — isolates channel count alone), and `alt_1ch` (seeded random initial condition, `solve_ivp`/RK45, x-component only — isolates initial condition and integrator, independent of channel count). **[OBS]** All three protocols failed the gate for both conditions, at both H=96 and H=336 (6/6 combinations). The channel-count/IC/integrator diagnostic therefore does not explain the FAIL — it is not a fragile result contingent on one arbitrary evaluation choice.

7.2.3 *Recomputed 50k Reference Contradicts the Anecdotal Figures*

Both 50k checkpoints (Experiment 28) were reloaded and evaluated under the identical harness used for the 100k checkpoints, across all three Lorenz protocols and both horizons. **[OBS]** The recomputed 50k values do not match the anecdotal 0.275 (baseline) / 0.617 (ablation) figures, and — more importantly — **reverse their ordering**: ablation beats baseline in 7 of 8 tested protocol/horizon/system combinations at 50k (later extended to 8 with held-out systems; see below), not the other way around. This is consistent with the log’s own prior caveat (Section 8) that the anecdotal figures were “referenced informally... underlying run artefacts were not available,” and confirms that caveat was warranted: **[HYP] Confidence: Medium.** the anecdotal figures were produced under a still-unrecoverable protocol, distinct from all three tested here. A checkpoint-identity concern was raised and directly ruled out: **[OBS]** both 50k checkpoints were independently confirmed by the user as correctly saved and retrievable, and `config.json`’s `use_dynamics_embedding` field was asserted to match the expected arm before every evaluation in this campaign.

A cross-session numerical discrepancy was also noted (baseline_100k’s `gate_3ch` H=96 value differed between two separate Kaggle sessions, 0.5647 vs. 0.6951, while ablation’s matched exactly across the same sessions) and is flagged as a curiosity rather than a threat to the results reported here: **[OBS]** R1 (Experiment 34) already confirmed within-session determinism, and all numbers in this subsection were computed within a single consolidated session, making them mutually comparable regardless of what a different, now-abandoned session produced. **[HYP] Confidence: Low.** A plausible mechanism, not verified, is cuDNN/cuBLAS algorithm-selection non-determinism specific to the dynamics embedding’s extra compute path, which the ablation arm lacks.

7.2.4 *Held-Out Systems: Rossler and SprottB*

Two systems confirmed absent from the actual loaded `skew40` corpus (Rossler and Sprott System B — Sprott, 1994 — verified against the printed list of 1150 unique skew-product source names, whose 110 unique atomic components do not include either) were simulated (`solve_ivp`/RK45, seeded IC) and evaluated identically. This tests generalization to genuinely different dynamics, not just protocol variants of one system (Lorenz). **[OBS]** Ablation beat baseline in 6/6 system/horizon combinations (Rossler H=96/336, SprottB H=96/336, plus the matching `gate_3ch` Lorenz rows). Effect size varies substantially: Rossler shows large, decisive gaps at both horizons (47% and 39% lower MAE for ablation);

SprottB’s gap is moderate at H=96 (25%) and small at H=336 (5.7%, the weakest single result in an otherwise consistent pattern).

7.2.5 Significance Testing

A direct paired Wilcoxon test (baseline_100k vs. ablation_100k, same windows, same session) was run across all five systems (`gate_3ch`, `gate_1ch`, `alt_1ch`, Rossler, SprottB) at both horizons (10 tests total). **[OBS]** 6/10 nominally significant ($p < 0.05$) in ablation’s favor. Applying a strict Bonferroni correction ($\alpha = 0.005$), 5/6 survive: `gate_3ch` H=96 ($p = 0.0007$), `gate_1ch` H=96 ($p = 0.0036$), Rossler H=96 ($p = 7 \times 10^{-6}$), Rossler H=336 ($p = 2 \times 10^{-6}$), SprottB H=96 ($p = 1 \times 10^{-4}$). Only `alt_1ch` H=336 ($p = 0.045$) does not survive strict correction.

[OBS] Effect size shrinks from H=96 to H=336 in 4 of 5 systems (consistent, in weakened form, with the published paper’s claim that the dynamics embedding specifically aids autoregressive rollout error — see the methodological comparison below). Rossler is the exception: its gap *widens* slightly at H=336 and remains the most significant result of the whole campaign at both horizons. No explanation for Rossler’s divergent behaviour is currently available.

[PAT] Confidence: High. Ablation significantly outperforms baseline at H=96 across four independent systems (Lorenz via `gate_3ch/gate_1ch`, Rossler, SprottB), robust to multiplicity correction. This is a substantially stronger evidentiary basis than any single-protocol or single-system result would provide.

7.2.6 50k→100k Trend: Gap Grew, Did Not Shrink

Given a confirmed $4\times$ smaller training batch size than the published setup (see below), a natural question is whether baseline is simply “catching up” and would overtake ablation with more training. This was tested directly: for each of the 8 protocol/system/horizon combinations with both 50k and 100k values, the gap (baseline MAE – ablation MAE) at 50k was compared against the gap at 100k. **[OBS]** The gap grew in 5/8 combinations, shrank in 2/8, and flipped sign entirely in 1/8 (`alt_1ch` H=336, where baseline was ahead at 50k and ablation overtook by 100k). If baseline were on a slower learning curve destined to close the gap, shrinking should dominate; it does not. **[HYP] Confidence: Medium.** The directional finding (ablation ahead) is unlikely to be a transient artifact of baseline simply needing more gradient steps to catch up, within the 50k–100k window observed. **[OBS]** This check rests on only two time points per condition and cannot rule out a crossover beyond 100k steps, or a non-monotonic trajectory between the two measured points.

7.2.7 Methodological Comparison Against the Published Setup

The published paper’s Appendix B (“Training Details”) was consulted directly rather than assumed. **[OBS]** Architecture (d_model=512, 8 layers, 8 heads, N_poly=120, N_rff=256), context length (512), patch size/stride (16), loss function (MSE), augmentation rate (20% of trajectories), and forecasting-model training length (100,000 iterations) all match this project’s setup exactly. Two confirmed differences stand out:

1. **[OBS] Batch size: 1024 (paper) vs. 256 (this project), a $4\times$ difference,** forced by hardware (the paper used $4\times$ AMD MI100X GPUs with 32GB HBM2 each;

this project used a single Kaggle T4). At matched iteration count (100k=100k), this project’s checkpoints have seen approximately one quarter of the total training examples the published checkpoint saw at its own 100k-step mark.

2. **[OBS] Training channel count is fixed to exactly 3** in the published setup (“we fix the dimensions of each input trajectory to 3 only during training by randomly sampling 3 channels... every system in our dataset has at least 3 dimensions”). This confirms `gate_3ch` is the only one of the three Lorenz protocols tested here that evaluates in-distribution channel count; `gate_1ch` and `alt_1ch` were testing genuinely out-of-distribution channel count, which is consistent with both showing uniformly higher MAE for both models than `gate_3ch` did.

A third, minor difference: the published ground-truth trajectories use adaptive integration with tight tolerances ($\text{rtol}=10^{-9}$, $\text{atol}=10^{-10}$); this project’s `gate_3ch` trajectory uses fixed-step manual RK4 with no tolerance control. **[HYP] Confidence: Low.** This could introduce a numerical-precision noise floor on the ground-truth trajectory itself, but has no identified mechanism for explaining the *differential* baseline-vs-ablation gap specifically, since it would affect both models’ error against the same (slightly imprecise) target equally. LR schedule and warmup are not stated in the published appendix and remain unknown; no comparison is possible on this point.

7.2.8 Interpretation: Two Separate Claims, Not One

[OBS] A uniform “baseline is undertrained” account predicts uniform degradation across evaluation systems. This is not what was observed: OOD results (below) show baseline *beating* ablation on Burgers $\nu = 1.0$, and roughly tied on Van der Pol/Duffing, while losing decisively on every chaotic ODE system tested. A pure compute-throughput explanation has no mechanism to produce this selective, dataset-dependent pattern. This motivates separating two claims that were previously being discussed as one:

[HYP] Confidence: Medium. Claim 1 (magnitude): the overall size of baseline’s underperformance is likely inflated by the $4\times$ smaller training batch relative to the published setup. Absolute numbers from this campaign should not be read as directly contradicting the paper’s own reported ablation figures — the two are not comparable at matched compute.

[HYP] Confidence: Medium. Claim 2 (direction, dataset-dependence): the qualitative pattern — Koopman lifting helps on Burgers $\nu = 1.0$ (aperiodic, non-chaotic PDE), is roughly neutral on periodic systems (Van der Pol, Duffing), and hurts on every tested chaotic ODE (Lorenz, Rossler, SprottB) — is not explained by the compute gap alone (Section above: gap grew, not shrank, from 50k to 100k) and replicates independently across two experiments (Experiment 28’s original 50k point-estimate table and this significance-tested 100k campaign). A theoretical account, offered at the same confidence level as the pattern itself: Koopman/eDMD-style linearization is mathematically motivated for weakly nonlinear or dissipative-but-non-chaotic systems, where a fixed-dimensional linear approximation of the dynamics is not fundamentally obstructed. A genuinely chaotic attractor has positive Lyapunov exponents — sensitive dependence on initial conditions that no *fixed, non-adaptive* lift (`rff_trainable: false` in this architecture) can compensate for beyond the Lyapunov horizon. If the lift is unhelpful specifically where this theoretical obstruction applies, and helpful specifically where it does not, that is a substantive, theoretically coherent claim rather than an artifact

— though it remains at **Confidence: Medium**. pending further testing (more OOD systems per category, multiple independent trajectories/seeds per system, and ultimately a continuously-trained control run without the compute and segment-discontinuity confounds present in this campaign). **This theoretical account is now directly tested in Section 8 (Koopman Feature-Space Geometry, A3); see that section for the result.**

7.2.9 OOD Evaluation

Five OOD systems (Weather, Burgers $\nu = 1.0$, Van der Pol, Duffing, Harmonic) were evaluated at $H \in \{96, 192, 336\}$ via paired Wilcoxon test (15 tests total, raw predictions saved for future CRPS/Hellinger re-scoring per the project’s July 2026 retention policy).

Dataset	H	Baseline MAE	Ablation MAE	Δ (abl–base)
Weather	96	0.7312	0.6942	−0.0370
Weather	192	0.7983	0.8483	+0.0500
Weather	336	0.9204	0.9984	+0.0780
Burgers $\nu=1$	96	0.0474	0.0686	+0.0212 ($p = 0.011$)
Burgers $\nu=1$	192	0.0764	0.0815	+0.0052
Burgers $\nu=1$	336	0.1666	0.1628	−0.0037
Van Der Pol	96	0.1330	0.1167	−0.0163
Van Der Pol	192	0.1781	0.1856	+0.0076
Van Der Pol	336	0.2277	0.2001	−0.0275
Duffing	96	0.8749	0.8721	−0.0028
Duffing	192	1.0135	0.8961	−0.1175
Duffing	336	0.8395	1.0026	+0.1631
Harmonic	96	0.3808	0.2383	−0.1426 ($p < 0.001$)
Harmonic	192	0.4673	0.4385	−0.0288
Harmonic	336	0.7165	0.5864	−0.1302 ($p < 0.001$)

Table 22: OOD paired evaluation, baseline_100k vs. ablation_100k, n=20 windows per row.

[OBS] Only 2/15 rows survive strict Bonferroni correction on their own (Harmonic $H=96$ and $H=336$, both $p < 0.001$, ablation better). Burgers $\nu = 1$ $H=96$ ($p = 0.011$, ablation *worse*) does not survive strict correction in isolation.

[HYP] Confidence: Medium. The Burgers $H=96$ result deserves more credence than its raw p -value alone suggests, because it is not an isolated result among 15 uncorrected tests — it is a specific, directional prediction already on record from Experiment 28’s independent, cruder (unpaired, non-significance-tested) 50k point-estimate table, which found “ablation worse than baseline at every horizon on Burgers $\nu = 1.0$.” An independently-predicted result being confirmed by a properly-controlled follow-up test is a different epistemic situation than a $p = 0.011$ appearing with no prior expectation among many comparisons.

[PAT] Confidence: Medium. Combining in-distribution and OOD results: the lift helps on Burgers $\nu = 1$ (aperiodic, non-chaotic PDE, the system class most aligned with the eDMD/Koopman motivation), hurts on chaotic ODEs (Lorenz, Rossler, SprottB), and is roughly neutral on periodic/weakly chaotic systems (Van der Pol, Duffing) and Weather (no consistent direction across horizons). This pattern cuts across the in-distribution/OOD

boundary rather than tracking it — the relevant axis appears to be aperiodic-PDE-like vs. periodic/chaotic-ODE-like, not whether the system was in the training corpus.

7.2.10 *Caveats Applying Throughout This Subsection*

1. Single trajectory (one seed, one initial condition) per system. Windows within a trajectory are not independent draws — this is the same structural weakness that caused the heterogeneity finding (Section 12.4) to collapse at proper sample size. No result in this subsection has been checked against multiple independent trajectories per system.
2. The training pipeline (discontinuous segment resumes, fresh optimizer state at every boundary, reactively-chosen rather than pre-registered warmup schedules, $4\times$ smaller batch than the published setup) means every claim here is a statement about *this specific implementation*, not a clean confirmation or refutation of the published paper’s architecture.
3. Pre-registration was compromised partway through this campaign (Section above). All results are exploratory re-analysis.

8 Koopman Feature-Space Geometry (A3)

8.1 Background and Motivation

Section 7 (Koopman Lifting Ablation, 100k campaign) established a behavioural pattern via controlled ablation: the fixed Koopman lift (`rff_trainable: false`) significantly improves forecasting on in-distribution/held-out chaotic ODEs (Lorenz, Rossler, SprottB) and on Harmonic, while significantly worsening it on Burgers $\nu = 1.0$. Alongside this **[OBS]**, a **[HYP]** theoretical account was offered at the same medium confidence: Koopman/eDMD-style linearization is mathematically motivated for weakly nonlinear or non-chaotic dynamics, and a fixed, non-adaptive lift cannot compensate for the sensitive dependence (positive Lyapunov exponents) of genuinely chaotic attractors. This section reports A3, designed specifically to test that theoretical account directly against the frozen lift’s own representational geometry, independent of any forecasting-accuracy measurement. **A3 uses only the baseline_100k checkpoint** (`use_dynamics_embedding=True`); it does not touch the ablation checkpoint and therefore cannot speak to the baseline-versus-ablation comparison itself, only to the theoretical account layered on top of it.

Two independent operationalisations were used, chosen to test different halves of the theoretical claim: (1) eDMD reconstruction residual, which tests whether the lift’s dictionary is well-approximated by one linear operator (the “linearization” half), and (2) Jacobian sensitivity of the lift, which tests whether the fixed lift amplifies input perturbations differently across system classes (the “cannot track sensitive dependence” half). Architecture confirmed directly from source (`PatchTSTKernelEmbedding.forward()`): the lift concatenates the raw patch, polynomial features at two degrees ($\{2, 3\}$, each contributing `num_poly_feats`), and random Fourier features, then applies a single learned 512×512 projection (`bias=False`) shared identically across every input regardless of channel or system type. All raw-dictionary components (`freq_weights`, `freq_biases`, polynomial index sets) are confirmed non-trainable, consistent with the `rff_trainable` config flag.

8.2 Experiment 35: eDMD Reconstruction Residual

8.2.1 Method

Five classes: Lorenz (`gate_3ch` protocol, verbatim from Section 7’s evaluation harness), Rossler, SprottB (both verbatim seeded-IC `solve_ivp` generators), Burgers $\nu = 1.0$ (top-3 PCA channels of a 128-mode spectral solve; see the methodological detours below for why top-3 and not the full 16), and Harmonic (single-channel). All trajectories fixed at exactly 512 steps (the model’s context length), extracted via one non-sliding forward pass per trajectory. Both Φ_{pre} (the pre-projection 512-dim dictionary) and Φ_{post} (the learned projection’s output) were captured via forward hooks and analysed separately. A single ridge-regularised linear operator K (`RidgeCV`-selected λ) was fit on a class-balanced pool of patch-to-patch pairs (40 fit trajectories/class, 20 held-out/class, 3 random resamples), with pairing performed strictly within (`trajectory`, `channel`) groups (never across a channel boundary, since the lift and downstream temporal attention treat channels independently at this stage). Normalised residual $\|\Phi_{t+1} - K\Phi_t\|/\|\Phi_{t+1}\|$ was compared against a persistence baseline ($K = \text{Identity}$) as the primary correctness check, per this log’s estimator-validation rule (Section 1.2): a fitted operator that does not beat trivial persistence is not evidence of genuine linear structure, regardless of its absolute residual value.

8.2.2 Methodological Detours (Recorded in Full, Per Section 1.2 Convention)

Three confounds were found and corrected during this experiment, each via direct verification rather than assumption; recorded here rather than silently absorbed into the final method, consistent with this log’s practice in Sections 7, 8, and 10.

1. **Input-orientation error.** An initial implementation called the model directly with an assumed `(batch, channels, time)` tensor layout, causing a shape failure in the patchifier. Reading `PatchTSTModel.forward()` directly confirmed the required layout is `(batch, sequence_length, num_channels)`. Corrected before any features were extracted for analysis.
2. **Cross-channel patch-pairing contamination.** An initial pairing implementation flattened the `(channels, patches)` dimensions before constructing consecutive-patch pairs, which silently paired the last patch of one channel with the first patch of the next at trajectory boundaries. Corrected by grouping strictly within `(trajectory, channel)` before pairing.
3. **Burgers dead-channel dilution.** [OBS] Direct per-channel inspection of the Burgers PCA representation found 11 of 16 channels essentially frozen for the full 512-step window (median patch-to-patch ratio $\sim 10^{-6}$ to 10^{-4} versus 0.09–0.21 for the top 4 channels). Pooling all 16 channels into the per-class residual statistic diluted the comparison with near-constant, uninformative channels. **Corrected by restricting Burgers to its top-3 PCA channels**, matching the channel count of the three chaotic ODE classes. **This is an explicit, acknowledged deviation from Section 7’s 16-channel load.burgers_nu1 protocol**; A3’s “Burgers” class is not identical to Section 7’s OOD Burgers evaluation object, though it is built from the same underlying simulator and viscosity. An intermediate, since-corrected version of this experiment also mistakenly discarded the first 200 steps of each Burgers trajectory (an addition not present in Section 7’s protocol, introduced and then removed within this session after the near-total absence of variance across the full 16-channel PCA representation – prior to isolating the true dead-channel cause – was initially, incorrectly, attributed to a decayed transient rather than channel dilution).

8.2.3 Observations

Class	Channels used	Φ_{pre} median resid.	Φ_{post} median resid.	Persistence baseline	
Lorenz	3	0.83–0.97	0.69–0.76	~ 1.2 – 1.3	
Rosler	3	0.47–0.53	0.44–0.48	~ 1.0 – 1.2	
SprottB	3	0.52–0.59	0.52–0.53	~ 0.9 – 1.0	
Harmonic	1	0.43–0.48	0.42–0.45	~ 1.1 – 1.2	
Burgers	3 (top PCA)	0.29–0.31	0.31–0.32	0.16 (Φ_{pre}), 0.16 (Φ_{post})	No – fitted

Table 23: Experiment 35 summary, median values across 3 random fit/held-out splits. Burgers uses the corrected top-3-channel representation throughout (see confound 3 above). Ranges reflect min–max across the 3 splits within each feature space.

[OBS] Across all 3 splits and both feature spaces, ordering by median residual is consistent: Lorenz > SprottB > Rosler $\approx$ Harmonic > Burgers. [OBS] For Lorenz, Rosler, SprottB, and Harmonic, the fitted K produces a substantially lower residual

than the persistence baseline in every case, indicating the pooled linear operator is capturing genuine, exploitable structure in the lift’s dictionary for these four classes. **[OBS]** For Burgers, the relationship inverts: the fitted K ’s residual (0.29–0.32) is *higher* than the persistence baseline (0.16), meaning the shared linear operator performs worse than assuming no change at all. This held after two independent corrections (dead-channel dilution and the transient-window question), and was checked directly via the same persistence-comparison methodology applied to the other four classes. **[PAT]** **Confidence: Medium.** Burgers’ apparently “low” absolute residual in the uncorrected pooled statistic is not evidence of good linear-fittability; once compared against its own persistence baseline, it is the *worst*-fit class of the five, not the best. This directly contradicts the naive reading of the theoretical account in Section 7 (that Burgers, as the aperiodic non-chaotic class, should show the cleanest linear structure).

8.2.4 Competing Explanations

1. **[HYP]** **Confidence: Medium.** $\nu = 1.0$ Burgers dynamics are dominated by the linear diffusive term of the underlying PDE rather than the nonlinear advective term across essentially the whole trajectory (strongly diffusive regime, established in Section 3’s viscosity sweep as the non-chaotic end of the range). If the true dynamics are already close to linear at the PDE level, there may be very little patch-to-patch *change* for any operator, linear or not, to meaningfully predict beyond persistence – consistent with Burgers’ low persistence baseline (0.16) itself, though this does not by itself explain why the *fitted* K underperforms even that baseline.
2. **[SPEC]** The lift’s dictionary, though shared across all five classes architecturally, may be tuned (via its fixed random index sets and frequencies, sampled once at initialisation and never revisited) in a way that happens to be poorly matched to Burgers’ specific spatial-PCA-derived channel statistics, independent of any chaos-related property – a representation-specific rather than dynamics-specific account. Not distinguished from the diffusion-dominance explanation by this experiment.
3. **[HYP]** **Confidence: Low.** The class-balanced pooled fit (equal-weighting all five classes’ pairs before fitting one shared K) may bias K toward the majority behaviour of the other four classes, actively disadvantaging Burgers specifically, rather than Burgers having no linear structure of its own. A per-class-only K (fit and evaluated on Burgers alone) was not run and would partially distinguish this from the two explanations above.

8.3 Experiment 36: Jacobian Sensitivity of the Fixed Lift

8.3.1 Method

For one representative patch per trajectory (15 trajectories/class), the full Jacobian of that patch’s lifted output with respect to the *entire* 512-step input context was computed via `autograd.grad`, one call per output dimension, and summarised by Frobenius and spectral norm. This required bypassing the pipeline’s `predict()` method (confirmed via source inspection to be wrapped in `@torch.no_grad()`) and calling the inner `PatchTSTModel.forward()` directly via its documented `past_values` keyword argument.

8.3.2 Methodological Detours

Three further confounds were found and corrected, in addition to the `@torch.no_grad()` bypass above.

1. **Hook-detach conflict.** The forward hooks used for Experiment 35’s feature extraction call `.detach()` on capture (correct for that purpose, since a large retained autograd graph is unnecessary and wasteful there). Reusing the same hooks for this experiment silently broke gradient flow (`RuntimeError: element 0 of tensors does not require grad`). Corrected by registering a separate, non-detaching hook pair specifically for this experiment.
2. **Raw-input scale confound.** **[OBS]** An initial run showed Burgers’ Frobenius norm ($\sim 33,000$) $400\text{--}2800\times$ larger than every other class ($\sim 12\text{--}80$), a magnitude gap inconsistent with a genuine sensitivity effect. Direct inspection of `PatchTSTStdScaler.forward()` confirmed the scaler divides by each window’s own per-channel standard deviation before patching; by the chain rule, differentiating with respect to the *raw* input introduces a $1/\text{scale}$ term that mechanically inflates the gradient for any class with small raw-signal variance (Burgers’ PCA channels have raw magnitude $\sim 0.01\text{--}0.3$ versus $\sim 1\text{--}30$ for the other four classes). **Corrected by differentiating with respect to the scaler’s output (the scaled input the model actually operates on) rather than the raw input**, via an additional forward-pre-hook on the patchifier. This reduced the gap to $5.6\text{--}11\times$ – smaller, but still class-dependent (see next item).
3. **Patch-position (transient) confound.** **[OBS]** With the scale confound corrected, Burgers remained the clear outlier ($\text{fro_norm} \approx 512$ vs. $45\text{--}91$ for the other four classes), using a fixed choice of patch index 0. A position-sweep diagnostic (patch indices $\{0, 5, 10, 15, 20, 25, 30\}$, 5 trajectories/class) found **[OBS]** Burgers elevated *only* at patch position 0 (523 vs. $78\text{--}92$ at positions $5\text{--}30$, indistinguishable from the other four classes’ range of $46\text{--}107$ across all positions). This is consistent with Burgers’ sharp early-trajectory transient (already characterised in Experiment 35’s confound-hunting) coinciding specifically with the arbitrary fixed choice of patch 0; the position-sweep result is the one reported as final.

8.3.3 Observations

Class	Frobenius norm, scaled-input Jacobian, by patch position						
	0	5	10	15	20	25	30
Lorenz	80.6	77.9	78.4	82.3	92.7	78.2	79.9
Rossler	79.0	85.9	94.6	106.7	78.7	86.8	83.8
SprottB	84.2	84.3	97.4	81.1	83.9	79.3	78.1
Harmonic	47.5	48.5	46.8	46.5	61.1	68.5	51.3
Burgers	523.4	92.1	77.5	77.2	77.9	78.4	78.7

Table 24: Experiment 36, median Frobenius norm across 5 trajectories/class, scaled-input Jacobian, by patch position within the 512-step context.

[OBS] Once patch position 0’s transient-affected value for Burgers is excluded (positions $5\text{--}30$), Jacobian sensitivity of the lift is statistically indistinguishable across all five classes: chaotic ODEs (Lorenz $\sim 78\text{--}93$, Rossler $\sim 79\text{--}107$, SprottB $\sim 78\text{--}97$), Burgers ($\sim 77\text{--}92$),

and Harmonic (consistently somewhat lower, $\sim 46\text{--}69$, but not dramatically so). **[PAT] Confidence: High.** No support is found for the “fixed lift cannot track sensitive dependence” half of the Section 7 theoretical account: the lift’s own local sensitivity to input perturbation does not differentiate chaotic from non-chaotic input classes.

8.4 Combined Interpretation

[PAT] Confidence: Medium. Neither operationalisation of the Section 7 theoretical account (eDMD linearizability, Jacobian sensitivity) shows the pattern that account predicts. Two independent lines of evidence, not one, point the same direction: eDMD residual shows Burgers as the *worst*-fit class relative to its own persistence baseline, not the best; Jacobian sensitivity shows no meaningful separation between chaotic and non-chaotic classes at all. A further dissociation, not predicted by the theoretical account in either direction, strengthens this reading: Harmonic – an exactly linear system in its true dynamics – shows *good* eDMD linear-fittability (fitted K clearly beats persistence) while Section 7 found the lift *hurts* Harmonic forecasting. Burgers shows the opposite pairing (poor linear-fittability, lift *helps* forecasting). Both directions of dissociation being present in the same small set of five classes is difficult to reconcile with a single linearizability-based mechanism in either its helping or hurting form.

What this does and does not affect. A3 never touches the ablation checkpoint and therefore has no logical bearing on Section 7’s own controlled-ablation MAE/significance results, which stand as reported. What A3 weakens is specifically the *theoretical account* layered onto those results (Section 7, “Interpretation: Two Separate Claims, Not One”, Claim 2’s explanatory mechanism) – not the underlying behavioural observation that the lift helps on Burgers and hurts on Harmonic/chaotic ODEs. That behavioural pattern (Section 7’s Claim 2) is unaffected and requires a different explanation than the one currently attached to it.

8.5 What Remains Open

1. Why the fitted K underperforms persistence specifically on Burgers is not resolved by this section (Experiment 35, competing explanations). A per-class-only (non-pooled) K fit, and a direct measurement of what fraction of Burgers’ PDE dynamics is diffusion-versus-advection-dominated across the evaluated window, would partially distinguish the explanations offered.
2. Given A3’s negative/mixed result on the lift’s own geometry as an explanation, the mechanism behind Section 7’s Claim 2 pattern remains unestablished. This directly elevates the priority of **A2a** (cheap temporal-attention probes, no retrain) as the next most promising untested architectural locus – see the Research Tree addendum (Section 13).
3. This section’s channel counts differ substantially across classes by construction (Lorenz/Rosler/SprottB: 3; Burgers (corrected): 3; Harmonic: 1) and, prior to correction, Burgers’ original 16-channel representation was heavily diluted by near-dead channels. Neither eDMD nor Jacobian analysis in this section isolates chaoticity from channel count/representation as the driving axis; a system that is chaotic but low-dimensional/non-ODE, or non-chaotic but matched in channel count and representation type to the chaotic classes, would be needed to fully disentangle this, and is not currently planned.

Confidence: Medium. for the section’s central negative finding (neither operationalisation supports the Section 7 theoretical account). **Confidence: High.** for the Jacobian result specifically (clean, position-swept, scale-corrected). **Confidence: Low.–Confidence: Medium.** for any explanation of the Burgers eDMD anomaly, which remains an open, unresolved side-finding rather than a settled account.

9 Temporal-Attention Probes (A2a)

9.1 Background and Motivation

A3 (Section 8) tested and did not support the leading theoretical account for Section 7’s Claim 2 pattern (Koopman/eDMD linearization). Channel attention was independently ruled out four times (Experiments 9, 22, 27, 33). This left temporal attention as the last major untested architectural locus for explaining why the fixed Koopman lift helps on chaotic ODEs and Harmonic but hurts on Burgers $\nu = 1.0$. A2a tests this cheaply, at inference time on the existing `baseline_100k` and `ablation_100k` checkpoints, with no retraining, via three probes: patch-order shuffling (Probe 1, the pre-registered gate for whether the expensive A2b retrain is justified), context-length truncation (Probe 2), and attention-map inspection (Probe 3).

Pre-registered escalation gate (Probe 1 only, fixed before running): escalate to A2b (temporal-attention ablation retrain) if shuffle-induced MAE degradation is significantly larger on the chaotic classes (Lorenz, Rossler, SprottB) than on Burgers $\nu = 1.0$ (16-channel, matching Section 7’s protocol) in ≥ 2 of 3 horizons ($H \in \{96, 192, 336\}$). Probes 2 and 3 are supporting/descriptive, not gating. Classes evaluated: Lorenz (`gate_3ch`), Rossler, SprottB, Burgers $\nu = 1.0$ (16-channel PCA), and Harmonic, on both checkpoints, $n_{\text{windows}} = 20$, seed 0.

9.2 Experiment 37: Probe 1 — Patch-Order Shuffling

9.2.1 Method

Within each context window, patches are randomly shuffled in temporal order before being passed to the model (position embeddings, if any, remain at their original slots; only the content ordering is permuted). MAE is compared against the unshuffled (vanilla) forecast on the same window, for both checkpoints, all five classes, all three horizons. `degradation_median` is defined as the per-class, per-horizon median of (shuffled MAE – vanilla MAE) across windows; significance via paired Wilcoxon.

9.2.2 Observations

[OBS] Shuffling significantly degrades MAE in every class, every horizon, both checkpoints (p ranging from $\approx 10^{-7}$ to $\approx 2 \times 10^{-3}$). Temporal order matters for all five classes tested; this part of the result is not in dispute.

[OBS] Running the pre-registered gate literally, on `baseline_100k` absolute degradation: at every one of the three horizons, all three chaotic classes (Lorenz, Rossler, SprottB) show larger absolute degradation than Burgers, all individually significant. **Mechanical gate verdict: ESCALATE (3/3 horizons, 3/3 chaotic classes each).**

9.2.3 Critical Finding: Scale Confound in the Gate Metric

[OBS] Burgers’ vanilla MAE (≈ 0.0013 – 0.0026 across horizons) is roughly 1000 – $4000\times$ smaller in raw units than the chaotic classes’ vanilla MAE (≈ 0.6 – 7.3), for reasons unrelated to temporal attention (differing native scale of the underlying signal, already documented in Section 7’s own OOD table). The pre-registered gate compares absolute MAE degradation across these wildly different scales. This means the gate can pass largely

by construction: a class with a native MAE floor near 0.002 will show smaller absolute degradation than a class with a floor near 1–7 almost regardless of whether shuffling disproportionately harms it.

Recomputing degradation as a *relative* quantity (`degradation_median / vanilla_mae_median`) — not part of the pre-registered gate, but a natural scale-corrected alternative — reverses the verdict:

Horizon	Burgers (rel.)	Chaotic classes (rel.)	Classes larger
96	0.80	Lorenz 1.11, Rossler 2.06, SprottB 0.66	2/3
192	0.58	Lorenz 0.58, Rossler 1.34, SprottB 0.29	1/3
336	0.65	Lorenz 0.17, Rossler 0.73, SprottB 0.10	1/3

Table 25: Experiment 37, relative-degradation reanalysis of the Probe 1 gate, `baseline_100k`. 0 of 3 horizons meet the pre-registered $\geq 2/3$ -classes-larger bar under this scale-corrected metric.

[OBS] Under relative degradation, 0 of 3 horizons meet the $\geq 2/3$ - chaotic-classes-larger bar. **Relative-degradation verdict: DO NOT ESCALATE.**

[PAT] Confidence: Medium. The escalate/don’t-escalate decision from Probe 1 is not robust to a scientifically natural, if not pre-specified, choice of normalisation. By the literal letter of the pre-registration (absolute MAE difference), the verdict is ESCALATE; under the scale-corrected alternative, it is not. This is recorded as a genuine ambiguity, not resolved by fiat: the pre-registered text is honored as written, but the result cannot be treated as strong, unambiguous evidence for a chaos-specific temporal-attention mechanism, since a large part of the absolute-scale signal is attributable to Burgers’ pre-existing, previously documented MAE-scale gap rather than to shuffle-specific sensitivity.

[OBS] Under relative degradation, Harmonic (not part of the formal Lorenz/Rossler/SprottB-vs-Burgers gate, but part of A1’s broader “lift helps” grouping together with the three chaotic classes) shows the *largest* relative degradation of any class tested: 2.16, 1.25, 0.91 at H=96/192/336 on `baseline_100k` — exceeding Rossler at two of three horizons. This is directionally consistent with, not contradictory to, A1’s four-class “lift helps” grouping (Lorenz/Rossler/SprottB/ Harmonic) versus Burgers alone, even though it falls outside the narrower three-class formal gate.

9.3 Experiment 38: Probe 2 — Context-Length Truncation

9.3.1 Method

For each class and horizon, context is truncated from the front (most recent `ctx_len` steps of the original 512-step context are kept, i.e. `context[-ctx_len:]`) at `ctx_len` $\in \{512, 256, 128, 64, 32\}$, and MAE against the fixed target is recorded, per checkpoint.

9.3.2 Observations

[OBS] Burgers is the only one of the five classes where `baseline_100k`’s MAE *improves* as context shrinks from 512 to 32 (e.g. $H = 96$: $0.00167 \rightarrow 0.00092 \rightarrow 0.00052 \rightarrow 0.00042 \rightarrow 0.00041$). Every other class (Lorenz, Rossler, SprottB, Harmonic) shows MAE *worsening* as context shrinks, consistent with normal reliance on longer history.

[PAT] Confidence: Medium. This asymmetry is a genuinely scale-free finding — it is a statement about the *direction* of the context-length curve, not its magnitude, so it sidesteps the scale confound that undermined Probe 1’s gate. It is the single cleanest piece of evidence in this probe set for “Burgers does not need long-range temporal structure the way the other four classes do.”

9.4 Experiment 39: Probe 3 — Attention-Map Inspection

9.4.1 Method

Layer-3 self-attention maps were extracted for one representative window per class on `baseline_100k` and visually/summarily compared (attention entropy, qualitative heatmap structure). This probe is explicitly descriptive, not statistically tested, and is based on a single trajectory per class.

9.4.2 Observations

[OBS] Qualitatively, Lorenz’s layer-3 attention shows a clear local diagonal band plus a strong final-column pattern (consistent with a register/sink-token-like structure). Rossler and SprottB show more diffuse attention with a similar final-column feature. Burgers shows a smooth, strongly banded diagonal (locally-Toeplitz-like) structure without the sink-token feature. Harmonic is diffuse with an isolated bright off-diagonal spot.

Confidence: Low. for any interpretation drawn from this probe alone: single trajectory per class, no statistical test, and no pre-registered criterion for what would count as supporting or against any hypothesis. Recorded as a descriptive observation to be weighed alongside Probes 1 and 2, not as independent evidence on its own.

9.5 Experiment 40: Addendum — Trivial-Baseline Control for Probe 2 (H1 vs. H3)

9.5.1 Motivation

Probe 2’s asymmetry is ambiguous between two explanations: **H1 (attention-mechanism)** — Panda’s temporal attention makes genuinely differential use of long-range order depending on class; and **H3 (intrinsic-predictability confound)** — Burgers $\nu = 1.0$ is a smooth, strongly diffusive PDE with short effective memory, and any predictor, transformer or not, would show the same context-length asymmetry on it. This addendum adjudicates by running the identical truncation sweep (same `CONTEXT_LEN`, `HORIZONS`, $n_{\text{windows}} = 20$, `TRUNCATION_LENGTHS`, seed 0, and Burgers/Lorenz simulators, copied verbatim from the A2a notebook) against non-learned baselines, with Lorenz included as a chaotic-system control.

9.5.2 Method

Five baselines were implemented: `persistence` (repeat last context value) and `tail_linear` (linear fit to the last 10 context points) as *documented negative controls*, expected to be structurally blind to context truncation since truncation only removes points from the front of the window and these two baselines only read its tail; and three genuinely context-length-sensitive baselines: `full_context_linear` (linear fit to the entire truncated context), `context_mean` (mean of the entire truncated context), and `ar5_ridge`

(ridge-regularised order-5 autoregressive model fit by least squares on all consecutive lag-5 windows within the truncated context, then rolled forward autoregressively).

9.5.3 Methodological Note: A Design Flaw Caught by a Smoke Test

A reduced-scale smoke test (before the full run) found that `persistence` and an initially-specified fixed-tail `linear_extrap` baseline returned bit-for-bit identical predictions across all tested `ctx_len` values, since truncation removes only the front of the context and these baselines only ever read its last k points. This is not informative about context-length sensitivity and was corrected before the full run by (a) explicitly retaining these two as documented negative controls rather than treating their flatness as a finding, and (b) adding `full_context_linear` and `ar5_ridge`, both of which use the entire truncated context and are therefore mechanically capable of responding to the manipulation.

9.5.4 Observations

[OBS] `persistence` and `tail_linear` are exactly flat across `ctx_len` in the full run as well (Spearman correlation undefined, `ConstantInputWarning`), confirming they behave as the documented negative controls predicted.

[OBS] `full_context_linear` and `context_mean` reproduce `baseline_100k`’s own sign pattern with high fidelity: on Burgers, both show Spearman $\rho = +1.0$ ($p < 10^{-24}$) between context length and MAE at all three horizons (longer context $\rightarrow$ worse MAE, matching Panda’s own direction exactly). On Lorenz, both show strongly negative correlation at H=192/336 ($\rho \approx -0.9$ to -1.0 for `full_context_linear`; -0.6 to -0.9 for `context_mean`, with a near-zero, likely noise-driven exception at Lorenz H=96 for `context_mean`, $\rho = +0.1$), also matching Panda’s own direction (more context helps a chaotic system).

[OBS] `ar5_ridge` did not reproduce a clean pattern and is assessed as broken/uninformative rather than as evidence against H3: on Burgers it is non-monotonic (U-shaped, minimum near `ctx_len`=256, Spearman ρ collapses to exactly 0.0 at all three horizons, masking rather than measuring the relationship); on Lorenz it shows a catastrophic numerical blow-up at `ctx_len`=32 (MAE of 241.9, 496,751, and 63,387 at H=96/192/336 respectively), consistent with an under-regularised order-5 autoregressive fit on ~ 27 lag-windows of a high-variance chaotic signal producing an unstable difference equation that diverges when unrolled autoregressively over a long horizon.

9.5.5 Interpretation

[PAT] Confidence: Medium.–Confidence: High. Two of the three context-length-sensitive, non-learned, non-attention baselines reproduce Panda’s own Burgers/Lorenz context-length asymmetry almost exactly (identical sign, near-perfect Spearman correlation on Burgers). This is difficult to reconcile with H1 (attention-mechanism-specific) as the primary explanation for Probe 2’s asymmetry: a linear fit or a running mean has no attention mechanism, no positional encoding beyond “how many points am I given,” and no notion of temporal order beyond raw arithmetic on the values it sees, yet it shows the identical qualitative pattern. **This substantially favours H3 (intrinsic-predictability/data property) over H1** as the explanation for Probe 2’s asymmetry specifically. The third baseline’s failure to replicate this (or anything else) is attributed to

its own numerical fragility (U-shape, autoregressive instability), not treated as evidence against H3, since its failure mode does not resemble a principled null result.

9.6 Combined Interpretation

[PAT] Confidence: Medium. Putting Experiments 37–40 together: Probe 1’s gate, taken at face value, mechanically says ESCALATE, but that verdict is not robust to a natural scale correction and is best read as weak, ambiguous evidence at most. Probe 2’s asymmetry is the one genuinely scale-free signal in this probe set, but Experiment 40 shows it is substantially reproducible by non-learned linear/mean baselines, undercutting its value as evidence specifically for temporal-attention involvement. Probe 3 is descriptive and $n=1$, and cannot bear much independent weight either way. **None of the three probes, individually or combined, provides a robust basis for concluding that temporal attention is the differentiating mechanism behind Section 7’s Claim 2 pattern.** Combined with A3’s negative/mixed result on the Koopman-lift theoretical account (Section 8) and the four independent channel-attention nulls (Experiments 9, 22, 27, 33), this is the third of three plausible architectural-component explanations tested for Section 7’s behavioural pattern, and the third to fail to provide clean supporting evidence. The behavioural pattern itself (Section 7, Claim 2) is unaffected by any of this and stands as reported; what remains unestablished is why it occurs.

9.7 Decision: A2b Not Pursued at This Time

Given the weak and partly self-undermining evidence base above, the decision was made not to proceed to A2b (the expensive full temporal- attention ablation retrain) at this time. This is a deprioritisation, not a formal rejection of the temporal-attention hypothesis: A2a’s cheap inference-time probes were never capable of ruling temporal attention in or out with the same rigor as an actual ablation (the standard applied to channel attention, Experiments 9/22/27/33, and to the Koopman lift itself, Section 7). What A2a establishes is that the cheap evidence available does not clear the bar to justify the expensive test, given competing, unaddressed priorities (Branch B3c, the downsampled-Weather intervention, which is the last remaining thread on the project’s original Weather- advantage question and has been unblocked and ready to run since before A2a began). The Research Tree (Section 13) is updated accordingly.

Confidence: Medium. for the overall conclusion that none of the three tested architectural components (channel attention, Koopman-lift geometry, temporal attention) provides a clean mechanistic explanation for Section 7’s behavioural pattern. **Confidence: High.** for the specific finding that Probe 2’s asymmetry is substantially reproducible by non-learned baselines (Experiment 40, near-perfect Spearman correlations, directly computed).

10 Structure Statistic Investigation (Persistent Homology)

This section replaces the retired correlation-dimension analysis (Section 5, Experiment 21 auxiliary). It develops a gate-validated instrument before touching real data, per the Section 1.2 estimator validation rule, and the validation process itself caught three pipeline failures that would otherwise have produced an uninterpretable or misleading real-data result.

10.1 Instrument Scope (Read Before Interpreting Any Result in This Section)

The validated statistic is **H1 persistence of delay reconstructions** — specifically, the dominance of the longest-lived loop (`top1`) in a Vietoris–Rips filtration of a Takens delay-embedded scalar channel, relative to a per-channel shuffle-surrogate null. **[OBS]** A dedicated control (Gate G, Section 10.2) found that this statistic is *stronger* on the delay-embedded Lorenz x series (`top1/floor` = 1.97) than on the true Lorenz (x, y, z) state-space cloud itself (`top1/floor` = 1.68, below the validation bar). The instrument is therefore **not validated as a faithful measure of attractor homology**, and no claim in this section is phrased as “attractor topology” or “geometric structure.” The validated claim is narrower: the instrument detects persistent, smooth, rotational/oscillatory deterministic structure as it survives Takens delay reconstruction at the tested embedding parameters. This narrowing is itself a finding, not merely a caveat: it constrains what any downstream interpretation of Weather or ETTh results is permitted to claim.

10.2 Experiment 29: Gate Validation (v1 through v3)

10.2.1 Motivation

Two prior estimator failures in this investigation (Rosenstein λ_1 , Experiments 3/15; correlation dimension, Section 5) were deployed on real or synthetic-unknown data before validation against ground truth. The estimator validation rule (Section 1.2) requires control signals with known topology to be run through the pipeline before any real dataset is interpreted.

10.2.2 Method (v1, Failed)

Signals with known structure: pure sine (period 100, one H1 loop expected), quasiperiodic 2-torus (golden-ratio frequency ratio, two H1 loops expected), Lorenz (x, y, z) channels (dominant loop(s) expected, per the prior synthetic pilot), white noise and AR(1) (no loops expected, negative controls). Delay embedding dimension $d = 3$, $n_{\text{points}} = 800$, τ selected as the first strict local minimum of mutual information, floor from a single shared white-noise null. Pre-registered pass criteria: dominance ratio $\text{top1/top2} \geq 3$ and $\text{top1/floor} \geq 3$ for structured signals.

10.2.3 v1 Observations and Diagnosis

[OBS] Hard gate failures on A (sine: $\text{top1/top2} = 141.89$, $\text{top1/floor} = 2.61$, $\tau = 5$) and C (Lorenz: best channel $\text{top1/floor} = 2.59$, just under the 3.0 bar). Gates D, E, F (noise, AR(1), seed stability) passed cleanly. **[HYP]** H1 (τ under-selection, medium-high confidence): the MI first-strict-local-minimum rule is brittle to estimation noise; $\tau = 5$ on a period-100 sine is diagnostic of a spurious early dip, compressing the embedding

and shrinking absolute lifetimes while dominance survives. **[HYP]** H2 (invalid cross-signal floor, medium confidence): Vietoris–Rips lifetimes are not comparable across point clouds of different intrinsic dimension; a shared white-noise floor systematically penalises low-dimensional structured signals relative to the higher-dimensional noise cloud. No pre-registered threshold was relaxed in response to these failures; the method was revised and rerun against the same bar (Section 1.2 policy).

10.2.4 Method (v2)

Two changes: robust tau selection (MI curve smoothed, minimum required to hold over a window, autocorrelation-zero fallback) and per-signal shuffle-surrogate floors (10 surrogates per signal, 95th percentile), replacing the single shared white-noise floor. Thresholds unchanged.

10.2.5 v2 Observations

[OBS] Gate A passed ($\text{top1/floor} = 6.89$, $\tau = 22$) but the sine control was found to be degenerate: only 858/1500 embedding points were unique, because the integer period (100) causes exact revisitation under integer sampling. Gate B (torus) failed entirely: $\text{top1/floor} = 1.06$, indistinguishable from its own null. Gate C (Lorenz) failed, and worsened relative to v1 (best channel 1.69 vs v1’s 2.59). **[EST]** Takens’ theorem guarantees a faithful delay embedding only for $d \geq 2 \cdot \dim + 1$. The Lorenz attractor dimension is ≈ 2.06 and a 2-torus has dimension 2; both require $d \geq 5$. The $d = 3$ failures on Gates B and C are consistent with operating below this theoretical guarantee, not with a further pipeline defect. This diagnosis was derived from theory before the v3 rerun, not fitted to the v2 failure after the fact.

10.2.6 Method (v3)

d raised to 5 (primary; $d = 3$ retained as a diagnostic arm), n_{points} raised to 1500, sine period changed to a non-integer value (96.7, removing the v2 degeneracy), Gate F moved from the degenerate sine to Lorenz x , and a new diagnostic Gate G added: H1 persistence computed directly on the true Lorenz (x, y, z) state-space point cloud (no delay embedding), with its own shuffle null, to distinguish “the delay embedding loses information present in the true attractor” from “the statistic is weak even on the true attractor.” Thresholds unchanged from v1/v2; Gate B’s **top1** criterion was promoted from soft to hard at $d = 5$, justified by the Takens bound rather than by the v2 result.

10.2.7 v3 Observations — All Hard Gates Passed

[OBS] The Lorenz z channel is markedly weaker (1.50) than x or y (3.07, 2.66). **[EST]** This is consistent with the known $(x, y, z) \rightarrow (-x, -y, z)$ symmetry of the Lorenz system: a delay embedding from z alone cannot distinguish the two wings of the attractor, a textbook non-generic-observable case. The x/y -versus- z asymmetry is evidence for, not against, the pipeline behaving as reconstruction theory predicts. **[OBS]** Gate G: state-space **top1** (0.9967) is lower in absolute terms than the delay-embedded Lorenz- x **top1** (1.9658); the delay embedding shows *more* apparent loop structure than the true attractor. **[HYP]** The delay embedding of an oscillatory signal may manufacture a large loop from the orbital period itself (at $\tau = 16$, $d = 5$ spans 64 samples, close to one Lorenz

Gate	Criterion	Observed	Pass
A (sine)	$\text{top1/top2} \geq 3, \text{top1/floor} \geq 3$	$\text{top1/floor} = 6.89, \tau = 22$	Yes
B (torus, top1)	$\text{top1/floor} \geq 3$	$\text{top1/floor} = 3.76, \tau = 19$	Yes
B (torus, top2)	$\text{top2/floor} \geq 2$ (soft)	$\text{top2/floor} = 2.73$	Yes
C (Lorenz)	$\text{median} \geq 2, \text{best} \geq 3$	$x=3.07, y=2.66, z=1.50$	Yes
D (white noise)	no dominant loop	$\text{top1/top2}=1.06, \text{top1/floor}=0.84$	Yes
E (AR(1))	no dominant loop	$\text{top1/top2}=1.07, \text{top1/floor}=1.04$	Yes
F (seed stability)	$\text{CV} < 0.20, > 0$	$\text{CV} = 0.0513$	Yes
G (state-space, diagnostic)	$\text{top1/floor} \geq 3$	$\text{top1/floor} = 1.68$	No

Table 26: Experiment 29, v3 gate verdicts. Gate G is diagnostic, not a pass/fail gate for proceeding to real data; its failure sets the instrument-scope statement in Section 10.1.

orbit), independent of the true wing-hole geometry. **Confidence: Medium.. [HYP]** Alternatively the state-space shuffle null may be mildly inflated by the bimodal Lorenz- x marginal (corner-cluster effect under independent-coordinate permutation). **Confidence: Low.:** floor values show only mild inflation (0.59 vs 0.53 white-noise) and do not fully account for the gap.

10.2.8 Verdict and Scope

All hard gates pass at $d = 5$, $n_{\text{points}} = 1500$. The configuration is validated to detect persistent rotational/oscillatory deterministic structure in delay reconstructions relative to a per-channel shuffle null. It is **not** validated as a faithful measure of intrinsic attractor geometry (Gate G). All downstream claims in Section 10.3–10.4 are phrased accordingly. **Confidence: Medium.** on the instrument as scoped; **Confidence: Low.** on any geometric interpretation of its output.

10.3 Experiment 30: Real-Data Structure Statistic (Weather, ETTh1, ETTh2, Lorenz)

10.3.1 Method

The v3-validated configuration ($d = 5$, $n_{\text{points}} = 1500$, robust tau, per-channel shuffle floors, $M = 10$ surrogates, 95th percentile) applied per channel to Weather (21 channels), ETTh1 (7), ETTh2 (7), and Lorenz (x, y, z) as the in-run reference. Structure threshold ($\text{top1/floor} \geq 3$) inherited unchanged from the gates. Degenerate channels (near-constant, e.g. **rain**, **raining**; flagged by low unique-point count) are excluded from summary statistics. Statistical caution: channels within a dataset share physical drivers and are not independent samples; only medians, IQRs, and structured- channel fractions are reported, with no cross-dataset significance test attempted.

10.3.2 Observations

[OBS] Dataset medians order: Lorenz (2.66) > Weather (1.49) > ETTh1 (0.76) > ETTh2 (0.66). Both ETTh boxes sit at or below their own permutation nulls. **[PAT]** Weather splits internally along a clean physical boundary: the thermodynamic/humidity channels (**rh**, **VPdef**, **H2OC**, **sh**, **VPact**, **Tpot**, **Tdew**, **Tlog**, **T**, **VPmax**; top1/floor 1.9–3.4) score well above the wind/precipitation/radiation/pressure channels (mostly at or below

Dataset	n channels	frac. structured	top1/floor median	IQR	n_H1 median
Lorenz	3	0.333	2.66	0.78	456
ETTh1	7	0.000	0.76	0.40	1014
ETTh2	7	0.000	0.66	0.19	1132
Weather	21	0.048	1.49	1.17	251

Table 27: Experiment 30. Structured = top1/own-floor ≥ 3 .

≈ 1.5). **[OBS]** Two radiation channels (SWDR, PAR), known to be strongly diurnal, score low (0.39–0.73) despite clear periodicity. **[HYP]** Half-rectified signals (flat at zero overnight) trace an out-and-back path in delay space rather than a rotation around a loop, producing no H1 signal despite genuine periodicity. **Confidence: Medium.:** this example demonstrates the instrument detects smooth rotational oscillation specifically, not periodicity in general, reinforcing the Section 10.1 scope statement.

10.3.3 Terminology Correction

The word “structured” in this section describes the delay-reconstruction statistic only, not the presence or absence of temporal structure in the domain sense. ETTh electricity-load channels are known on independent domain grounds to carry daily/weekly cyclical structure; the correct reading of the ETTh result is that this structure does not register as a smooth loop under Takens reconstruction at hourly sampling and $N = 3000$ points, not that the channels are dynamically featureless.

10.4 Experiment 31: Weather Downsampling Control (Sampling-Density Confound)

10.4.1 Motivation

Weather is sampled at 10-minute resolution (144 samples/day); ETTh is hourly (24 samples/day), a $6\times$ difference in how densely a daily loop is traced in the point cloud. Coarse sampling can fragment loops in a Vietoris–Rips filtration independently of any intrinsic property of the dynamics. This experiment tests whether the Section 10.3 dataset-level separation survives matching Weather’s sampling rate (and, at the decisive arm, its window span) to ETTh’s.

10.4.2 Method

Weather downsampled by factors $\{2, 3, 6\}$ (factor 6 = hourly, matching ETTh1’s sampling rate; with $N_{\text{points_timeseries}} = 3000$ fixed, factor 6 also matches ETTh1’s ~ 125 -day window span). Identical gate-validated pipeline. **Pre-registered contrast, fixed before running:** thermo-group median top1/floor at factor 6 $\geq 2\times$ the ETTh1 median (≥ 1.52) $\Rightarrow$ INTRINSIC (component survives rate and span matching); $\leq 1.2 \Rightarrow$ ARTIFACT (dataset-level claim not supportable); intermediate $\Rightarrow$ inconclusive, dose-response reported without a categorical claim.

Factor	Weather thermo median	Weather other median	ETTh1 median
1 (native)	2.26	0.98	0.76 (ref.)
2	1.36	0.72	—
3	1.21	0.71	—
6 (hourly)	1.12	0.92	0.76

Table 28: Experiment 31. Pre-registered decision variable is the factor-6 thermo median against the 1.52/1.20 bounds above.

10.4.3 Observations

[OBS] Thermo-group median at factor 6 is 1.12, below the 1.2 ARTIFACT threshold. **Pre-registered verdict: ARTIFACT.** The dataset-level Weather-versus-ETTh structure-statistic separation reported in Experiment 30 does not survive matching sampling rate and window span, and is not claimable as evidence of an intrinsic Weather/ETTh difference.

[OBS] The decline is front-loaded: $2.26 \rightarrow 1.36$ at factor 2 alone (a sampling-rate effect), with a slower slide to 1.12 by factor 6. **[OBS]** Pressure (p (mbar)) moves in the *opposite* direction under downsampling: $1.32 \rightarrow 2.40 \rightarrow 2.29 \rightarrow 2.37$. **[HYP]** Downsampling simultaneously extends the window span (21 to 125 days at fixed $N = 3000$); pressure’s synoptic-scale oscillations (period $\sim 3\text{--}7$ days) go from ~ 4 cycles in-window to ~ 25 at still-adequate resolution, so its loop structure strengthens rather than fragments. **Confidence: Medium.:** directly consistent with the counter-trend and with τ diagnostics (below). **[OBS]** At factor 6, channels whose tau selector locks onto the daily cycle ($\tau \approx 13\text{--}14$, i.e. 24 samples/cycle) score 0.84–1.13, while channels locking onto synoptic scales ($\tau \approx 59\text{--}60$) score 1.28–1.68 and hold up best.

10.4.4 Interpretation

[HYP] The structure statistic is a function of measurement conditions (samples-per-cycle, cycles-per-window, signal smoothness), not an intrinsic property of the underlying dynamics. **Confidence: Medium.:** three independent behaviours (decline under rate reduction, increase under span extension, tau-dependent channel survival) are jointly consistent with this account and none is individually explained by an intrinsic- geometry account. A caveat is registered rather than resolved: the gate control signals were validated at ~ 97 samples/cycle and never tested near 24 samples/cycle, so the diurnal-channel decline is plausibly density-and-noise jointly rather than density alone; a dedicated micro-gate (dose–response of a noisy synthetic sine over samples/cycle) would isolate the mechanism and is filed as optional methods hygiene, not on the critical path.

10.4.5 Consequence for the Research Direction

The intrinsic-attractor-geometry form of the boundary hypothesis (originally Branch B1) is retired. It merges into the signal-family / structure-as-sampled candidate (Branch B3). The one result that survives this control is the *within-Weather* channel split at native sampling (Experiment 30), where sampling rate is held constant across all 21 channels by construction and cannot explain the split.

10.4.6 *The Surviving Falsifiable Prediction*

If Panda’s Weather advantage depends on resolvable smooth oscillation of the kind this instrument detects, then **Panda’s advantage over Chronos on Weather should shrink when Weather is downsampled to hourly**, independent of what the structure statistic itself does. This is an intervention on model behaviour, not a further correlational reading of the statistic, and is the strongest test available in the boundary-hypothesis branch. Two competing accounts of a shrinking advantage are distinguished by decomposing which model moves (Section 1.2 policy): if Panda’s MAE rises while Chronos is stable, the structure hypothesis is supported (H-i); if Chronos’s MAE falls while Panda is stable, the result instead reflects Chronos’s training corpus being denser in hourly/daily-frequency data, i.e. hourly Weather becoming more in-distribution for Chronos specifically (H-ii). This experiment is queued (Section 13) and will also serve, in the same run, as the per-channel data source for testing whether the Section 10.3 thermo/other split predicts per-channel Panda advantage (Section 13).

11 Confirmatory Discriminator Experiment: B2a

11.1 Background and Motivation

Two open items from Sections 6 and 10 converged onto a single experiment. First, the Section 6 revision (Chronos Heterogeneity Calibration Experiment) identified an unrun discriminator: whether the sensor heterogeneity bottleneck (Experiments 24–26) reflects joint channel attention being corrupted by mixing dynamically incompatible channels (**H-attention**) or simply reflects that the “heterogeneous” subset happens to contain individually harder channels (**H-signal**), with no cross-channel interaction required to explain the degradation. Second, the Research Tree’s Cross-Branch Check (Section 13) flagged a risk: the Section 10.3 structure statistic and the Section 6 heterogeneity subsets were developed independently but might be measuring overlapping things.

11.2 Experiment 32: Structure–Heterogeneity Unification Check

11.2.1 Motivation

Before designing a discriminator for H-attention versus H-signal, it was necessary to check whether the existing `homo_matched` and `hetero_controlled` subsets (Experiments 25–26) could even support such a discrimination cleanly, or whether they were already confounded with the independently-derived structure statistic (Section 10.3).

11.2.2 Method

The per-channel structure statistic (top1/floor) from Experiment 30 was cross-referenced against the verbatim channel indices of `homo_matched` and `hetero_controlled` as saved in the Experiment 25–26 notebook outputs. No new model runs were required.

11.2.3 Observations

[OBS] `homo_matched` = channels [VPact, sh, VPmax, Tdew, VPdef, Tpot, T] — 7 of 7 fall in the Section 10.3 thermodynamic/high-structure group (top1/floor range 1.94–2.69, minimum 1.94). **[OBS]** `hetero_controlled` = channels [SWDR, rh, Tlog, VPdef, max.wv, Tpot, OT] — 4 of 7 are thermodynamic/ high-structure, 3 of 7 are low-structure (top1/floor range 0.72–1.05). **[OBS]** The original (pre-difficulty-matched) stratification subsets from Experiment 24 show the same pattern: the homogeneous subset is 6 of 7 thermodynamic; the heterogeneous subset mixes 4 high-structure with 3 low-structure channels.

11.2.4 Critical Finding

[PAT] As operationalised in Experiments 24–26, “heterogeneous” means, to a substantial and previously unquantified degree, *mixing high-structure with low-structure channels*, and “homogeneous” means *uniformly high-structure*. This overlap is partly expected, since the heterogeneity index (Experiments 24–26) and the structure statistic (Section 10.3) were both built from overlapping classes of dynamical features. The consequence is not merely descriptive: **the existing subset design cannot distinguish “mixing incompatible channels corrupts joint attention” from “low-structure channels are individually harder for Panda,”** because both explanations predict the

same observed pattern on the existing subsets. The heterogeneity finding (Section 6) and the structure finding (Section 10.3) are substantially entangled rather than two independent lines of evidence, confirming the risk flagged in the Research Tree’s Cross-Branch Check.

Confidence: High. for the entanglement observation (a direct index cross-reference, not an estimated quantity). **Confidence: Medium.** for its consequence for interpretability of Experiments 24–26, since the degree of overlap (4/7, not 7/7 or 0/7) leaves some room for partial separation.

11.3 Experiment 33: B2a Three-Arm Discriminator

11.3.1 Motivation

Given Experiment 32, a clean discriminator between H-attention and H-signal requires a third arm that breaks the mixing/structure confound: a subset that is *uniformly low-structure* but involves *no mixing*. H-attention predicts this arm should recover toward homogeneous-level performance (nothing incompatible is being attended to jointly). H-signal predicts this arm should degrade to heterogeneous-level performance regardless (individual channels are simply harder). Combined with a univariate-vs-multivariate ablation on each arm, the design discriminates in both directions simultaneously.

11.3.2 Method

Three arms, all $N_{\text{channels}} = 7$, $H = 96$, $n_{\text{windows}} = 20$ (upgraded from the $n = 8$ used in Experiments 24–26, per the Section 1.2 confirmatory-status policy):

- **homo_matched** — unchanged channel set from Experiment 25 (uniformly high-structure).
- **hetero_controlled** — unchanged channel set from Experiment 26 (mixed structure).
- **homo_low** (new) — channels [wv, max.wv, SWDR, PAR, wd, OT, (+1)], pre-registered as all having $\text{top1/floor} \leq 1.49$, i.e. uniformly low-structure with no mixing.

For each arm: (a) multivariate Panda vs. Chronos, (b) univariate Panda vs. Chronos, (c) a direct paired Wilcoxon test of univariate vs. multivariate Panda (the H-attention readout). A full 21-channel Weather arm (Arm 0) was also run with per-channel recording, to compute a structure-vs-advantage Spearman correlation with a difficulty-controlled partial correlation.

Pre-registered recovery metric and thresholds (fixed before running): $\text{gap}(\text{mode}) = \text{MAE}_{\text{hetero_controlled}} - \text{MAE}_{\text{homo_matched}}$ for that mode (uni or multi); $\text{recovery} = 1 - \text{gap}_{\text{uni}}/\text{gap}_{\text{multi}}$. $\text{Recovery} \geq 0.5$ with paired $p(\text{uni} < \text{multi}) < 0.05 \Rightarrow \text{H-attention}$; $\text{recovery} \leq 0.2 \Rightarrow \text{H-signal}$; otherwise mixed/inconclusive. **homo_low** predictions: H-attention predicts its advantage should approximately match **homo_matched** (no mixing penalty despite low structure); H-signal predicts its advantage should be small, closer to **hetero_controlled**, despite zero mixing. Structure-advantage correlation: $|\rho| \geq 0.5$ pre-registered as supportive of a structure-driven account.

Arm	Comparison	H	Mode A	Mode B	Adv/Diff	p
Arm 0 (full, 21ch)	multi vs. Chronos	96	0.6386	0.7920	+0.1534	0.001*
homo_matched	multi vs. Chronos	96	0.5412	0.8368	+0.2956	0.001*
homo_matched	uni vs. Chronos	96	0.6150	0.9296	+0.3146	0.000*
homo_matched	uni vs. multi	96	0.6150	0.5412	−0.0738	0.763
hetero_controlled	multi vs. Chronos	96	0.5636	0.8979	+0.3342	0.000*
hetero_controlled	uni vs. Chronos	96	0.6212	0.7630	+0.1418	0.001*
hetero_controlled	uni vs. multi	96	0.6212	0.5636	−0.0576	0.923
homo_low	multi vs. Chronos	96	0.6101	0.8689	+0.2588	0.000*
homo_low	uni vs. Chronos	96	0.6104	0.7912	+0.1808	0.000*
homo_low	uni vs. multi	96	0.6104	0.6101	−0.0003	0.378

Table 29: Experiment 33 (B2a), $n_{\text{windows}} = 20$. Mode A/Mode B columns report MAE for the two models/modes named in the Comparison column, in that order. Adv/Diff = Mode-B-MAE − Mode-A-MAE for Chronos comparisons (positive = Panda better); Mode-A-MAE − Mode-B-MAE for uni-vs-multi rows (positive = uni better). * denotes $p < 0.05$.

11.3.3 Observations

[OBS] $\text{gap}_{\text{multi}} = \text{MAE}_{\text{hetero_controlled, multi}} - \text{MAE}_{\text{homo_matched, multi}} = 0.5636 - 0.5412 = +0.0224$. $\text{gap}_{\text{uni}} = 0.6212 - 0.6150 = +0.0062$. $\text{recovery} = 1 - 0.0062/0.0224 = 0.724$. **[OBS]** The pre-registered recovery threshold ($0.724 \geq 0.5$) is met numerically, but the paired significance requirement is not: paired $p(\text{uni} < \text{multi} \text{ on hetero_controlled}) = 0.923$, far from the $p < 0.05$ requirement. By the pre-registered rule as literally written this triggers neither the H-attention nor the H-signal bucket cleanly; the notebook’s own automatic verdict printed “MIXED / partial recovery.” **[OBS]** All three uni-vs-multi paired tests are individually non-significant: homo_matched $p = 0.763$, hetero_controlled $p = 0.923$, homo_low $p = 0.378$. This is a fourth independent null result on channel attention as a driver of any of these effects (after Experiments 9, 22, and 27). **[OBS]** homo_low’s multi-vs-Chronos advantage (+0.2588) sits between homo_matched (+0.2956) and hetero_controlled (+0.3342) — but hetero_controlled, the mixed-structure arm, has the *largest* advantage of the three, not the smallest. Neither H-attention (which predicts hetero_controlled should be worst) nor H-signal (same prediction) anticipated this ordering. **[OBS]** Structure–advantage correlation (Arm 0, full Weather, per-channel): $\rho = +0.11$, $p = 0.64$. Difficulty-controlled partial correlation: $\rho = -0.075$, $p = 0.76$ (wrong sign relative to the structure hypothesis). Neither approaches the pre-registered $|\rho| \geq 0.5$ bar.

11.3.4 The Result That Matters More Than the Discriminator

[PAT] Comparing homo_matched and hetero_controlled at $n = 20$ against the same two subsets’ $n = 8$ values from Experiments 25–26 (same hardcoded channel indices, same $H=96$): at $n = 8$, Panda MAE was 0.331 (homo) $\rightarrow$ 0.605 (hetero), an 83% increase, reported at the time as “MAE nearly doubles.” At $n = 20$, Panda MAE is 0.5412 (homo) $\rightarrow$ 0.5636 (hetero), a 4% increase. **These do not replicate each other.** The effect this entire discriminator (and, retrospectively, the Section 6 heterogeneity narrative) was built to explain has shrunk by roughly a factor of 20 in absolute terms between $n = 8$ and $n = 20$ on identical channel sets. The ordering has also partially inverted:

`hetero_controlled`’s advantage (+0.3342) now exceeds `homo_matched`’s (+0.2956), the opposite of the direction the Section 6 series reported. **[OBS]** One reassuring control: Arm 0 (full 21-channel Weather, $n = 20$, H=96: Panda = 0.6386, Chronos = 0.7920) lands almost exactly on the independently-run Experiment 8 result at $n = 20$, H=96 (Panda = 0.6378, Chronos = 0.8115). The evaluation harness itself is not broken in general; the discrepancy is specific to the small 7-channel subsets.

11.3.5 *Reading of the Recovery Metric*

The pre-registered recovery number (0.724) should not be treated as evidence for H-attention despite clearing its numeric threshold. It is computed as a ratio of two differences, *neither of which is individually statistically distinguishable from zero* ($p = 0.763$, $p = 0.923$), and both are small relative to the ~ 0.55 – 0.62 MAE scale of the subsets. Dividing two noise-scale quantities produces a high-variance ratio that can clear a threshold by chance; a 72% “recovery” computed this way is not informative. This is a flaw in the pre-registered design itself (the recovery metric’s denominator was not required to be individually significant), not a flaw in the run, and is recorded here so the same design is not reused uncritically.

11.3.6 *Competing Explanations for the Non-Replication*

1. **[HYP]** Small-sample noise inflated the original $n = 8$ result. With only 8 windows, a handful of unusually favourable or unfavourable windows for one subset can produce a large apparent effect that regresses toward a much smaller true effect at $n = 20$. **Confidence: Medium.**: consistent with the direction and rough magnitude of the shrinkage, and with the general pattern in this log of $n = 8$ results not surviving confirmatory reruns (Section 12).
2. **[HYP]** Window-sampling differences: the $n = 20$ windows may span the time series more evenly (covering more of the seasonal cycle) than the original $n = 8$ windows, which may have clustered in a particular season where the homo/hetero gap happened to be larger. **Confidence: Medium.**: plausible, not yet checked against window timestamps.
3. **[HYP]** An implementation difference between the original Experiment 25–26 evaluation function and the Experiment 33 harness (e.g. in window selection or any stochastic component of `panda_forecast`) could produce systematically different per-window MAE distributions. **Confidence: Low.**: no such difference has been identified, and the harness was reused verbatim per the project’s code-reuse convention, but it has not been positively ruled out either.

11.3.7 *What This Means for the Heterogeneity Finding*

The Section 6 heterogeneity bottleneck — previously characterised in Section 12 as the strongest convergent mechanistic finding of the investigation — is not currently confirmed at proper sample size and should not be cited at its previous confidence level until this is resolved. This is a direct application of Rule 7 to the project’s own prior work: the finding was explicitly flagged in the Section 6 revision as pending an $n = 20$ confirmatory rerun, and the rerun, arriving as a byproduct of this discriminator experiment rather than as a dedicated replication, does not confirm it. The H-attention- versus-H-signal question is downstream of, and less important than, this

replication question: there is limited value in discriminating between two mechanisms for an effect whose existence at proper sample size has not been established.

11.3.8 What Is Needed Before Interpreting Further

1. A determinism check on `panda_forecast`: run the identical window through the model twice and confirm identical output, to rule out unaccounted stochasticity as a contributor to the $n = 8/n = 20$ discrepancy.
2. A direct, controlled side-by-side replication of `homo_matched` vs. `hetero_controlled` at both $n = 8$ and $n = 20$ in the same run, with window positions logged and compared against the original Experiment 25–26 windows, to distinguish competing explanations 1 and 2 above.

Neither the pre-registered H-attention/H-signal verdict nor the `homo_low` result should be promoted to a Summary-of-Findings entry until these two checks are complete. The null channel-attention results (a fourth independent null across Experiments 9, 22, 27, 33) and the structure–advantage null correlation are treated differently: these do not depend on the disputed heterogeneity effect being real, and are recorded as standalone findings below.

11.4 Standalone Findings From This Section, Independent of the Replication Question

[OBS] Channel attention shows no detectable effect (all $p > 0.35$) in a fourth independent test, across three different subset compositions, at $n = 20$. Combined with Experiments 9, 22, and 27, channel attention has now failed to show a significant effect in every direct test attempted in this investigation. **[OBS]** The per-channel structure statistic (Section 10) does not predict per-channel Panda advantage on full Weather ($\rho = +0.11$, $p = 0.64$; difficulty-controlled $\rho = -0.075$, $p = 0.76$), independent of and in addition to the Experiment 31 downsampling ARTIFACT verdict. This further weakens Branch B3 (signal family/structure) as a boundary explanation, on top of the sampling-rate confound already identified.

Confidence: High. for the channel-attention null (four independent replications at increasing rigor). **Confidence: High.** for the structure-advantage null correlation (direct computation, no estimator-validity concerns of the kind that affected Sections 5 and 10). **Confidence: Medium.**, pending resolution of the two checks above, for anything concerning the heterogeneity effect’s existence at $n = 20$.

11.5 Experiment 34: Independent Replication (R1 + R2) and Closing Verdict on the Heterogeneity Bottleneck

11.5.1 Motivation

Section 11.4 identified two checks needed before the heterogeneity non-replication in Experiment 33 could be interpreted: a determinism check on `panda_forecast` (R1), and a controlled $n = 8$ -vs- $n = 20$ replication with window positions logged against the original Experiment 25/26 windows (R2). This experiment reports both, run in a notebook independent of Experiment 33’s implementation (separate harness code, reusing only the original model-loading and evaluation primitives verbatim).

11.5.2 Method

R1: `panda_forecast` called twice on an identical window; outputs compared directly. R2: window start positions recomputed exactly via `np.linspace(0, max_start, n_windows, dtype=int)` for both $n = 8$ and $n = 20$ at $H \in \{96, 336\}$, confirmed non-nested (the two window sets share only their first and last start index). Both `homo_matched` and `hetero_controlled` evaluated on both window sets in the same session. Channel indices (`homo_matched` = [6, 8, 5, 3, 7, 2, 1], `hetero_controlled` = [16, 4, 19, 7, 12, 2, 20]) were verified directly against the source cells of the original `new_experiments.ipynb` notebook (Cells 42–43) rather than assumed from memory or prior chat transcription, after an initial concern that they might not match.

11.5.3 Observations

Subset	Window set	H	Panda MAE	Chronos MAE	Adv (p)
<code>homo_matched</code>	$n=8$	96	0.3309	0.7024	+0.3715 (0.0078)
<code>hetero_controlled</code>	$n=8$	96	0.6126	0.8561	+0.2435 (0.0078)
<code>homo_matched</code>	$n=20$	96	0.5539	0.8173	+0.2634 (0.0060)
<code>hetero_controlled</code>	$n=20$	96	0.5609	0.9247	+0.3638 (0.0001)
<code>homo_matched</code>	$n=8$	336	0.8331	1.1787	+0.3456 (0.0039)
<code>hetero_controlled</code>	$n=8$	336	0.8669	1.0580	+0.1911 (0.0273)
<code>homo_matched</code>	$n=20$	336	0.9220	1.1806	+0.2586 (0.0002)
<code>hetero_controlled</code>	$n=20$	336	0.7084	1.0852	+0.3768 (0.0000)

Table 30: Experiment 34 (R1+R2). Independent re-implementation of the `homo_matched`/`hetero_controlled` comparison at matched horizons.

[OBS] R1: `panda_forecast` is bit-for-bit deterministic across repeated calls on an identical window (max abs difference = 0.0). Per-call stochasticity is ruled out as a contributor to any n -dependent discrepancy. **[OBS]** Channel indices for both subsets were confirmed byte-identical to the source notebook’s Cell 42/43 definitions. An initial concern, raised before this direct check, that the hardcoded indices used in this replication might not match the original (based on an earlier, superficially similar but distinct clustering printout from a different, earlier exploratory cell) is resolved: the indices match exactly. **[OBS]** $H=96$: Panda MAE increases 85% from `homo_matched` to `hetero_controlled` at $n = 8$ ($0.331 \rightarrow 0.613$) versus 1.3% at $n = 20$ ($0.554 \rightarrow 0.561$) — closely reproducing Experiment 33’s finding (83% at $n = 8$ vs. 4% at $n = 20$) via an independent code path. **[OBS]** $H=336$: the effect does not merely shrink but reverses sign at $n = 20$. At $n = 8$, `homo_matched` (0.833) is better than `hetero_controlled` (0.867), a 4% degradation in the original direction. At $n = 20$, `homo_matched` (0.922) is *worse* than `hetero_controlled` (0.708), a 23% degradation in the opposite direction. **[OBS]** Seasonal/window-clustering check: $n = 8$ and $n = 20$ window sets both span the full year evenly by construction (`linspace`), day-of-year std 118.4 ($n = 8$) vs. 109.8 ($n = 20$) — not markedly different, and in the direction opposite to what a clustering account would predict (the smaller sample has slightly higher, not lower, temporal spread). The seasonal-clustering explanation (competing explanation 2, Experiment 33) is not supported. **[OBS]** Reproducibility (Check 1) against the logged Experiment 25/26 $n = 8$ values: 2 of 4 subset/horizon combinations match to within 0.001; the other 2 mismatch by 0.0076–0.0079 absolute

(<1.3% relative to the $\sim 0.6\text{--}0.9$ MAE scale). With channel indices now confirmed correct, this small residual is attributed to cross-session numerical drift (different Kaggle session, possible library/driver differences) rather than implementation error, given its magnitude is roughly two orders smaller than the $n = 8$ -vs- $n = 20$ effect being investigated and shows no systematic pattern by subset or horizon.

11.5.4 *Closing Verdict*

[PAT] High confidence. The sensor-heterogeneity bottleneck as characterised in Experiments 24–26 does not hold at proper sample size. This is now a twice-replicated non-replication: two independently implemented evaluations (Experiment 33 and this experiment), on source-verified identical channel indices, with per-call model stochasticity (R1), seasonal window clustering, and channel-index mismatch each directly checked and ruled out as alternative explanations. The H=336 sign reversal is the single strongest piece of evidence against “real but smaller effect”: a genuine architectural sensitivity to heterogeneity should not invert in direction between horizons. **[HYP] High confidence.** By elimination of the other three competing explanations from Experiment 33 (small-sample noise, seasonal clustering, implementation drift), the original $n = 8$ finding is best explained as small-sample variance in the choice of evaluation windows, not a real property of the model.

12 Summary of Findings (Revised, July 2026)

12.1 Solid Observations (High Confidence)

1. Panda has statistically significant MAE advantage over Chronos on Weather at $H = 96, 192, 336$ ($n = 20, p \leq 0.001$), advantage magnitude 0.17–0.24, with a strikingly stable relative skill ($1.27\text{--}1.33\times$ across horizons). This asymmetry cannot be explained by Chronos training-corpus contamination: both Weather and ETT are confirmed zero-shot for Chronos.
2. Channel attention does not drive Panda’s advantage. Univariate Panda performs comparably to or marginally better than multivariate Panda on Weather (Experiment 9), on Burgers at non-chaotic viscosities (Experiment 27), and on all three heterogeneity-series subsets at $n = 20$ (Experiment 33) — four independent null results in total.
3. Panda wins on all tested chaotic **dysts** and Lorenz systems, and on the Harmonic oscillator, directly contradicting chaos-necessity. Under relative skill (July 2026 revision), the Burgers advantage-versus-viscosity trend inverts entirely: no advantage grows with chaos in relative terms, and relative skill peaks at the non-chaotic $\nu = 1.0$ ($3.06\times$).
4. Destroying deterministic temporal structure while exactly preserving the power spectrum (phase surrogate, Experiment 14) degrades Panda by $13\times$ versus Chronos’s $2\times$ (July 2026 revision) — currently the strongest evidence that Panda exploits deterministic structure rather than second-order statistics.
5. Prediction head fine-tuning worsens Panda MAE, ruling out the head as a lightweight recoverable bottleneck (for the tested configuration).

12.2 Medium Confidence Observations

1. The Experiment 20 finding that both models degrade equally under decomposition was itself an overcorrection (July 2026 re-revision): a large shared projection-error component ($\approx +0.27$ MAE, both models) coexists with a smaller Panda-specific full-signal-dependence component ($\approx +0.20$ at $H = 96$), strongest at short horizon.
2. The complexity continuum’s non-monotonicity is real but its headline framing overstated: under relative skill, cross-system variance is mostly Chronos-side (near-ceiling Panda on 4/5 systems) plus one both-fail case (Duffing), not a genuine anomaly in Panda’s behaviour.
3. Diversity subsampling advantage over stratified subsampling is not reproducible across seeds; Panda absolute MAE is invariant to subsampling method, Chronos drives the observed differences.
4. The Koopman lifting ablation at 50k steps (preliminary, Section 7) shows ablation consistently worse than the matched-steps baseline on Burgers and roughly neutral-to-better on periodic systems, but this is confounded by undertraining and lacks a direct significance test; not citable until the 100k rerun, which is launched but not yet gated (Section 7 revision).
5. The per-channel structure statistic and the heterogeneity subset construction are substantially entangled (Experiment 32): **homo_matched** is 7/7 high-structure channels and **hetero_controlled** mixes 4 high-structure with 3 low-structure channels. Experiments 24–26 cannot, on their own, distinguish a joint-attention-corruption mechanism from a per-channel-difficulty mechanism.

6. *Addendum (July 2026): Koopman Feature-Space Geometry (A3)*. [OBS] Neither of two independent tests of the Section 7 theoretical account (eDMD linearizability, lift Jacobian sensitivity) supports it. Burgers $\nu = 1.0$ – the class Section 7’s account predicts should show the cleanest linear structure – is instead the class where a fitted linear operator performs *worse* than trivial persistence; the lift’s Jacobian sensitivity shows no separation between chaotic and non-chaotic classes once a patch-position artifact is controlled for. This does not affect Section 7’s own ablation MAE/significance results, which are a different, unaffected measurement (no ablation checkpoint was used in A3); it specifically weakens the mechanistic explanation attached to those results. See Section 8 for full detail. **Confidence: Medium..**
7. *Addendum (July 2026): Temporal-Attention Probes (A2a)*. [OBS] Probe 1’s pre-registered escalation gate (Section 9, Experiment 37) mechanically returns ESCALATE under its literal absolute-MAE specification, but reverses to DO NOT ESCALATE under a scale-corrected relative-degradation reanalysis — the verdict is not robust to this choice. Probe 2’s scale-free asymmetry (Burgers uniquely improves with less context) is substantially reproduced by non-learned linear/mean baselines (Experiment 40), favouring a data-property explanation over an attention-specific one. A2b (the full ablation retrain) was not pursued given this weak evidence base. Temporal attention is now the third of three tested architectural-component candidates (with channel attention and Koopman-lift geometry) to fail to provide clean supporting evidence for Section 7’s Claim 2 pattern. **Confidence: Medium..**
8. *Addendum (July 2026): Downsampled-Weather Intervention (B3c)*. [OBS] The one experiment in this log using the real Panda-vs-Chronos advantage metric (rather than a self-referential ablation) to test a candidate mechanism found no support for resolution-dependency: under the convention that isolates resolution from horizon-length confounds, neither model’s native-vs-hourly MAE change is statistically significant (Panda $p = 0.55$, Chronos $p = 0.18$, $n = 20$, paired). Held at modest confidence given limited power and a weaker-than- Experiment-8 native anchor (a diagnosed window-selection effect, Section 14.2). **The underlying Weather advantage is unaffected.** This is the fourth of four tested candidate mechanisms (channel attention, Koopman-lift geometry, temporal attention, resolution-dependency) to fail to explain the advantage; see Section 14. **Confidence: Medium..**
9. *Addendum (July 2026): Burgers Representation Test (B3b)*. [OBS] The Burgers advantage is representation-robust at $\nu = 0.05$ across three encodings (PCA, Sub-sample, Fourier; two cited from Experiment 12, one fresh). At $\nu = 1.0$, formally MIXED under the pre-registered rule at both $n = 8$ and a confirmatory $n = 20$ rerun, but the direction is consistent across all three representations at $n = 20$ (Fourier’s sign flipped positive from the $n = 8$ run), with significance limited by that cell’s high inherent noise rather than a genuine disagreement between representations. A3’s Burgers dead-channel finding (Section 8) is re-characterised as reflecting $\nu = 1.0$ ’s intrinsic low-dimensionality rather than PCA-specific damage: effective rank is statistically indistinguishable between PCA and Fourier at this viscosity despite very different literal dead-channel counts. **Confidence: Medium..**
10. *Addendum (July 2026): Chronos Horizon-Mismatch Check (G5)*. [OBS] Chronos-t5-small’s documented ≤ 64 -step recommended prediction length (confirmed active in this project’s own pipeline, Section 16) does not explain Panda’s Weather advantage: relative skill is essentially unchanged moving Chronos from 32 steps out of spec (H=96) to exactly in spec (H=64), and moves in the opposite direction at H=32.

This closes the immediate concern for Weather specifically; other datasets and the project’s longer horizons ($H > 96$) remain untested for this confound. **Confidence: Medium.** for Weather; untested elsewhere.

12.3 Under Active Dispute (New, July 2026) — Resolved Below

1. **The sensor heterogeneity bottleneck (Experiments 24–26), previously listed as the strongest convergent mechanistic finding in this investigation, did not replicate at $n = 20$ on the identical channel subsets (Experiment 33).** Panda MAE increased 83% from homo_matched to hetero_controlled at $n = 8$ versus 4% at $n = 20$; the relative ordering of the two subsets’ advantage also inverted. This is downgraded from Section 12.1 (Solid Observations) pending the determinism check and controlled $n = 8$ -vs- $n = 20$ replication described in Section 11.4. Until resolved, no claim about sensor heterogeneity as an architectural bottleneck should be cited as established, and the downstream motivation for node embeddings / G-SWaN / XXLTraffic (Section 13, Branch B2) is correspondingly weakened rather than strengthened by Experiment 33.

Resolution (July 2026, Experiment 34): R1 (determinism) and R2 (controlled $n = 8$ -vs- $n = 20$ replication) are complete, run independently of Experiment 33 with channel indices verified against source. The non-replication is confirmed, not merely still-pending: an 85%-to-1.3% collapse at $H=96$ and a full sign reversal at $H=336$ (see Experiment 34 for full results). This entry is retained here as the historical record of the dispute as it stood before resolution; the resolved status and its consequences for Branch B2 are recorded in Section 12.4 below. The heterogeneity bottleneck is no longer merely under dispute — it is retired.

12.4 Retired or Downgraded (July 2026)

1. **Sensor heterogeneity bottleneck (Experiments 24–26):** retired, per the resolution above and Experiment 34’s independent replication. **High confidence.** The effect does not survive proper sample size ($n = 20$), confirmed twice independently with per-call stochasticity, seasonal window clustering, and channel-index mismatch each directly checked and ruled out as alternative explanations. **[HYP]** The original $n = 8$ result is best explained as small-sample variance in window selection, not a real architectural sensitivity, by elimination of the other competing explanations. **Confidence: Medium.** on this specific causal account, though **Confidence: High.** that the original finding itself does not hold. Consequence for Branch B2: node embeddings (B2b), XXLTraffic (B2c), and any further heterogeneity-motivated intervention lose their entire motivation, not merely their gating condition — both the effect and its proposed attention- based mechanism (independently ruled out four times: Experiments 9, 22, 27, 33) are now unsupported. Branch B2 is retired in full; see Section 13 Research Tree update.
2. **Intrinsic attractor-geometry hypothesis (Branch B1):** retired. The pre-registered downsampling control (Experiment 31) returned an ARTIFACT verdict on the dataset-level Weather-versus- ETTh structure-statistic separation; it does not survive sampling-rate and window-span matching. Merged into the signal-family/structure-as-sampled candidate (Branch B3). The within-Weather channel split (Experiment 30), measured at native sampling with rate held constant, survives

as the residual finding, though it is now also known to be entangled with the (disputed) heterogeneity subsets (Experiment 32) and unsupported as a per-channel advantage predictor (Experiment 33).

3. **Permutation entropy (order 3) and correlation dimension:** both retired as complexity/geometry discriminators. PE cannot separate periodic from chaotic regimes at any tested order; correlation dimension fails its own ground truth on Lorenz ($\hat{d} \approx 0.86$ vs. literature ≈ 2.05), so the previously reported Weather–Lorenz clustering is uninterpretable.
4. **Scalar-offset node-identity proxy (Experiment 22):** retired as a test of the sensor-identity hypothesis; too weak a proxy to distinguish identity from interaction-structure explanations. The univariate-on-heterogeneous discriminator (Experiment 33) that was to supersede it is itself now moot, since the effect it was built to attribute does not survive replication (Experiment 34).
5. **Rosenstein λ_1 on Burgers PCA components:** retired; estimator invalid for high-dimensional PDE observables at the tested embedding dimension.

12.5 Not Established

1. The positive mechanism behind Panda’s Weather and Burgers advantage. Channel attention is now ruled out with high confidence (four independent nulls); the prediction head is ruled out as a lightweight fix; the Koopman lifting ablation completed its 100k evaluation campaign (Section 7) with a dataset-dependent, significance-tested result, not a clean mechanism finding; the lift’s own representational geometry (Section 8, A3) does not support the theoretical account attached to that result either; temporal attention (Section 9, A2a) was tested via cheap inference-time probes and also failed to provide clean supporting evidence, though not with the same rigor as a full ablation (A2b was not pursued). All three tested component-level candidates have now failed to explain the pattern; the mechanism remains open.
2. Whether any architectural modification recovers OOD performance. No intervention tested to date has done so.
3. **Correction (July 2026):** the item previously here read “Distributional (CRPS/Hellinger) validity of any chaotic-regime claim in this log,” stated as a blanket requirement covering every MAE-based result regardless of horizon. This overstated the case. The concern originated from feedback in a meeting with Prof. Salim’s group questioning whether MAE is appropriate for chaotic systems beyond the Lyapunov horizon — a legitimate question, but the blanket framing did not distinguish *which* results it actually applies to. Checking directly against the published paper’s own stated evaluation philosophy (Section 5, and the paper’s explicit framing of Panda as a “weather” model — short-term pointwise accuracy — as distinct from a “climate” model that prioritizes long-run invariant statistics over pointwise accuracy): **[OBS]** the paper’s own primary metrics are MSE, MAE, sMAPE, and Spearman correlation, all pointwise; its one distributional/attractor-fidelity metric is correlation dimension (Grassberger-Procaccia), applied specifically to the masked-completion (MLM) task, not to the standard forecasting benchmark. The sharper, corrected framing: **MAE at trained or near-trained horizons ($H \leq 128$, single-pass, no autoregressive rollout) is an adequate metric per the paper’s own design philosophy and does not require distributional revalidation to be cited.** The legitimate open question is narrower: **whether long-horizon autoregressive rollout**

forecasts (H>128, chained passes) preserve correct attractor structure, which pointwise MAE genuinely cannot answer and correlation dimension (matching the paper’s own method, not a generic CRPS/Hellinger sweep) is the appropriate tool for. This affects a specific subset of results in this log — primarily the H=336 rollout claims in Section 7 (the Rossler significance result, the Burgers/Harmonic OOD rows) — not every MAE number that has ever been reported. **[OBS]** The early benchmark run’s preview (MAE-winner and Hellinger-winner disagreeing in 9/33 rows, Double Pendulum and Lorenz $\rho = 10$ flagged as at risk) remains a real, unretracted observation — it is not invalidated by this correction. What is corrected is the inference drawn from it: this observation alone does not establish that *all* MAE claims in this log are suspect, only that horizon-stratified revalidation (short vs. rollout) of those specific flagged rows is worth doing before treating them as settled.

13 Research Tree and Next Experiments (Restructured, July 2026)

The tree below replaces the original data-type-organised proposed- directions list. It is organised by hypothesis and gated by outcome rather than by dataset, since the central finding of this investigation — that the Weather/ETTh boundary does not track data type — makes a type-organised tree the wrong ontology for planning further work. Status tags: **[done]** completed, **[queued]** designed and scheduled, **[blocked]** blocked on a resource or upstream result, **[waiting]** awaiting external input (Prof. Salim’s group), **[conditional]** conditional on another node’s outcome, **[retired]** retired.

13.1 Layer 0 — Validity Gates

The “standardising methodology” direction. These gate every chaotic-regime claim above.

[done] G3 — Contamination check. ETT and Jena-Weather confirmed absent from the Chronos training corpus (Section 1.1).

[queued] G1 — Correlation-dimension revalidation (corrected scope, July 2026). Previously scoped as a blanket “CRPS/Hellinger revalidation” of every chaotic-regime MAE claim in this log; that scope is corrected (Section 12, Not Established, item 3) to match the published paper’s own evaluation philosophy and actual distributional metric (correlation dimension via Grassberger-Procaccia, not a generic distributional sweep). **Narrowed scope:** rollout-horizon ($H > 128$) claims only — specifically the Section 7 Rossler significance result and the Burgers/Harmonic OOD rows at $H=336$ — plus horizon-stratification of the early benchmark’s flagged 9/33 MAE/Hellinger disagreement rows (Double Pendulum, Lorenz $\rho = 10$) to check whether the disagreement is rollout-specific. Short-horizon ($H \leq 128$) MAE claims, which constitute most of this log’s significance-tested results, do not require this revalidation. **Priority downgraded** from “highest remaining priority in Layer 0” to standard queue position, reflecting the narrowed scope. Raw predictions for the Section 7 campaign are already saved (Section 1.2 policy), so this remains a re-scoring task, not a rerun.

[done] [conditional] G2 — Confirmatory $n = 20$ reruns. Partially complete via Experiment 33 (heterogeneity chain), with a materially important negative outcome (Section 12.3). Complexity continuum and surrogate control reruns remain **[queued]**. ETTh2 $H=336$ anomaly ($p = 0.013$) remains unreplicated.

[queued] G4 — Classical baseline column. Seasonal-naive/DLinear on Weather and ETTh. No result in this log currently rules out a classical model beating both Panda and Chronos on ETTh, which would reframe “Panda fails to generalise to ETTh” as “ETTh favours simpler models entirely.”

13.2 Branch A — Positive Mechanism

[done] A1 — Koopman lifting, 100k matched-steps rerun. Correction (July 2026): this entry previously read **[blocked]** → “in progress,” describing a pre-completion state (baseline segment 3 running, ablation resume pending, gate not yet reported). This was stale — it had not been updated after Section 7’s “100k Evaluation” subsection was written up and completed, and the two were directly contradicting each other within

this same document. **Corrected here to match Section 7’s actual reported content, rather than silently edited.** Both arms reached 100k steps and were fully evaluated (Section 7): the pre-registered blind gate was compromised (exploratory re-analysis throughout, stated explicitly), the original gate FAILED for both conditions, and a full campaign followed — three Lorenz protocol variants, two held-out systems (Rossler, SprottB), paired significance testing (6/10 nominally significant, 5/6 surviving strict Bonferroni), a 50k→100k trend check, a methodological comparison against the published setup, and a 5-system/3-horizon OOD evaluation. **Result: ablation significantly beats baseline on in-distribution/held-out chaotic ODEs; baseline beats ablation on Burgers $\nu = 1.0$; ablation beats baseline on Harmonic.** This behavioural pattern is established (medium-high confidence, per Section 7’s own caveats: single trajectory per system, window-overlap in the significance tests — see the held caveat below). The theoretical account originally offered for *why* this pattern occurs is separately tested and not supported (A3, Section 8).

[done] A2a — Cheap temporal-attention probes (no retrain). Complete (Experiments 37–40, Section 9). Patch-order shuffling (Probe 1, the pre-registered gate), context-length truncation (Probe 2), attention-map inspection (Probe 3), plus an addendum testing Probe 2’s asymmetry against non-learned baselines. **Result: weak and partly self-undermining.** Probe 1’s mechanical gate says ESCALATE under its literal (absolute-MAE) specification, but reverses to DO NOT ESCALATE under a scale-corrected relative-degradation reanalysis — the verdict is not robust to this choice, so the gate cannot be treated as clean evidence either way. Probe 2’s asymmetry (Burgers uniquely improves with less context) is genuinely scale-free, but the addendum (Experiment 40) found that non-learned linear/mean baselines reproduce it almost exactly, substantially favouring a data-property explanation (H3) over an attention-mechanism-specific one (H1). Probe 3 is descriptive and $n=1$. **Net: none of the three probes, individually or combined, provides a robust basis for concluding temporal attention explains A1’s Claim 2 pattern.** This is the third of three tested architectural-component candidates (channel attention, Koopman-lift geometry, temporal attention) to fail to provide clean supporting evidence for that pattern.

[queued] A2b — Temporal attention ablation (retrain). Not pursued at this time (deprioritised, not formally rejected). A2a’s evidence base does not clear the bar to justify the expensive retrain, and B3c (below) is unblocked, ready, and more directly targets the project’s original Weather- advantage question. Remains available to revisit if a more decisive cheap test or new motivation for temporal attention specifically emerges.

[done] A3 — Koopman feature-space geometry. Complete (Experiments 35–36, Section 8). Result: negative/mixed. Neither eDMD reconstruction residual nor lift Jacobian sensitivity supports the theoretical account attached to A1 (linearization of non-chaotic dynamics as the mechanism behind the lift’s selective helping/hurting pattern). Burgers shows the *worst* eDMD linear-fittability of the five tested classes, not the best; Jacobian sensitivity shows no chaotic/non-chaotic separation once a patch-position artifact is controlled for. Does not affect A1’s ablation MAE findings directly (different measurement, ablation checkpoint not used here); weakens only the theoretical account layered on top of them.

13.3 Branch B — Generalisation Boundary

Organised by candidate boundary variable, not by dataset.

[retired] **B1 — Intrinsic attractor geometry.** Retired (Experiment 31, ARTIFACT verdict). Merged into B3.

[done] → [retired] **B2a — Univariate Panda on the heterogeneous subset.** Completed as Experiment 33; superseded by the Experiment 34 resolution below. The recovery-metric ambiguity noted at completion (0.724 nominal but resting on two individually non-significant differences) is now moot, since Experiment 34 confirmed the underlying heterogeneity effect does not replicate at $n = 20$ at all — there is no effect left for B2a’s discriminator to attribute to either mechanism.

[done] **B2 unification check.** Completed as Experiment 32: the heterogeneity subsets and the structure statistic are substantially entangled (7/7 vs. 4-high/3-low composition), confirming the risk flagged in the original Cross-Branch Check. Retained as a methodological finding even though the subsets it was checking are now retired.

[done] **Replication check (R1+R2).** Completed as Experiment 34. **Verdict: the heterogeneity bottleneck does not replicate at $n = 20$.** Confirmed independently of Experiment 33, with channel indices verified against source. See Section 12.4 for the full retirement entry.

[retired] **B2b — Proper learned node embeddings (G-SWaN-style). Retired.** Not merely unmet-gate as previously stated — Experiment 34 closes the question the gate was waiting on, in the negative. No further motivation remains for this direction absent a new, independent reason to suspect a heterogeneity-related bottleneck.

[retired] **B2c — XXLTraffic. Retired.** Same basis as B2b.

[queued] **B3 — Signal family / structure-as- sampled.** Primary boundary candidate after B1’s retirement, now further weakened by Experiment 33’s null structure-advantage correlation ($\rho = 0.11$, difficulty-controlled $\rho = -0.075$). Elevated relevance for B3b specifically, given A3’s Burgers eDMD side-finding (Section 8) – see B3b below.

- B3a — Chronos-alone attribution arm (Harmonic with amplitude/frequency/sampling variants), testing the tokenisation-failure account of the continuum result.

[queued] ↑ **B3b — Burgers representation test:** PCA-16 vs. raw-spatial-16 vs. Fourier-16 channels at $\nu \in \{1.0, 0.05\}$, resolving whether the Burgers claim is representation-specific. **Relevance elevated (July 2026):** A3 (Section 8) found 11/16 of Burgers’ PCA channels near-frozen across the full evaluated window and the fitted eDMD operator underperforming trivial persistence – suggestive, independent motivation for checking whether the original Burgers advantage (Experiment 10) is itself representation-specific rather than a genuine PDE-generalisation result.

[done] **B3c — Downsampled-Weather model intervention** (Experiment 31’s surviving prediction). **Complete (Experiment 41, Section 14).** Advantage decomposition (Section 1.2) plus a paired significance test on native-vs-hourly MAE per model. **Result: no support for H-i.** Under the convention that isolates resolution without confounding it with horizon length, neither Panda’s nor Chronos’s MAE changes significantly. The large, significant changes under the horizon-confounded convention are fully explained by that convention’s known difficulty confound. Held at modest confidence: power is limited at $n = 20$, off a `native_H96` anchor weaker than Experiment 8’s own (a diagnosed, understood window-selection effect, not a bug). The underlying Weather

advantage itself is unaffected and remains established at high confidence by Experiment 8 and Experiment 33.

[conditional] **B4 — Interventions**, all gated on A and B outcomes: node embeddings (gated on B2a positive — **now permanently unmet**, see B2a/B2b above, not merely pending), LoRA/full fine-tune on ETTh [waiting], learned Panda/Chronos gating hybrid (ungated, runnable anytime), mixed pretraining/diversity expansion (gated on mechanism being known — an arbitrary mixture design is not motivated before then).

13.4 New: Replication Lane (added July 2026)

Not present in the original tree; added as a direct consequence of Experiment 33's headline finding. Both items complete as of Experiment 34.

[done] **R1 — Determinism check on panda_forecast**. Complete (Experiment 34): bit-for-bit deterministic, max abs difference = 0.

[done] **R2 — Controlled $n = 8$ vs. $n = 20$ side-by-side replication** of homo_matched vs. hetero_controlled. Complete (Experiment 34): non-replication confirmed, with channel indices verified against source and seasonal clustering / implementation drift ruled out as alternative explanations. **Verdict: heterogeneity bottleneck retired (Section 12.4)**. B2b and B2c, previously blocked pending this result, are now retired outright rather than unblocked.

13.5 Critical Path

Correction (July 2026): this paragraph previously described A1 as “in progress,” which is stale for the same reason noted at the A1 tree entry above; A1 is complete (Section 7). The Replication Lane (R1, R2) is also closed, resolving in the negative: Branch B2 (node embeddings, XXLTraffic) is retired rather than unblocked. A3 (Section 8) is complete, with a negative/mixed result. **A2a is now also complete (Section 9, Experiments 37–40)**, with a weak and partly-self-undermining result: temporal attention is the third of three tested architectural-component candidates (after channel attention and Koopman-lift geometry) to fail to provide a clean mechanistic explanation for Section 7's Claim 2 behavioural pattern. A2b is not pursued at this time (deprioritised, not formally rejected; see the A2a entry above).

With all three Branch A component-level mechanism candidates tested and none providing clean supporting evidence, and **B3c now also complete (Section 14) with no support for the resolution-dependency hypothesis**, the project has four consecutive null results on candidate mechanisms for the Weather/Burgers advantage — three architectural, one resolution-based, the last using the real advantage metric rather than a self-referential comparison. **The underlying advantage itself is unaffected and remains established at high confidence**. Priority order going forward: **B3b (Burgers representation test, foundational — checks whether the second half of the original advantage puzzle is even representation-independent, before further mechanism-hunting there)**, a confirmatory $n = 20$ rerun of the **Harmonic oscillator advantage** (currently resting on an $n = 8$ result, the same evidentiary tier the heterogeneity bottleneck sat at before its collapse), **B3a (Chronos-alone attribution arm)**, and **G1/G4 (validity gates, standard priority)**. A non-architectural alternative — whether the advantage tracks distributional similarity to the skew40 pretraining corpus

rather than any single component — is flagged as the leading candidate direction beyond the current queue, motivated by the pattern of four consecutive component/mechanism nulls, but is not yet formally queued.

Revision (July 2026): B3b and G5 Outcomes

B3b (Burgers representation test) is complete (Section 15, Experiments 42–43). **Result: REPRESENTATION-ROBUST** at $\nu = 0.05$; **MIXED** at $\nu = 1.0$ (formally, by the pre-registered rule, at both $n = 8$ and a confirmatory $n = 20$ rerun), though the $n = 20$ rerun materially strengthens the $\nu = 1.0$ reading from “one representation actively disagrees” to “all three representations agree in direction, one lacks power to prove it.” A secondary finding re-characterises A3’s Burgers dead-channel observation as reflecting intrinsic low-dimensionality of $\nu = 1.0$ dynamics rather than PCA-specific damage (Section 15.6). An unflagged methodological note (Chronos non-determinism across nominally identical repeated calls, Section 15.5) is recorded as an open item, not yet resolved.

G5 (Chronos horizon-mismatch confound check, new gate, not previously on this tree) is complete for its Weather arm (Section 16, Experiment 44). **Result: NOT SUPPORTED** – Chronos’s out-of-spec evaluation horizon does not explain Panda’s Weather advantage. Scope is explicitly limited to Weather at $H \leq 96$; other datasets and the project’s longer horizons (192/336/720) remain untested and are recommended for consolidation into G1’s already-queued rollout-horizon revalidation pass rather than a separate campaign.

Updated priority order: confirmatory $n = 20$ rerun of the Harmonic oscillator advantage (same risk tier the heterogeneity bottleneck sat at before its collapse) → B3a (Chronos-alone attribution arm) → G1/G4 (validity gates; G1 now also carries the horizon-mismatch question for its already-flagged rollout-horizon rows) → Branch B4 interventions (the Panda/Chronos gating hybrid remains ungated and runnable; a trainable Koopman lift, motivated directly by A1+A3’s joint result, is a newly-articulated candidate not previously on this tree).

14 Downsampled-Weather Model Intervention (B3c)

14.1 Background and Motivation

Experiment 31 (Section 10.4) established the ARTIFACT verdict on the structure statistic’s Weather-versus-ETTh separation, but left one surviving falsifiable prediction on the table: *if* Panda’s real advantage over Chronos depends on resolvable fine-grained temporal structure, downsampling Weather to hourly resolution should shrink that advantage. This section reports that intervention. It is deliberately different in kind from every Branch A experiment (A1–A3, A2a): those tested self-referential comparisons (Panda’s own baseline versus ablated checkpoints); this uses the actual Panda-vs-Chronos advantage metric and the published checkpoint, directly targeting the project’s original stated question rather than an internal architectural component.

Object of manipulation: the input data (Weather downsampled to hourly), not the model. No architectural component is touched. This means a result here can identify *whether* Panda’s advantage depends on resolution, but — unlike A1–A3 — cannot by itself identify *which* component is responsible if it does.

14.2 Experiment 41: Design

14.2.1 Method

Published Panda checkpoint (GilpinLab/panda, matching every other advantage-based result in this log; **not** `baseline_100k`, which Section 7 confirmed is undertrained relative to it) versus `amazon/chronos-t5-small`, on three conditions: native 10-minute Weather (`native_H96`, $H = 96$) and hourly-downsampled Weather (factor-6 stride decimation, matching Experiment 31’s method) under two horizon conventions — fixed sample-horizon (`hourly_H96_fixedsample`, $H = 96$ at both resolutions, i.e. 16h native vs. 4 days hourly) and fixed physical horizon (`hourly_H16_fixedphys`, $H = 16$ at hourly resolution, matching native’s 16h physical span exactly). Both conventions are run because they confound differently: fixed-sample-horizon holds step count constant but lets physical forecast length vary with resolution; fixed-physical-horizon holds physical length constant but lets step count (and therefore task difficulty) vary. $n_{\text{windows}} = 20$, `CONTEXT_LEN = 512` (Panda’s fixed native context length, architecture-determined, not a free choice — this means hourly context necessarily spans a much longer real-world period, ~ 21 days vs. native’s ~ 3.6 days, an acknowledged, unavoidable consequence of holding sample-count fixed across resolutions).

Context/target windows for the two resolutions are aligned by shared real-world start timestamp (not independently sampled per resolution), so a given window index corresponds to the same calendar moment at both resolutions.

Advantage decomposition (Section 1.2 policy) is the design’s central discriminator: **H-i (structure hypothesis)** — Panda’s MAE rises at hourly resolution while Chronos is stable; **H-ii (frequency-affinity confound)** — Chronos’s MAE falls while Panda is stable.

14.2.2 Methodological Detours

Six real implementation bugs and one real, non-bug harness discrepancy were found and fixed during this experiment, each via direct diagnosis rather than assumption, recorded here per this log’s established practice (Sections 7, 8, and 10) rather than silently absorbed into the final method.

1. **Datetime parsing.** The Weather file’s actual date format (ISO yyyy-mm-dd, unambiguous) was initially assumed to be the raw Jena dd.mm.yyyy format (day/month-ambiguous, requiring `dayfirst=True`); forcing `dayfirst=True` on an already-ISO string broke parsing outright (`ValueError`). Corrected by detecting the actual delimiter present in the data (`.` vs. `-`) rather than assuming a specific raw-download format.
2. **Duplicate timestamps.** A single duplicate-timestamp row in the real file broke `get_indexer(method='nearest')`, which requires a unique index (`InvalidIndexError`). Corrected by dropping duplicate timestamps (keep-first) at load time, with the drop count logged rather than silently absorbed.
3. **Panda input type.** `PatchTSTPipeline.predict()`’s `_prepare_and_validate_context` asserts `isinstance(context, torch.Tensor)`; an initial implementation passed a plain numpy array (`AssertionError`). Corrected by wrapping in `torch.tensor(...)` before the call. Confirmed via direct inspection of the pipeline’s own docstring that the expected shape is `[context_length x num_channels]` (time-major, matching this project’s window arrays directly — an earlier, now-corrected implementation had also applied an unnecessary transpose here).
4. **Denormalization before error computation.** **[OBS]** An early version denormalized Panda’s forecast back to raw physical units before computing MAE against the raw target, producing a mean MAE of ≈ 496.8 on `native_H96` despite a median of only ≈ 0.585 — a $\sim 850\times$ mean/median gap. Root cause: a single near-zero-variance channel (`rain (mm)`, exactly 0 for the full 512-step context of one window) produced a near-infinite normalized value when divided by `std + 1e-8`. **Two corrections:** (a) error is now computed entirely in normalized units (never denormalized), matching Experiment 8 and every other advantage-based result in this log; (b) a degenerate-channel guard (`std < 10-6` treated as unscaled rather than divided-by) prevents the near-infinite blowup at its source. 8 degenerate channel-windows were flagged and guarded across the full run (both models independently, matching counts). Results are reported both including and excluding degenerate channel-windows throughout this section, per Experiment 30’s own precedent of excluding near-constant channels from summary statistics rather than letting them dominate.
5. **Chronos keyword argument.** `ChronosPipeline.predict()`’s actual signature takes `inputs`, not `context`, as its first parameter (`TypeError: got an unexpected keyword argument 'context'`), confirmed via `inspect.signature(chronos_pipeline)` directly rather than assumed. Corrected.
6. **Chronos sample count.** `fixed_experiments.ipynb`’s own `chronos_forecast` (the function that produced Experiment 8’s Chronos numbers) uses `num_samples=1`, not `num_samples=20`; an initial implementation used 20 samples with median aggregation, plausibly a systematically more accurate (and therefore not comparable) point estimate than the single-draw protocol used everywhere else in this log. Corrected to `num_samples=1` once the true source was consulted directly.

7. **Aggregation statistic.** `fixed_experiments.ipynb`’s `evaluate()` aggregates per-window MAE across windows via `np.median`, not `np.mean`. An initial analysis notebook used the mean throughout, which is not robust to the skewed per-window MAE distributions present in this data (several Weather channels, e.g. `wd (deg)`, `SWDR`, are heavily non-Gaussian across windows) and was traced as a contributing — though not sole — cause of an initial sign reversal in the `native_H96` advantage. Corrected to match the established convention exactly.

One further, non-bug discrepancy, diagnosed and resolved rather than treated as an error: **[OBS]** after items 3–7 above were corrected, Panda’s `native_H96` median MAE (0.7531–0.7819, depending on degenerate-inclusion) remained noticeably above Experiment 8’s reference (0.6378) despite Panda’s own forecasting code being independently verified correct. **[HYP] Confidence: Medium.** A direct diagnostic — rerunning Panda on an independently-constructed, Experiment-8-style window set (same `linspace`-based selection logic applied without B3c’s hourly-context timestamp constraint) — returned a median MAE of 0.644, within 1% of the 0.6378 reference. **This isolates the residual gap to window selection, not to any remaining bug:** B3c’s three conditions share one timestamp-aligned window set (a deliberate design choice, required so that native and hourly windows correspond to the same real-world calendar period), and that shared set is constrained by the much longer real-time span hourly context requires (~ 21 days), which is not the same 20 dates Experiment 8’s own, less-constrained window selection landed on. Weather’s strong seasonality plausibly explains why a different specific 20-day sample produces a non-trivially different median MAE for Panda specifically. **This is an acknowledged, understood limitation of the shared-window design, not an error;** it means `native_H96` in this section is not a literal replication of Experiment 8 on identical data, only on the same protocol.

14.3 Results

Condition	n	Panda MAE	Chronos MAE	Advantage	Rel. skill	p
<i>All channels</i>						
<code>native_H96</code>	20	0.7654	0.8410	+0.0756	1.099	0.088
<code>hourly_H96_fixedsample</code>	20	0.8080	0.7620	−0.0461	0.943	0.905
<code>hourly_H16_fixedphys</code>	20	0.4962	0.4657	−0.0305	0.939	0.943
<i>Degenerate channel-windows excluded (8 flagged, both models)</i>						
<code>native_H96</code>	20	0.7819	0.8566	+0.0747	1.096	0.101
<code>hourly_H96_fixedsample</code>	20	0.8080	0.7620	−0.0461	0.943	0.905
<code>hourly_H16_fixedphys</code>	20	0.4962	0.4657	−0.0305	0.939	0.943

Table 31: Experiment 41 summary. p = Wilcoxon (Chronos > Panda, one-sided), per-window MAE averaged across channels before pairing. Advantage = $\text{MAE}_{\text{Chronos}} - \text{MAE}_{\text{Panda}}$.

[OBS] `native_H96` (clean): Panda 0.7819, Chronos 0.8566, advantage +0.0747 ($p = 0.101$) — both individual model MAEs are within ~ 4 –6% of Experiment 8’s reference values (Panda 0.6378, Chronos 0.8115; the Panda gap is the window-selection effect diagnosed above), but the combined *advantage* is roughly a tenth of Experiment 8’s +0.1534 ($p \leq 0.001$) and does not reach significance at $n = 20$. **[PAT] Confidence: Medium.** Two individually modest, mostly-understood per-model discrepancies compound into a much larger relative

error in their *difference*. The `native_H96` condition here should be treated as a materially weaker anchor than Experiment 8’s own result, not as a fresh independent confirmation of it; every interpretation below inherits this weakness.

14.4 Advantage Decomposition

Convention	Δ Panda	Δ Chronos	Δ Advantage
Fixed sample-horizon (H=96 both)	+0.0261	−0.0947	−0.1208
Fixed physical-horizon (16h both)	−0.2857	−0.3910	−0.1052

Table 32: Decomposition as difference-of-condition-medians (degenerate-excluded), native $\rightarrow$ hourly. Positive Δ = MAE worsens at hourly resolution.

Methodological note on two non-identical “delta” statistics, recorded rather than silently resolved: the table above computes Δ = median(hourly) − median(native), i.e. a difference of two independently-computed medians. The paired significance test below instead computes median(hourly_{*i*} − native_{*i*}) across matched windows *i* — the median of paired differences. These are not algebraically identical at $n = 20$ with skewed per-window MAE, and disagree numerically in this run (e.g. Panda’s fixed-physical-horizon Δ : −0.2857 above vs. −0.1811 in the paired-test table below). **The paired-differences version is treated as the primary, more principled read** throughout the interpretation in this section, since it is the statistic the significance test is actually computed on; the difference-of-medians version is retained here as the original, first-computed table rather than silently discarded, per this log’s revision convention.

14.5 Paired Significance Test

For each model, native-vs-hourly MAE was tested directly on matched windows (two-sided Wilcoxon on paired per-window differences), closing the gap left by the magnitude-only decomposition above.

Model	Convention	Native med.	Hourly med.	Δ (paired)	<i>p</i>
Panda	Fixed sample-horizon	0.7531	0.7185	−0.0196	0.546
Panda	Fixed physical-horizon	0.7531	0.4225	−0.1811	< 0.0001*
Chronos	Fixed sample-horizon	0.7696	0.6477	−0.0578	0.177
Chronos	Fixed physical-horizon	0.7696	0.3838	−0.3418	< 0.0001*

Table 33: Experiment 41, paired native-vs-hourly test, degenerate-excluded, $n = 20$ windows (matched by window index). * denotes $p < 0.05$. All four directions are negative (hourly $\leq$ native); none are in the direction H-i predicts for Panda (Panda *worsening* at hourly resolution).

[OBS] Under fixed-sample-horizon — the convention specifically designed to isolate resolution without also changing physical forecast length — **neither model’s native-vs-hourly change is statistically significant** (Panda $p = 0.546$, Chronos $p = 0.177$).

[OBS] Under fixed-physical-horizon, both models improve hugely and significantly (Panda −24%, Chronos −44% relative), but this convention has a large, anticipated confound by design: $H = 16$ steps is intrinsically an easier forecasting task than $H = 96$, independent of resolution, which the project’s own decomposition logic (Section 1.2) was built to guard against conflating with a genuine resolution effect.

14.6 Interpretation

[PAT] Confidence: Medium. No support is found for H-i (Panda’s advantage depends on resolvable fine temporal structure). Under the cleanest available comparison (fixed-sample-horizon), Panda’s MAE does not change significantly at hourly resolution; if anything the point estimate is a small *improvement*, the opposite of H-i’s prediction. The large, significant changes under fixed-physical-horizon are fully and more parsimoniously explained by that convention’s known horizon-length confound (both models improving on an easier, shorter-horizon task) without invoking resolution or structure at all.

[PAT] Confidence: Low.–Confidence: Medium. No clean support for H-ii either. Chronos’s fixed-sample-horizon change is also non-significant ($p = 0.177$), so this experiment does not cleanly establish a frequency-affinity account any more than a structure account.

This is a genuine negative result, not a null-power artifact to be dismissed, but it should be held at appropriately modest confidence. Two limitations compound: (a) statistical power is modest at $n = 20$, off (b) a `native_H96` anchor that is itself materially weaker than Experiment 8’s own result (the window-selection gap diagnosed in Section 14.2). A properly-powered, cleanly-anchored version of this same test could still turn up an effect this run was underpowered to detect.

What this section does and does not affect. The underlying Weather advantage itself is unaffected — it remains established independently at high confidence by Experiment 8 ($n = 20$, $p \leq 0.001$) and replicated near-exactly by Experiment 33’s Arm 0 (different code path, different session). The weaker `native_H96` anchor in this section is a property of B3c’s own shared-window design (diagnosed above as a real, understood, non-buggy effect of the timestamp-alignment requirement), not new evidence against Experiment 8’s finding.

Combined with Branch A. This is the fourth tested candidate explanation for Panda’s non-chaotic advantage (after channel attention, Koopman-lift geometry, and temporal attention, none of which found support either) and the first to use the real Panda-vs-Chronos advantage metric rather than a self-referential architectural comparison. **The advantage itself remains real and repeatedly replicated; no tested candidate mechanism, architectural or now resolution-based, explains it.**

14.7 What Remains Open

1. The delta-statistic labeling inconsistency (Section 14.4) should be fixed in the analysis notebook before any further use — the decomposition cell’s difference-of-medians and the paired test’s median-of-differences should not share ambiguous terminology.
2. B3b (Burgers representation test, Section 13, Branch B3) remains the standing recommended next foundational check: whether the Burgers $\nu = 1.0$ advantage (Experiment 10) survives a different channel representation, given A3’s independent finding that its PCA representation is heavily diluted by near-dead channels.
3. The Harmonic oscillator advantage (Experiment 19, $+0.370$, $p = 0.004$) has never been confirmed at $n = 20$ using the real Panda-vs-Chronos advantage metric — only via Section 7’s self-referential baseline-vs-ablation comparison. Given this project’s own precedent (the heterogeneity bottleneck’s 83%-to-4% collapse at $n = 20$, Section

11), this should not be treated as a third established advantage target without the same confirmatory rerun applied to Weather and Burgers.

4. The distribution-similarity hypothesis (whether Panda’s advantage tracks statistical proximity to the **skew40** training distribution, rather than any single architectural component) remains untested and is the leading candidate direction given four consecutive null results on component-level and resolution-level explanations.

Confidence: Medium. for the section’s central finding (no support for H-i under the cleanest tested convention). **Confidence: Low.**–**Confidence: Medium.** for treating this as a strong or final word on the resolution-dependency question, given the power and anchor-strength limitations discussed above. **Confidence: High.** that the underlying Weather advantage itself is unaffected by anything in this section.

15 Burgers Representation Test (B3b)

15.1 Background and Motivation

A3 (Section 8) found that 11 of 16 PCA channels of Burgers $\nu = 1.0$ are essentially frozen across the full evaluated window, and that a fitted eDMD operator underperforms trivial persistence on this representation. This raised an open question flagged explicitly in Experiment 10’s July 2026 revision: whether the original Burgers advantage (Experiment 10, Section 4) is a genuine PDE-generalisation result or an artifact of the PCA-16 representation specifically. B3b tests this directly.

Status of this experiment. B3b is a boundary/confound-elimination test, not a mechanistic one. It establishes whether the Burgers advantage survives a change in spatial-to-channel encoding; it does not probe Panda’s internals and cannot explain *why* the advantage exists where it does. This distinction is maintained throughout what follows.

15.2 Non-Redundancy Note

Experiment 12 (Section 4) already compared PCA against Uniform, Variance-Stratified, and Diversity spatial subsampling at $\nu \in \{0.05, 0.005\}$ and found the advantage direction survives across all of them, though magnitude differs substantially (PCA advantage $\approx 3\times$ the spatial-subsample advantage at $\nu = 0.05$). Re-testing PCA-versus-Subsample at $\nu = 0.05$ would reproduce an already-answered comparison. B3b is scoped accordingly:

- $\nu = 1.0$: PCA, Subsample, and Fourier all run fresh. This regime was never tested for representation dependence, and is the one A3 specifically implicates.
- $\nu = 0.05$: Fourier run fresh (the genuinely untested arm); PCA and Subsample **cited** from Experiment 12 (`fixed_subsampling_results.csv`), not re-simulated.

15.3 Method

Three channel representations of the identical underlying simulated field: **PCA-16** (as in Experiment 10); **Subsample-16** (variance-stratified uniform spatial subsampling, the Experiment 12 fix); **Fourier-16** (real+imaginary parts of the lowest 8 non-DC spatial wavenumbers, phase retained rather than magnitude-only, so as not to structurally handicap this arm relative to the other two). For each ν , the field is simulated once with a fixed seed and all three representations are derived from the same trajectory, isolating representation as the only varying factor. $H = 128$ throughout, matching Experiment 10’s protocol exactly for direct comparability.

Pre-registered decision rule (fixed before running): at a given ν , the advantage is **REPRESENTATION-ROBUST** only if significant ($p < 0.05$) and same-signed under both alternative representations relative to PCA. 1-of-2 survival is **MIXED**; 0-of-2 is **PCA-SPECIFIC**.

Consistency gate: the recomputed PCA arm at $\nu = 1.0$ must match Experiment 10’s logged value ($\text{adv} = +0.0382$, $n = 8$).

15.4 Experiment 42: Initial Run ($n = 8$)

15.4.1 Observations

ν	Representation	Source	Panda MAE	Chronos MAE	Adv (p)
0.05	PCA	Exp. 12 (cited)	0.1374	0.2915	+0.1541 (0.004)*
0.05	Subsample	Exp. 12 (cited)	0.0296	0.0736	+0.0440 (0.004)*
0.05	Fourier	fresh	0.0259	0.0839	+0.0580 (0.004)*
1.00	PCA	fresh	0.0188	0.0650	+0.0461 (0.004)*
1.00	Subsample	fresh	0.0235	0.0481	+0.0246 (0.020)*
1.00	Fourier	fresh	0.0162	0.0139	−0.0023 (0.727)

Table 34: Experiment 42, B3b at $n_{\text{windows}} = 8$. * denotes $p < 0.05$.

[OBS] Consistency gate: recomputed PCA @ $\nu = 1.0$, $\text{adv} = +0.0461$, matches Experiment 10’s logged $+0.0382$ within 0.008 . **Gate: PASS.** **[OBS]** At $\nu = 0.05$, all three representations (two cited, one fresh) are significant and same-signed. **Verdict: REPRESENTATION-ROBUST.** **[OBS]** At $\nu = 1.0$, PCA and Subsample are significant and positive; Fourier is not significant and negative-signed ($p = 0.727$). **Verdict: MIXED (1/2 alternative representations survive).**

15.4.2 Channel-Health Diagnostic: A Design Flaw Caught by Its Own Sanity Check

An input-side diagnostic (no model calls) was added alongside the MAE comparison, computing effective rank (participation ratio of singular values, basis-comparable across representations) and a near-dead-channel count (variance $< 1\%$ of the max-channel variance, **not** basis-comparable) for each representation matrix.

[OBS] A sanity check at $T = 200$, $\nu = 0.05$ found effective rank *identical* across all three representations (1.2/16 for PCA, Subsample, and Fourier alike) while near-dead-channel counts varied sharply (15, 1, and 9 respectively). This is expected, not anomalous: PCA is variance-ordered by construction, so its trailing channels are mechanically low-variance regardless of the field’s true complexity, while Subsample and Fourier are not variance-ordered and do not show the same artifact. **The original degeneracy verdict, which combined both measures ($\text{eff_rank} < 8$ OR $\text{n_dead} \geq 4$), was corrected to use effective rank only**, with near-dead-channel count retained as a descriptive-only field. This mirrors the log’s established practice (Sections 8, 10, 14) of catching and recording confounds during an experiment rather than absorbing them silently into the final method.

15.4.3 A Second, Independently-Discovered Calibration Problem

Once the full $T = 1000$ run (not the $T = 200$ sanity check) was scored: **[OBS]** effective rank across *every* tested $(\nu, \text{representation})$ cell fell in the range $1.42 - 2.11$ out of a possible $16 - 16$ — far below the $\text{eff_rank} \geq 8$ threshold used for the degeneracy flag. *Every cell was flagged degenerate; the flag (dependence) : the threshold itself was never calibrated to Burgers’ actual effective-rank range.* **Decision:**

One genuinely informative fact survives this correction. At $\nu = 1.0$, PCA and Fourier have *identical* effective rank (2.112 vs. 2.112) despite PCA’s MAE-advantage being significant and Fourier’s not. Since effective rank is the one basis-comparable measure available, this rules out the naive account that “Fourier channels are simply

lower-quality/more degenerate than PCA’s” as an explanation for Fourier’s weaker result at this ν – by the one representation- invariant measure available, the two representations carry the same intrinsic complexity.

15.5 Experiment 43: Confirmatory Rerun of the $\nu = 1.0$ Arm ($n = 20$)

15.5.1 Motivation

The Fourier cell at $\nu = 1.0$ ($p = 0.727$, $n = 8$) had two features consistent with an underpowered null rather than a genuine disagreement: `chronos_iqr` (0.0211) exceeded `chronos_mae` (0.0139) itself, and this project has direct, independent precedent (Experiment 34, the heterogeneity-bottleneck non-replication) for $n = 8$ producing misleading results. Since B3b uses only published checkpoints, this is an inference-only rerun – no retraining required.

15.5.2 Method

The $\nu = 1.0$ arm (all three representations) was rerun at $n_{\text{windows}} = 20$. The $\nu = 0.05$ /Fourier cell was left at $n = 8$, since it was not the ambiguous cell and was already REPRESENTATION-ROBUST.

15.5.3 Observations

ν	Representation	n	Panda MAE	Chronos MAE	Adv (p)
1.00	PCA	20	0.0190	0.0527	+0.0337 (0.000)*
1.00	Subsample	20	0.0223	0.0364	+0.0142 (0.000)*
1.00	Fourier	20	0.0146	0.0218	+0.0072 (0.115)

Table 35: Experiment 43, $\nu = 1.0$ confirmatory rerun at $n_{\text{windows}} = 20$. * denotes $p < 0.05$.

[OBS] Consistency gate ($n=20$): recomputed PCA adv = +0.0337 vs. Experiment 10’s logged +0.0382 ($n = 8$), diff = -0.0045 . **Gate: PASS**, a second independent confirmation of harness correctness at a different sample size.

[OBS] Fourier’s advantage **changes sign** between $n = 8$ and $n = 20$: $-0.0023 \rightarrow +0.0072$. Relative skill moves the same direction: $0.858 \rightarrow 1.492$. **[PAT] Confidence: Medium.** This is best read as “real but weak effect, underpowered at $n = 8$ to even recover the correct sign,” not as a genuine representation-specific reversal. The updated cell is directionally consistent with PCA and Subsample at $\nu = 1.0$; only its statistical strength differs, and it remains the noisiest of the three cells at both sample sizes (widest IQR-to-mean ratio throughout).

[OBS] The formal verdict at $\nu = 1.0$ is unchanged by the confirmatory rerun: Subsample clears significance, Fourier does not ($p = 0.115$). **Verdict: MIXED (1/2 alternative representations survive), by the pre-registered rule, at both $n = 8$ and $n = 20$.** The substantive reading is nonetheless materially improved: what looked at $n = 8$ like Fourier actively disagreeing with the other two representations now looks like Fourier weakly agreeing with them, underpowered to prove it.

15.6 An Unflagged Methodological Observation: Chronos Non-Determinism

[OBS] The $\nu = 0.05/\text{Fourier}/n = 8$ cell, held fixed across the initial run and the confirmatory rerun (identical seed, identical inputs, Panda’s own output identical both times per its known determinism – Experiment 34, R1), returned two different Chronos MAE values: 0.0839 in the initial run versus 0.0721 in the confirmatory-rerun session (10.2% relative difference). Advantage moved correspondingly, $+0.0580 \rightarrow +0.0461$.

[HYP] Confidence: Medium. R1 (Experiment 34) established `panda_forecast`’s bit-for-bit determinism but never tested `chronos_forecast`. Given this project’s established convention of `num_samples=1` (a single stochastic draw, not a seeded deterministic decode), Chronos’s own forecasts may not be reproducible run-to-run independent of any window-sampling variance. This is a plausible, previously unflagged contributor to the wide `chronos_iqr` values noted throughout this log (e.g. Section 8’s Jacobian-sensitivity work, and multiple B3b cells above), separate from the small- n window-sampling variance already well-documented (Section 11). This is recorded as an open methodological note, not yet directly tested with a repeated-call determinism check analogous to R1’s for Panda.

15.7 Combined Interpretation

[PAT] Confidence: Medium.–Confidence: High. The Burgers advantage at $\nu = 0.05$ is representation-robust across three independent encodings (two cited, one fresh), strengthening Experiment 10’s original claim at this viscosity beyond what a single representation could support. At $\nu = 1.0$, the advantage direction is now consistent across all three representations at $n = 20$, though only two of three clear significance – a materially better-supported position than the $n = 8$ MIXED verdict alone suggested, but not yet a clean REPRESENTATION-ROBUST result by the pre-registered rule.

[PAT] Confidence: Medium. A3’s “dead channel” finding for Burgers $\nu = 1.0$ PCA is re-characterised by this experiment. Effective rank – the one basis-comparable diagnostic available – is statistically indistinguishable between PCA and Fourier at this ν (2.112 vs. 2.112), despite PCA showing many literal near-dead output channels and Fourier showing far fewer. **This indicates A3’s finding reflects intrinsic low-dimensionality of $\nu = 1.0$ Burgers dynamics (very few true degrees of freedom), which PCA’s variance-ordering makes directly visible as dead channels, rather than damage specific to the PCA representation itself.** Subsample and Fourier carry the same underlying redundancy, just distributed across individually-nonzero-variance channels instead of concentrated into literal zeros.

Confidence: Medium. for the section’s central findings. **Confidence: Low.** for any mechanistic account of *why* Fourier is the weakest-significance arm at $\nu = 1.0$ specifically, which this experiment cannot address (it tests representation robustness of the behavioural pattern, not Panda’s internal processing of any given representation).

16 Chronos Horizon-Mismatch Confound Check (G5)

16.1 Background and Motivation

Chronos-t5-small’s own source (`chronos-forecasting/src/chronos/chronos.py`, `BaseChronosPipeline`) issues the following warning whenever `prediction_length > model.config.prediction_length`:
“*We recommend keeping prediction length $j = 64$. The quality of longer predictions may degrade since the model is not optimized for it.*” This was independently confirmed against the live source (GitHub, `amazon-science/chronos-forecasting`) rather than assumed. `chronos-t5-small`’s `config.prediction_length` is 64, confirmed directly in this project’s own environment (Cell 2, below).

This is a Chronos-specific ceiling, not Panda’s. Panda’s own documented native horizon is 128 (Section 1), with autoregressive rollout beyond that. Nearly every Panda-vs-Chronos MAE comparison in this log uses $H \geq 96$ – out of spec for Chronos across virtually the entire investigation, while Panda is evaluated at or near its own native range for most of the same comparisons. This is a previously unflagged candidate confound, upstream of essentially every headline result in the log, not any single experiment.

Scope of this check. This tests the confound’s direction and approximate size on one dataset (Weather, matching Experiment 8’s exact protocol). It is not a full revalidation of every affected MAE claim in the log; it is the initial dose-response check that determines whether that larger revalidation is warranted.

16.2 Design

Shrinking H makes the forecasting task easier for *both* models (the same confound B3c’s fixed-physical-horizon arm encountered, Section 14). To separate “task got easier for everyone” from “Chronos specifically was pushed out of spec,” the primary metric is **relative skill** (`chronos_mae/panda_mae`, Section 1.2 convention) rather than absolute advantage: if both models simply find shorter horizons easier by similar proportions, relative skill should stay roughly flat as H shrinks; if Chronos specifically closes the gap once H enters its own trained range, relative skill should move toward 1 disproportionately.

Pre-registered decision rule (fixed before running): Experiment 8’s Weather $H = 96$ result is cited as the out-of-spec anchor ($n = 20$, relative skill = 1.272). $H = 64$ (Chronos’s exact native ceiling) and $H = 32$ (extra margin) are run fresh, identical protocol (21 channels, per-window instance normalisation, $n_{\text{windows}} = 20$). **SUPPORTED:** relative skill at $H = 64$ is $\geq 15\%$ lower (relatively) than at $H = 96$, continuing in the same direction at $H = 32$. **NOT SUPPORTED:** relative skill at $H = 64$ is within $\sim 10\%$ of $H = 96$ ’s value. **INCONCLUSIVE:** anything between, reported as a dose-response without a categorical claim (matching Experiment 31’s convention for the same situation).

What this design does not establish, even if SUPPORTED. A clean SUPPORTED verdict would show relative skill moving with H in the predicted direction; it would not by itself prove the mechanism is Chronos’s internal chunking behaviour past 64 specifically, as opposed to some other H -dependent property. A stronger, not-yet-built version would compare Chronos generating $H = 64$ as one native-length call versus an artificially forced two-chunk $32 + 32$ call at the same total H , to isolate the chunking mechanism directly.

16.3 Experiment 44: Weather Dose-Response

16.3.1 Method

Published GilpinLab/panda and amazon/chronos-t5-small checkpoints, full 21-channel Weather, $n_{\text{windows}} = 20$, identical protocol to Experiment 8. $H = 96$ cited (Experiment 8, not rerun); $H = 64$ and $H = 32$ run fresh.

16.3.2 Verification (Cell 2/4a)

[OBS] `chronos_model.model.config.prediction_length = 64`, confirmed via direct introspection in this environment. [OBS] A logging-capture check around a real `chronos_forecast` call at $H = 96$ on an actual Weather context window confirmed the warning fires through this project’s specific pipeline wrapper, returning the identical text quoted above verbatim. This closes the gap between “the warning exists in the published source” and “the warning is actually active in this project’s evaluation calls” – both are now directly confirmed, not assumed.

16.3.3 Observations

H	Source	In spec?	Panda MAE	Chronos MAE	Rel. skill
96	Exp. 8 (cited)	No	0.6378	0.8115	1.272
64	fresh, $n = 20$	Yes	0.5623	0.7158	1.273
32	fresh, $n = 20$	Yes	0.4075	0.5411	1.328

Table 36: Experiment 44, Weather horizon dose-response. Advantage at $H = 64$: +0.1535 ($p = 0.009$). Advantage at $H = 32$: +0.1336 ($p = 0.007$).

[OBS] Relative skill: 1.272 ($H=96$, out-of-spec) $\rightarrow$ 1.273 ($H=64$, in-spec), a +0.08% change $\rightarrow$ 1.328 ($H=32$), a +4.4% change relative to $H=96$.

[PAT] **Confidence: High.** Moving Chronos from 32 steps out of spec to exactly at its trained ceiling produced no meaningful change in relative skill. Moving further into spec ($H=64 \rightarrow 32$) moved relative skill in the *opposite* direction from what the horizon-mismatch hypothesis predicts – Chronos became relatively *worse*, not better, as the horizon shrank further below its own ceiling. **Pre-registered verdict: NOT SUPPORTED**, and more than merely non-significant: the dose-response actively points away from the hypothesis rather than simply failing to confirm it.

[HYP] **Confidence: Low.** A plausible alternative account for the $H=32$ direction: Panda’s chaotic-ODE training may make it disproportionately effective at precise short-horizon dynamics for reasons unrelated to Chronos’s spec at all, which would produce exactly this pattern (relative skill rising as H shrinks) without the Chronos ceiling being involved. Not directly tested here.

16.4 Interpretation and Scope

The specific worry motivating this check – that Chronos’s out-of-spec evaluation was quietly inflating Panda’s Weather advantage – is not supported, on this dataset, at this horizon range. This closes the immediate concern for Weather’s headline result.

What remains open, and should not be conflated with the Weather result above: this check covers one dataset (Weather) and one horizon range ($96 \rightarrow 32$, a modest step). It does not cover Burgers, ETTh, Lorenz, or the `dysts` systems, and does not cover the project’s longer horizons (192/336/720), which sit considerably further out of Chronos’s spec than 96 did. Two of those longer-horizon results are already independently queued for revalidation under G1 for a different reason (the Rossler significance result and the Burgers/ Harmonic OOD rows, Section 7) – folding a horizon-mismatch check into that same pass, rather than treating G5 as closed project-wide, is the recommended next step rather than a new standalone campaign.

Confidence: Medium. that the horizon-mismatch confound does not materially affect the Weather result specifically. **Confidence: Low.** on any claim about its impact elsewhere in the log, which remains untested.